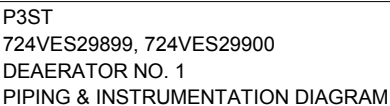




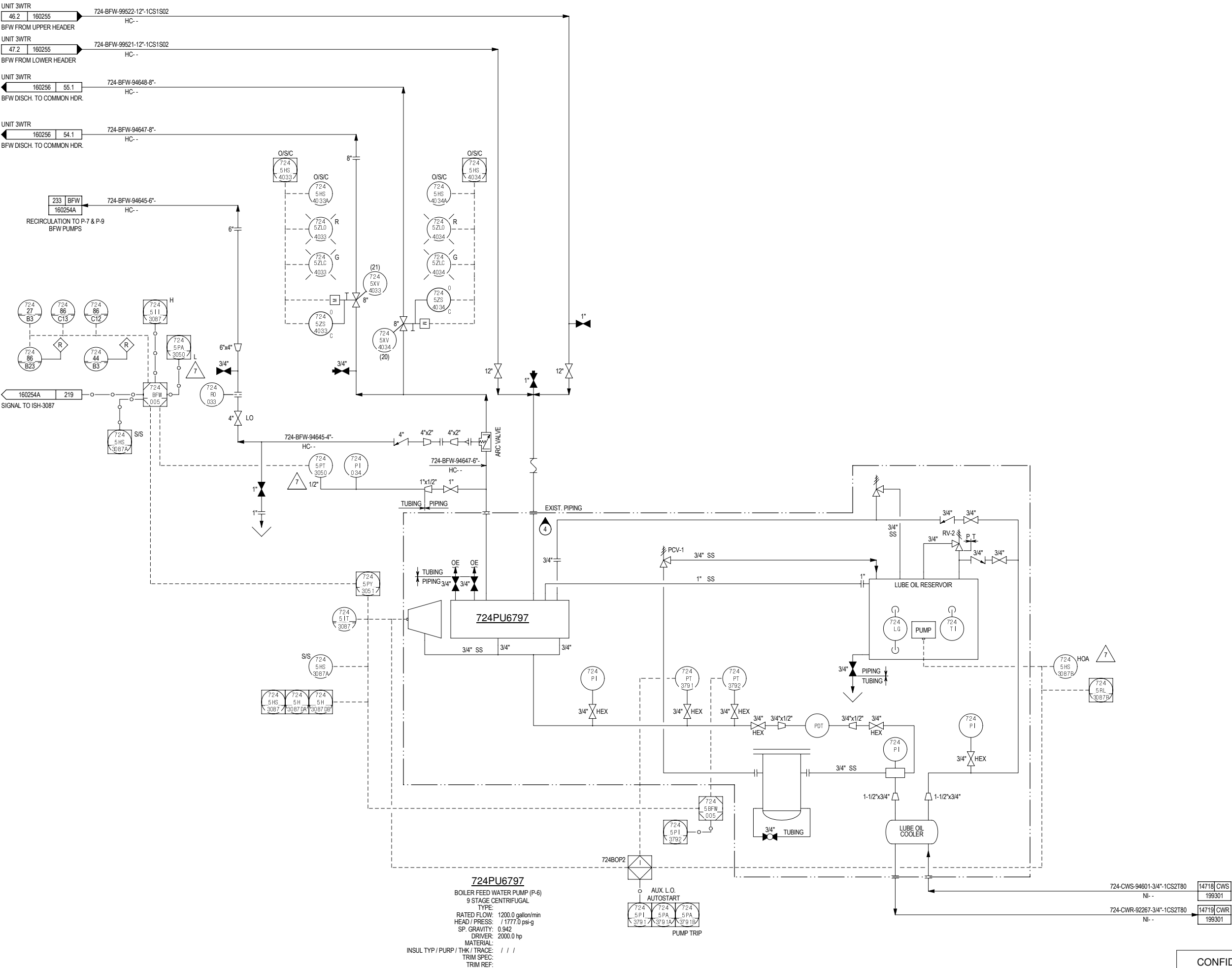
1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199999 TO YC-199999.
3. MINIMUM 1002 LOCKED OPEN AT ALL TIMES.
4. THE 8" GATE VALVES SHALL BE INSTALLED WITH THE STEAM HORIZONTAL.
5. RO-4151 IS SIZED AT 50 GPM.
6. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

00401706.XLSX LINE LIST

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YC- 160248 10



NOTES

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5". UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. FOR 724PU6797 CWS, CWR, PUMP SEALS, VIBRATION AND TEMPERATURE MONITORING SEE DWG. YC-199301.
4. FOR VENDOR SKID P&ID'S, SEE SULZER DWG. #AEP1002909530010-01 SHT. 3 OF 4 AND LUBE-POWER, INC. DWG. #L29222-F.
5. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.
6. YC-160253 HAS BEEN SPLIT INTO TWO DRAWINGS. PLEASE REFERENCE YC-160253A AND YC-160253B.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

8	09/30/20	REVISED PER P3ST WALK DOWN	WOT	MOC 19-1361
7	10/29/19	MOC REVISION (PER USMF0112)	JBB	19-432
6	09/24/14	MOC REVISION	JMC	M20142980
5S	08/05/14	SMARTPLANT CONVERSION		
4	03/14/08	AS BUILT	ARS	4372
3	10/18/07	GENERAL REVISION	ARS	4372
2	05/09/07	ISSUE FOR GENERAL MOV REV	ARS	4372
				3742
REV	DATE	DESCRIPTION	BY	CHARGE

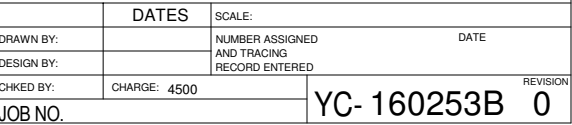
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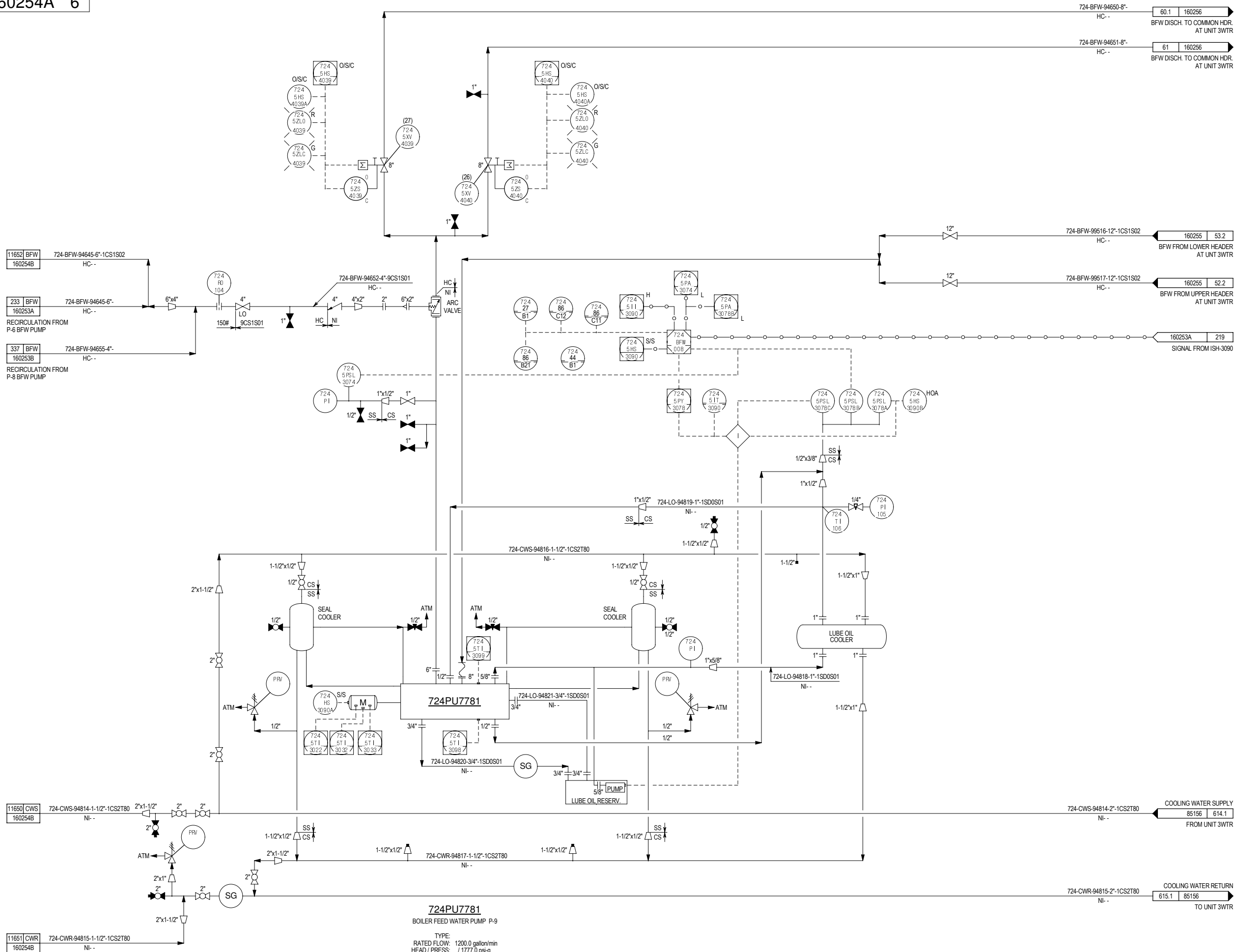
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P3ST
724PU6797
BOILER FEED WATER SHT 1 OF 2
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: TP	2/1/1996	NUMBER ASSIGNED	DATE
DESIGN BY: TJ	11/1/1995	AND TRACING	05/01/96
CHECKED BY: TJ	CHARGE: 4372	RECORD ENTERED	REVISION
JOB NO.		YC- 160253A	8





NOTES

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.
4. YC-160254 HAS BEEN SPLIT INTO TWO DRAWINGS. PLEASE REFERENCE YC-160254A AND YC-160254B.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

6	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
5S	08/05/14	SMARTPLANT CONVERSION		
4	03/14/08	AS BUILT	ARS	4372
3	10/23/07	GENERAL REVISION	ARS	4372
2	05/09/07	ISSUE FOR GENERAL MOV REV	ARS	4372
1	07/20/06	ISSUE FOR WTP3 INSTRUMENTATION	ARS	4372
				3742
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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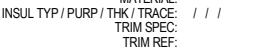


P3ST
724PU7781
BOILER FEED WATER
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: TP	2/1/1996	NUMBER ASSIGNED	DATE
DESIGN BY: TJ	5/1/1996	AND TRACING	
CHECKED BY: TJ	CHARGE: 4372	RECORD ENTERED	05/01/96
JOB NO.		YC- 160254A	REVISION 6

CONFIDENTIAL

YC- 160254A 6



1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.
4. YC-160254 HAS BEEN SPLIT INTO TWO DRAWINGS. PLEASE REFERENCE YC-160254A AND YC-160254B.

00401706.XLSX LINE LIST

NOTE

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P3ST
724PU6798
BOILER FEED WATER
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY:		NUMBER ASSIGNED	DATE
DESIGN BY:		AND TRACING	
CHECKED BY:	CHARGE:	RECORD ENTERED	
JOB NO.	YC- 160254B		0

1. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-1999989 TO YC-1999999.
2. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

00401706.XLSX LINE LIST

4	04/01/20	REVISED PER P3ST WALK DOWN		
3S	08/06/14	SMARTPLANT CONVERSION	WCT	MOC 19-1361
2	08/14/00	REVISED EMS SYSTEM	EM	3925
1	08/18/99	ADDED AT-3199	KL	3951
				3742
REV	DATE	DESCRIPTION	BY	CHARGE

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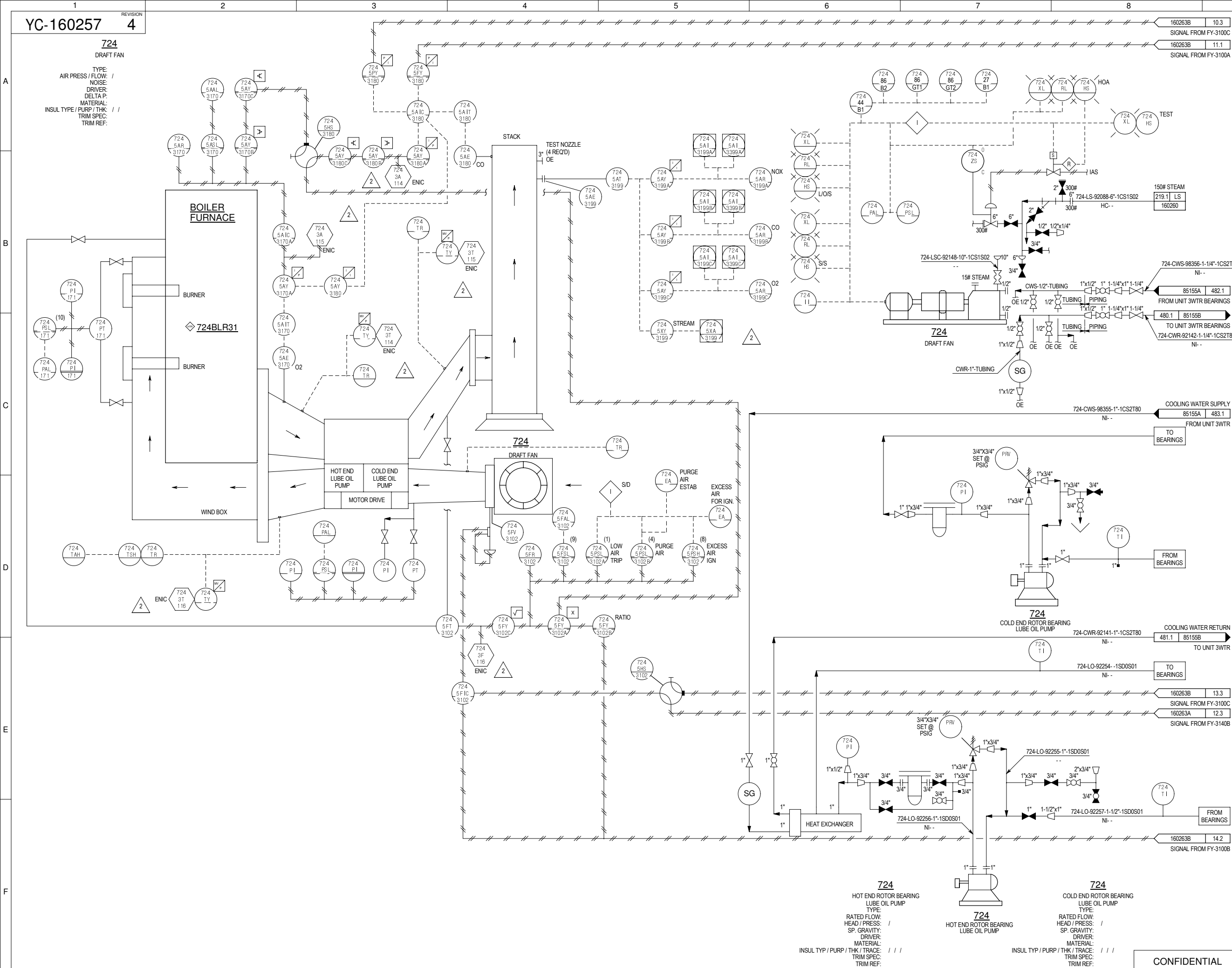
PORT ARTHUR
REFINERY

	DATES	SCALE: NONE	
DRAWN BY: JRC	2/1/1996	NUMBER ASSIGNED AND TRACING	DATE 5-1-96
DESIGN BY: TJ	2/1/1996	RECORD ENTERED	

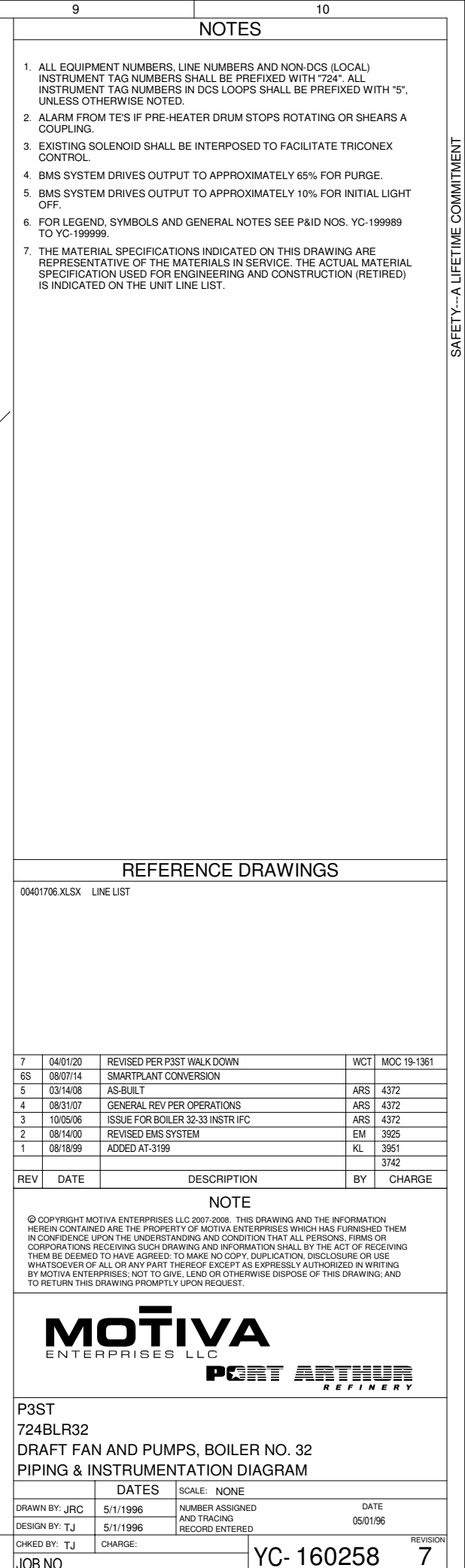
YC- 160257 4

SAFETY---A LIFETIME COMMITMENT

SPPID REV 0



CONFIDENTIAL

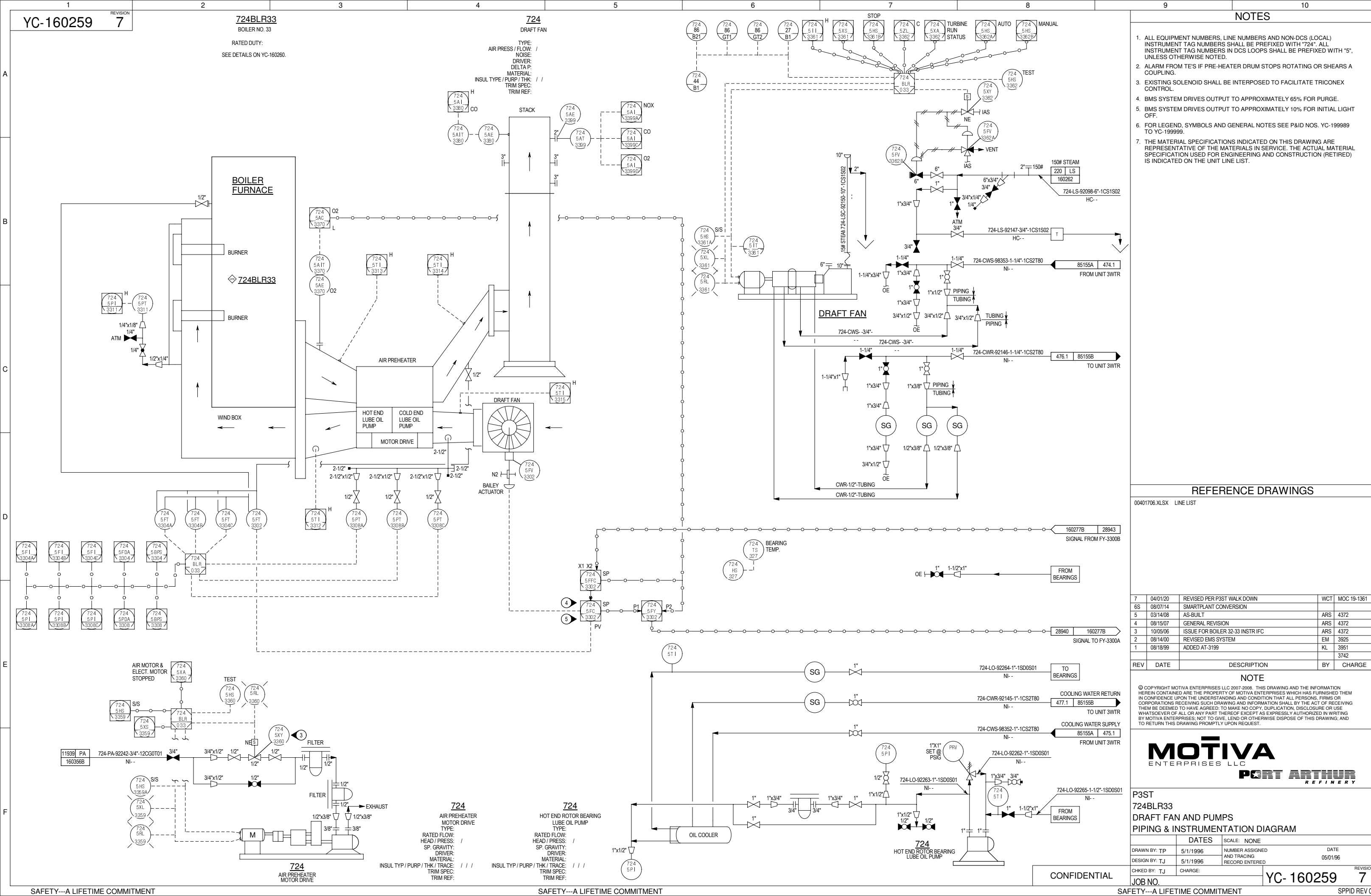


00401706.XLSX LINE LIST

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PIPING & INSTRUMENTATION DIAGRAM

CONFIDENTIAL



- ## SAFETY---A LIFETIME COMMITMENT



SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

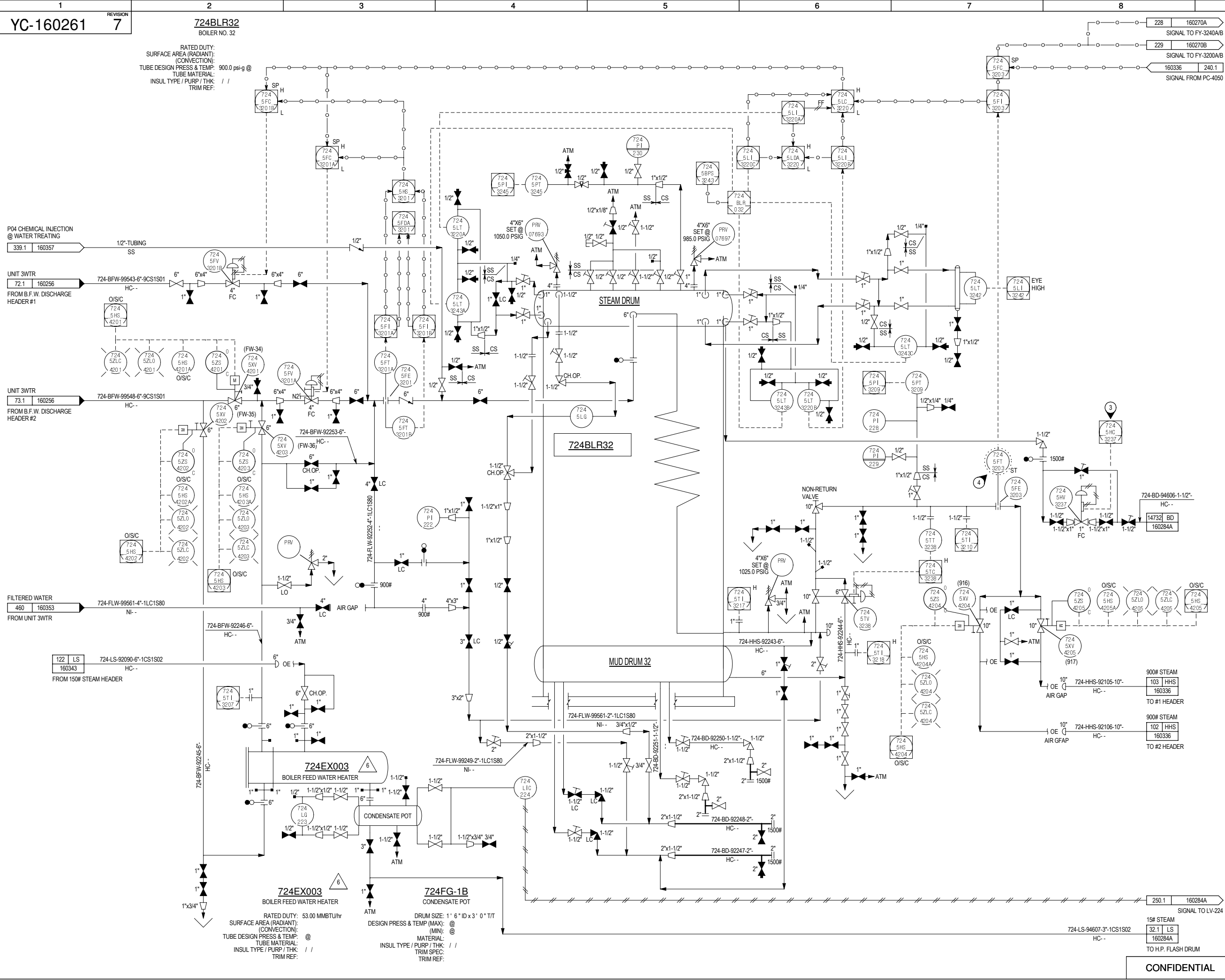
SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT



SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT



NOTES

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.

2. DELETED.

3. LOGIC OVERRIDE AUTOMATICALLY OPENS VALVE ON HIGH LEVEL AND CLOSSES VALVE ON LOW LEVEL.

4. STEAM TRACE HAS BEEN CUT.

5. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.

6. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX

LINE LIST

7	04/01/20	REVISED PER P&ST WALK DOWN	WCT	MOC 19-1361
6	04/14/16	MOC REVISION	RJB	M2016869
5S	08/14/14	SMARTPLANT CONVERSION		
4	03/14/08	AS BUILT	ARS	4372
3	08/14/07	GENERAL REVISION PER OPERATIONS	ARS	4372
2	10/05/06	ISSUE FOR BOILER 32-33 INSTR IFC	ARS	4372
1	08/14/00	REVISED EMS SYSTEM	EM	4372
				4372
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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MOTIVA

ENTERPRISES LLC

PORT ARTHUR

REFINERY

P3ST

724BLR32, 724EX003, 724FG-1B

BOILER FEED WATER SUPPLY & 900# STEAM OUTLET PIPING & INSTRUMENTATION DIAGRAM

DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED	DATE
DESIGN BY: TJ	5/1/1996	AND TRACING	05/01/96
CHKD BY: TJ	CHARGE:	RECORD ENTERED	

CONFIDENTIAL

YC- 160261

7

- ## SAFETY---A LIFETIME COMMITMENT



SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

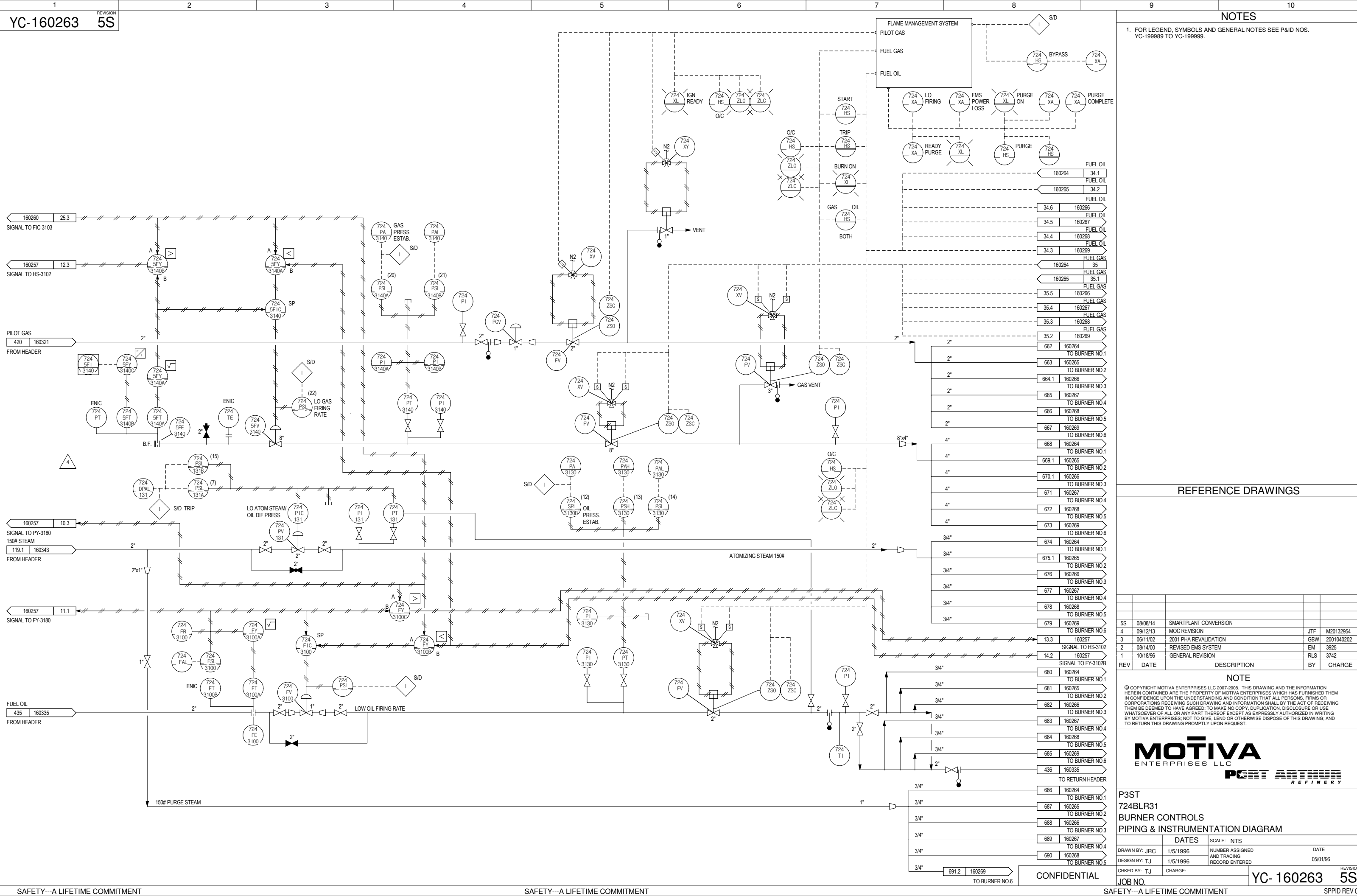
SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

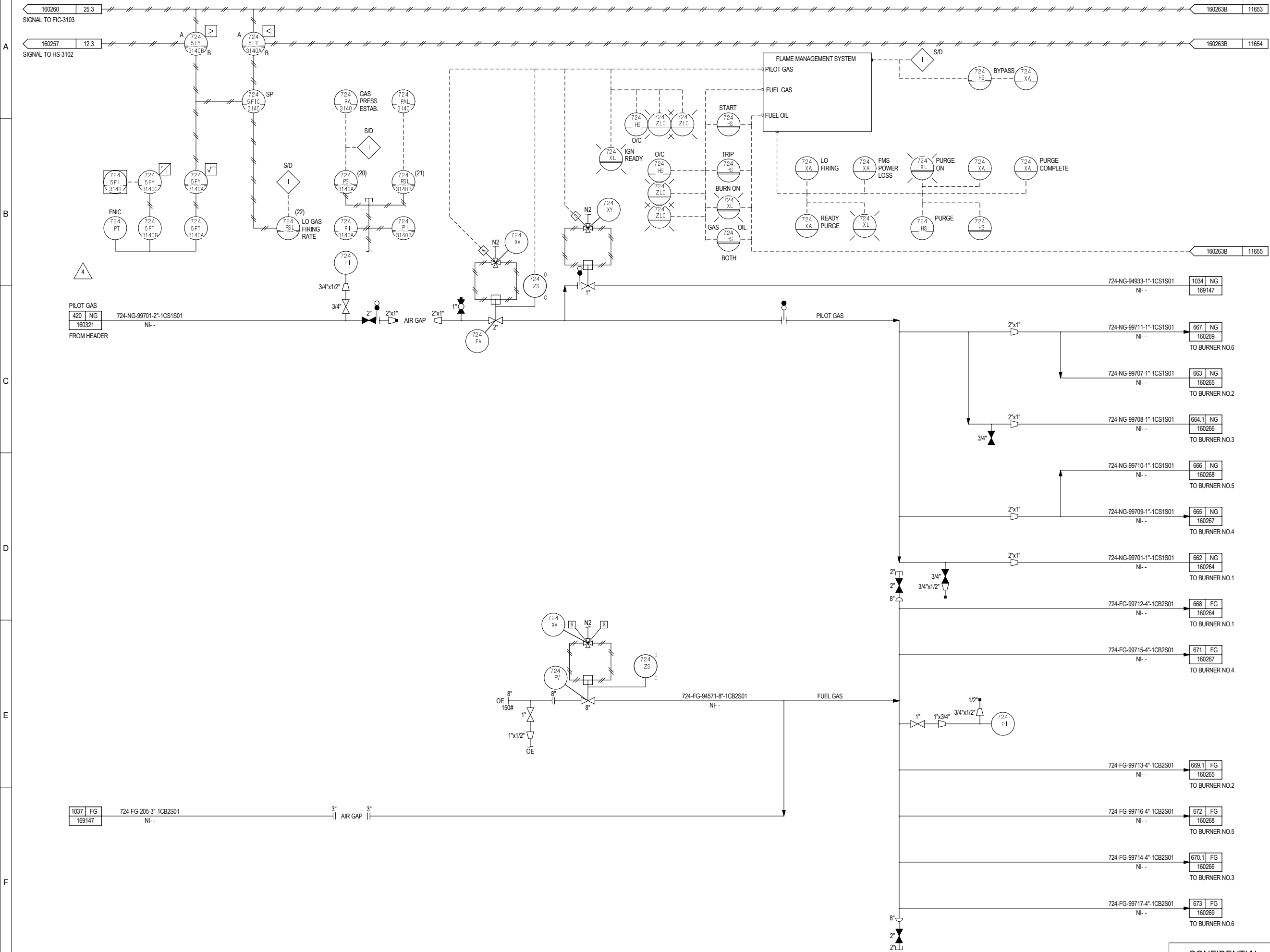
SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT



1. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
2. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.
4. YC-160263 HAS BEEN SPLIT INTO TWO DRAWINGS, PLEASE REFERENCE YC-160263A AND YC-160263B.



REFERENCE DRAWINGS

00401706.XLSX LINE LIST

6	04/01/20	REVISED PER P3ST WALK DOWN		WCT	MOC 19-1361
5S	08/08/14	SMARTPLANT CONVERSION			
4	09/12/13	MOC REVISION	JTF	M20132954	
3	06/11/02	2001 PHA REVALIDATION	GBW	2001040202	
2	08/14/00	REVISED EMS SYSTEM	EM	3925	
1	10/18/96	GENERAL REVISION	RLS	3742	
REV	DATE	DESCRIPTION	BY	CHARGE	

NOTE

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P3ST
724BLR31
BURNER CONTROLS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	1/5/1996	NUMBER ASSIGNED	DATE
DESIGN BY: TJ	1/5/1996	AND TRACING	05/01/96
CHKD BY: TJ	CHARGE:	RECORD ENTERED	
JOB NO.	YC- 160263A		REVISION 6

- ## REFERENCE DRAWINGS

00401706.XLSX LINE LIST

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724BLR31

BURNER CONTROLS

PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY:		NUMBER ASSIGNED	DATE
DESIGN BY:		AND TRACING	
CHKD BY:	CHARGE:	RECORD ENTERED	
JOB NO.		YC- 160263B	REVISION 0



1. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
2. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

00401706.XLSX LINE LIST

2	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-136
1S	07/24/14	SMARTPLANT CONVERSION		
				3742
REV	DATE	DESCRIPTION	BY	CHARGE

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P3ST
724BLR31
BURNER #1 CONTROLS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED AND TRACING DATE
DESIGN BY: TJ	5/1/1996	RECORD ENTERED 5-1-96

CHKD BY: TJ	CHARGE:	YC- 160264
JOB NO.		

SAFETY—A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SPPID REV 0

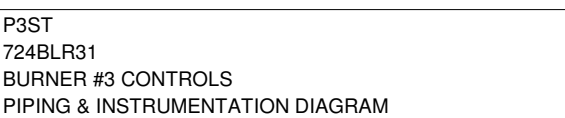
- ## SAFETY---A LIFETIME COMMITMENT

[illegible]

NOTE

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CHCKED BY: TJ	CHARGE:	YC- 160266	REVISION
JOB NO.			2

YC- 160266 2

SPPID REV 0



NOTES

1. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
2. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "1-". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

2	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
1S	08/13/14	SMARTPLANT CONVERSION		
REV	DATE	DESCRIPTION	BY	3742 NTS CHARGE

NOTE

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P3ST
724BLR31
BURNER #4 CONTROLS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED AND TRACING	DATE
DESIGN BY: TJ	5/1/1996	RECORD ENTERED	05/01/96

CHKD BY: TJ	CHARGE:	YC- 160267	2
JOB NO.			

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SAFETY---A LIFETIME COMMITMENT

SPPID REV

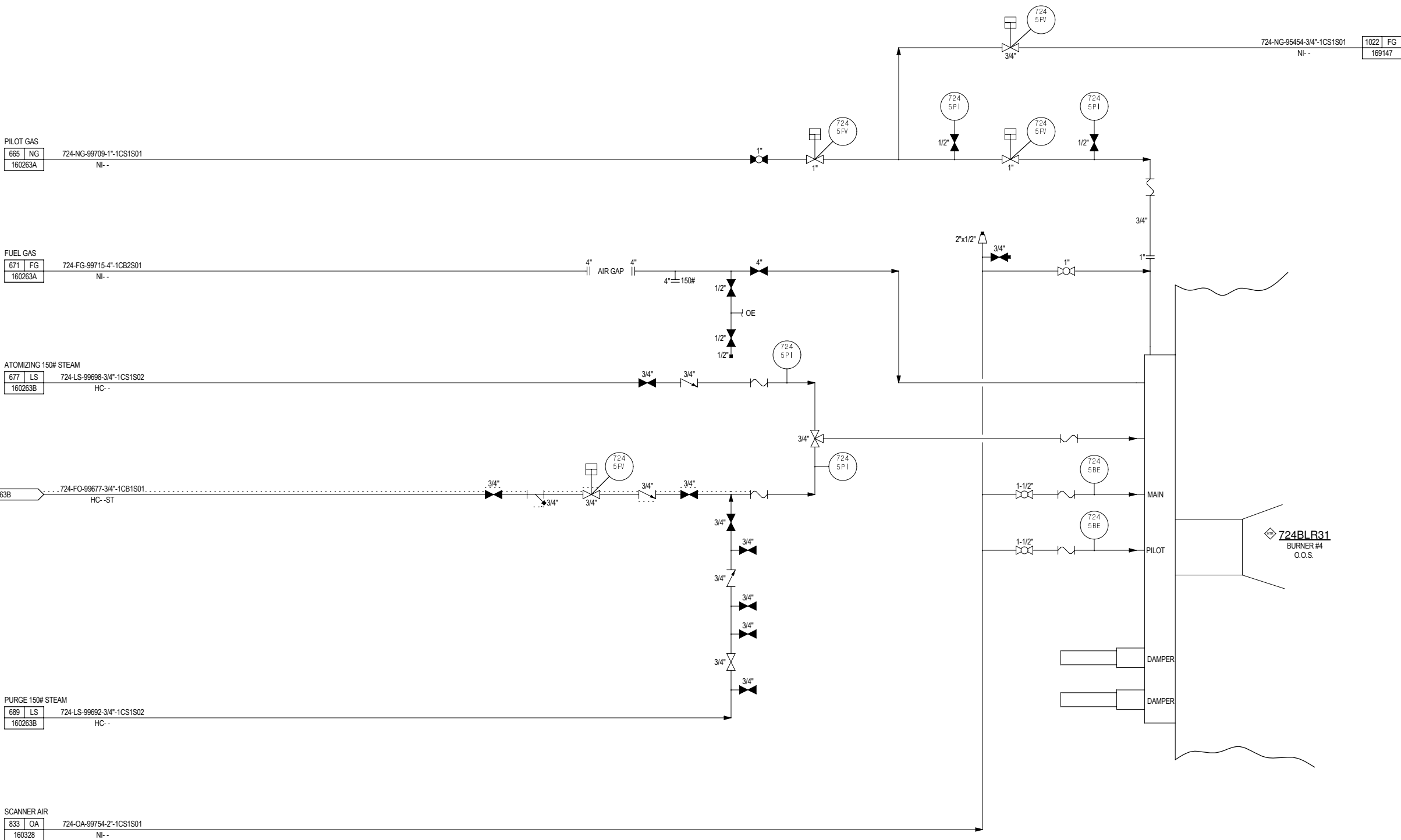
SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SPPID REV

SAFETY---A LIFETIME COMMITMENT



1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. AUTOMATE LEAK TEST.
3. PRESSURE OVERRIDE FROM PC-3244A AND PC-3244B.
4. TRIP TO MANUAL.
5. DELETED.
6. REFER TO SPECIFIC VALVE DETAILS ON DWG. YC-181653.
7. DELETED.
8. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
9. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.
10. YC-160270 HAS BEEN SPLIT INTO TWO DRAWINGS. PLEASE REFERENCE YC-160270A AND YC-160270B.

00401706.XLSX LINE LIST

7	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
6S	08/05/14	SMARTPLANT CONVERSION		
5	03/14/08	AS-BUILT	ARS	4372
4	10/05/06	ISSUE FOR BOILER 32-33 INSTR IFC	ARS	4372
3	06/03/02	2001 PHA REVALIDATION	JES	2001040714
2	08/14/00	REVISED EMS SYSTEM	EM	3925
1	18/96	GENERAL REVISION	RLS	3742
REV	DATE	DESCRIPTION	BY	CHARGE

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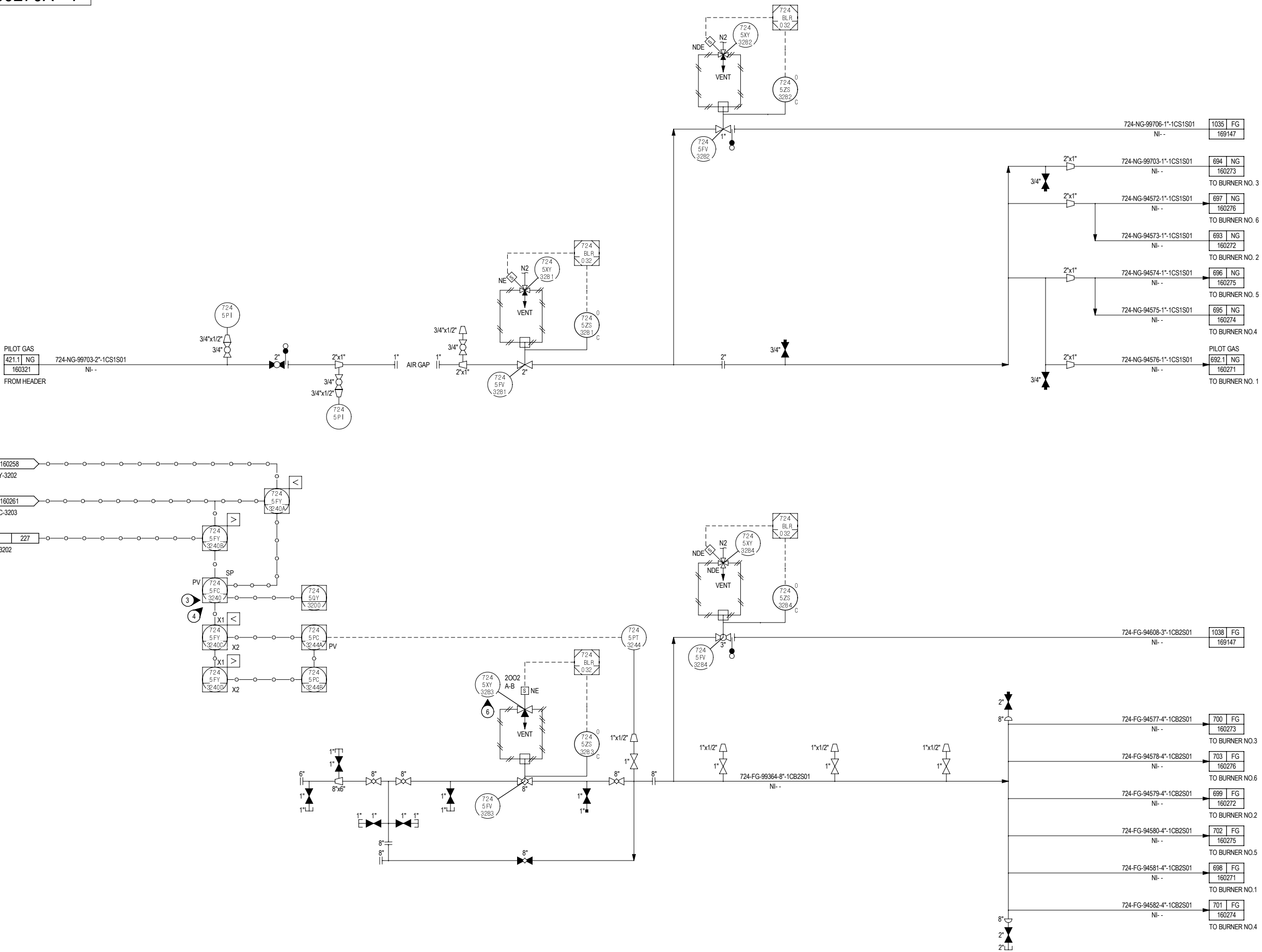
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DRAWN BY: JRC	1/5/1996	NUMBER ASSIGNED AND TRACING	DATE 05/01/96
DESIGN BY: TJ	1/5/1996	RECORD ENTERED	

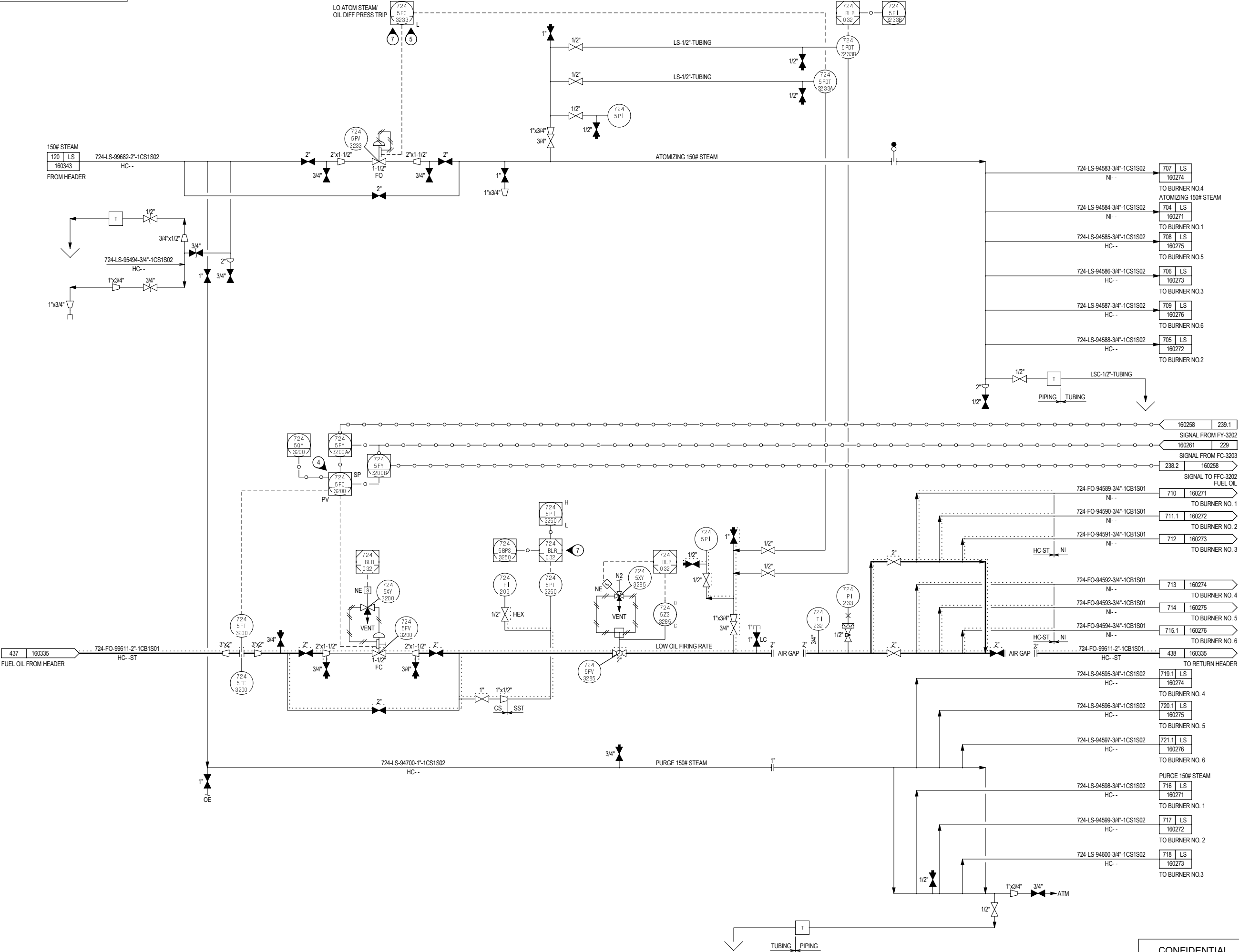
CONFIDENTIAL

JOB NO. YC-160270A /

SAFETY---A LIFETIME COMMITMENT

SPPID REV 0





1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. AUTOMATE LEAK TEST.
3. DELETED.
4. TRIP TO MANUAL.
5. TRIP TO NORMAL AFTER ONE MINUTE.
6. DELETED.
7. ALARMS ACTIVE IF FUEL OIL PUMP RUNNING (XS-3065 OR COMMANDED STATE XV-3110).
8. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
9. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.
10. YC-160270 HAS BEEN SPLIT INTO TWO DRAWINGS. PLEASE REFERENCE YC-160270A AND YC-160270B.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

0	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361	
REV	DATE	DESCRIPTION	BY	CHARGE	

NOTE

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P3ST
724BLR32
BURNER CONTROLS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY:		NUMBER ASSIGNED	DATE
DESIGN BY:		AND TRACING	
CHKD BY:	CHARGE:	RECORD ENTERED	
JOB NO.		YC- 160270B	0



- ## REFERENCE DRAWINGS

00401706.XLSX LINE LIST

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	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED	DATE
DESIGN BY: TJ	5/1/1996	AND TRACING	05/01/96
CHKD BY: TJ	CHARGE: 7342	RECORD ENTERED	REVISION
JOB NO.	YC- 160271		4

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

4	04/01/20	REVISED PER P3ST WALK DOWN		WCT	MOC 19-1361
3S	08/13/14	SMARTPLANT CONVERSION			
2	03/14/08	AS-BUILT		ARS	4372
1	10/05/06	ISSUE FOR BOILER 32-33 INSTR FC		ARS	4372
					3742
REV	DATE	DESCRIPTION		BY	CHARGE

NOTE

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P3ST
724BLR32
BURNER #2 CONTROLS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED	DATE
DESIGN BY: TJ	5/1/1996	AND TRACING RECORD ENTERED	05/01/96

YC- 160272	REVISION 4
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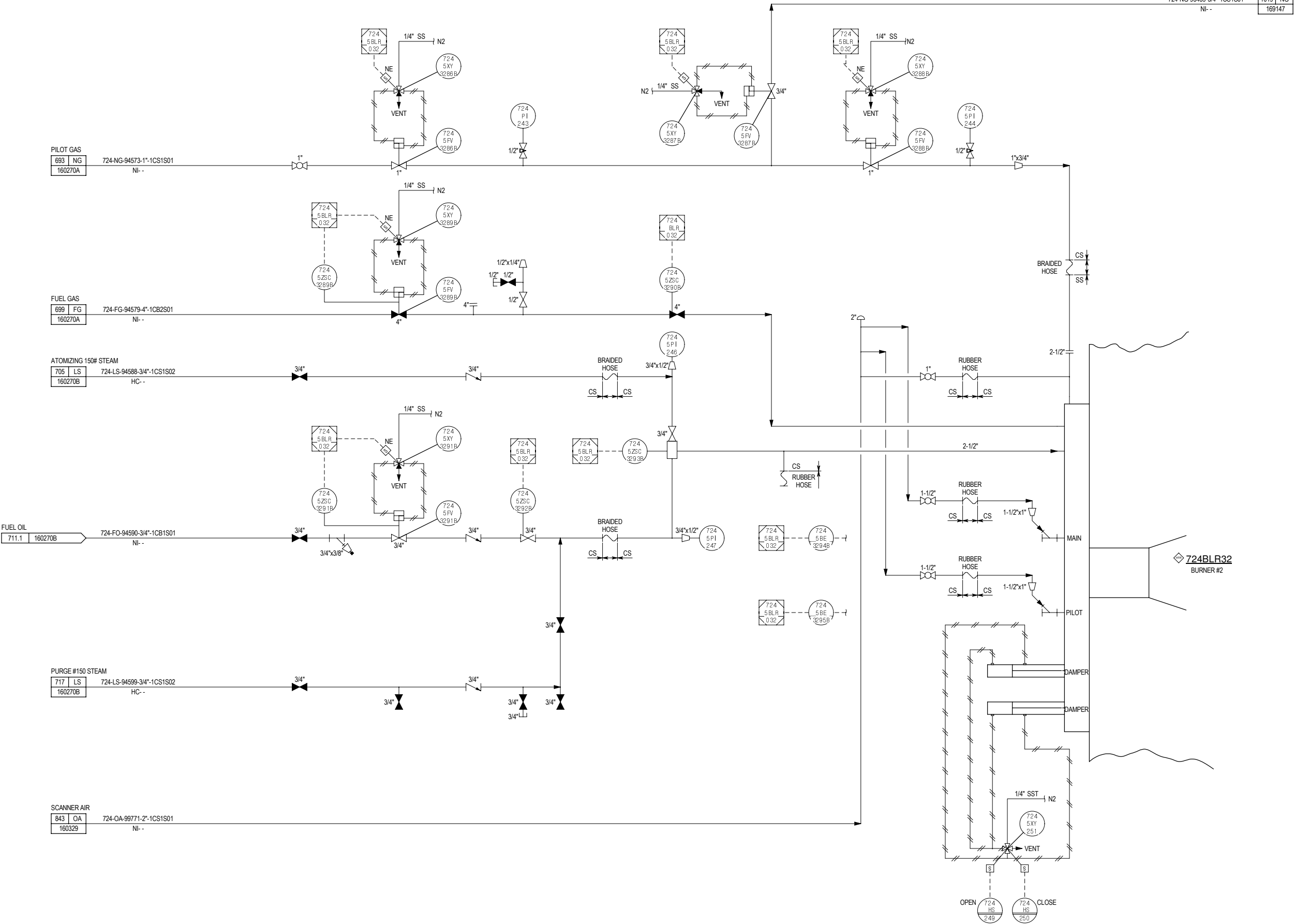
CONFIDENTIAL

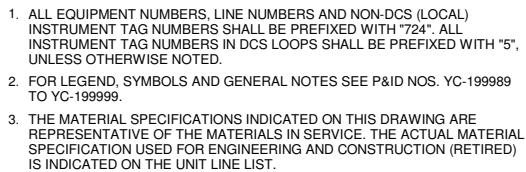
SAFETY---A LIFETIME COMMITMENT

SPPID REV 0

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT





00401706.XLSX LINE LIST

NOTE

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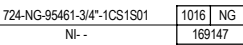
P3ST
724BLR32
BURNER #3 CONTROLS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED AND TRACING	DATE 05/01/96
DESIGN BY: TJ	5/1/1996	RECORD ENTERED	
CHKD BY: TJ	CHARGE: 3742	YC- 160273	
JOB NO.		REVISION 4	

CONFIDENTIAL

YC- 160273

REVISION



- ## REFERENCE DRAWINGS

00401706.XLSX LINE LIST

NOTE

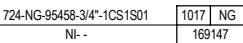
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P3ST
724BLR32
BURNER #4 CONTROLS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED	DATE
DESIGN BY: TJ	5/1/1996	AND TRACING	05/01/96
CHKD BY: TJ	CHARGE:	RECORD ENTERED	
JOB NO.		YC- 160274	REVISION 4

CONFIDENTIAL

YC- 160274	REVISION 4
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1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

00401706.XLSX LINE LIST

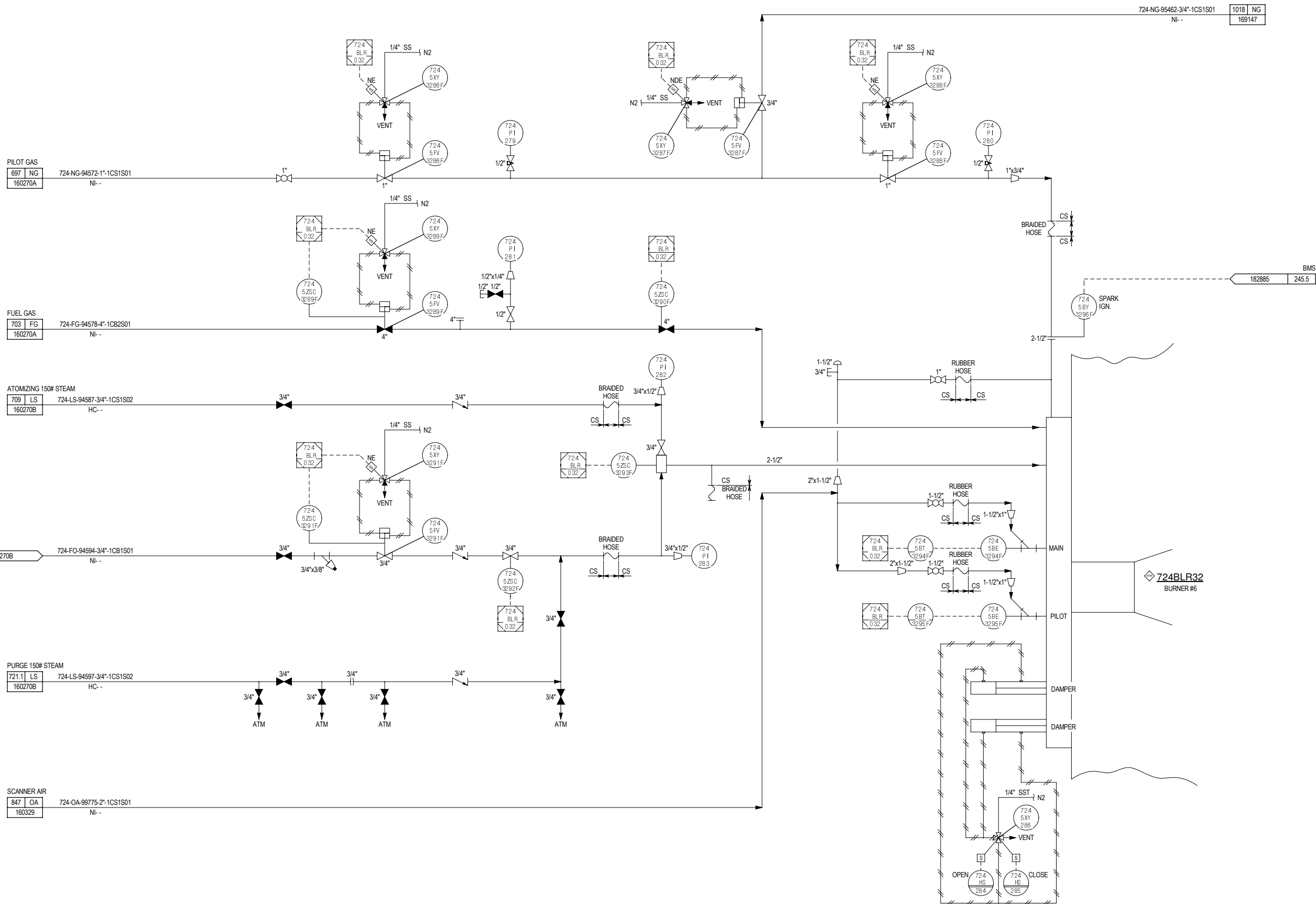
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3S	08/14/14	SMARTPLANT CONVERSION		
2	03/14/08	AS-BUILT	ARS	4372
1	10/05/06	ISSUE FOR BOILER 32-33 INSTR IFC	ARS	4372
				3742
REV	DATE	DESCRIPTION	BY	CHARGE

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	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED AND TRACING	DATE 05/01/96
DESIGN BY: TJ	5/1/1996	RECORD ENTERED	

YC- 160275 4



NOTES

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

4	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
3S	08/14/14	SMARTPLANT CONVERSION		
2	03/14/08	AS-BUILT	ARS	4372
1	10/05/06	ISSUE FOR BOILER 32-33 INSTR IFC	ARS	4372
				3742
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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P3ST

724BLR32

BURNER #6 CONTROLS

PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED AND TRACING	DATE 06/01/96
DESIGN BY: TJ	5/1/1996	RECORD ENTERED	

JOB NO.

YC- 160276

SAFETY... A LIFETIME COMMITMENT

1. ALL INSTRUMENTS IN DCS LOOPS ARE PREFIXED WITH "S". NON DCS (LOCAL) INSTRUMENTS ARE PREFIXED WITH "XXX".
2. AUTOMATE LEAK TEST.
3. PRESSURE OVERRIDE FROM PC3244A AND PC3244B.
4. TRIP TO MANUAL.
5. TRIP TO NORMAL AFTER ONE MINUTE.
6. REFER TO SPECIFIC VALVE DETAILS ON DWG. YC-181653.
7. ALARMS ACTIVE IF FUEL OIL PUMP RUNNING (XS-3065 OR COMMANDED STATE XV-3110).
8. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199889 TO YC-199999.

REFERENCE DRAWINGS

YC-181626	724BLR33 BURNER CONTROLS PAR IPF OVERVIEW
YC-181653	2002 VALVES & RCS PANELS

5S	08/06/14	SMARTPLANT CONVERSION			
4	03/14/08	AS-BUILT	ARS	4372	
3	10/05/06	ISSUE FOR BOILER 32-33 INSTR IFC	ARS	4372	
2	08/14/00	REVISED EMS SYSTEM	EM	3925	
1	11/01/96	GENERAL REVISION	RLS	3742	
REV	DATE	DESCRIPTION	BY	CHARGE	

NOTE

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MOTIVA
ENTERPRISES LLC

PORT ARTHUR
REFINERY

P3ST

724BLR3E

BURNER CONTROLS

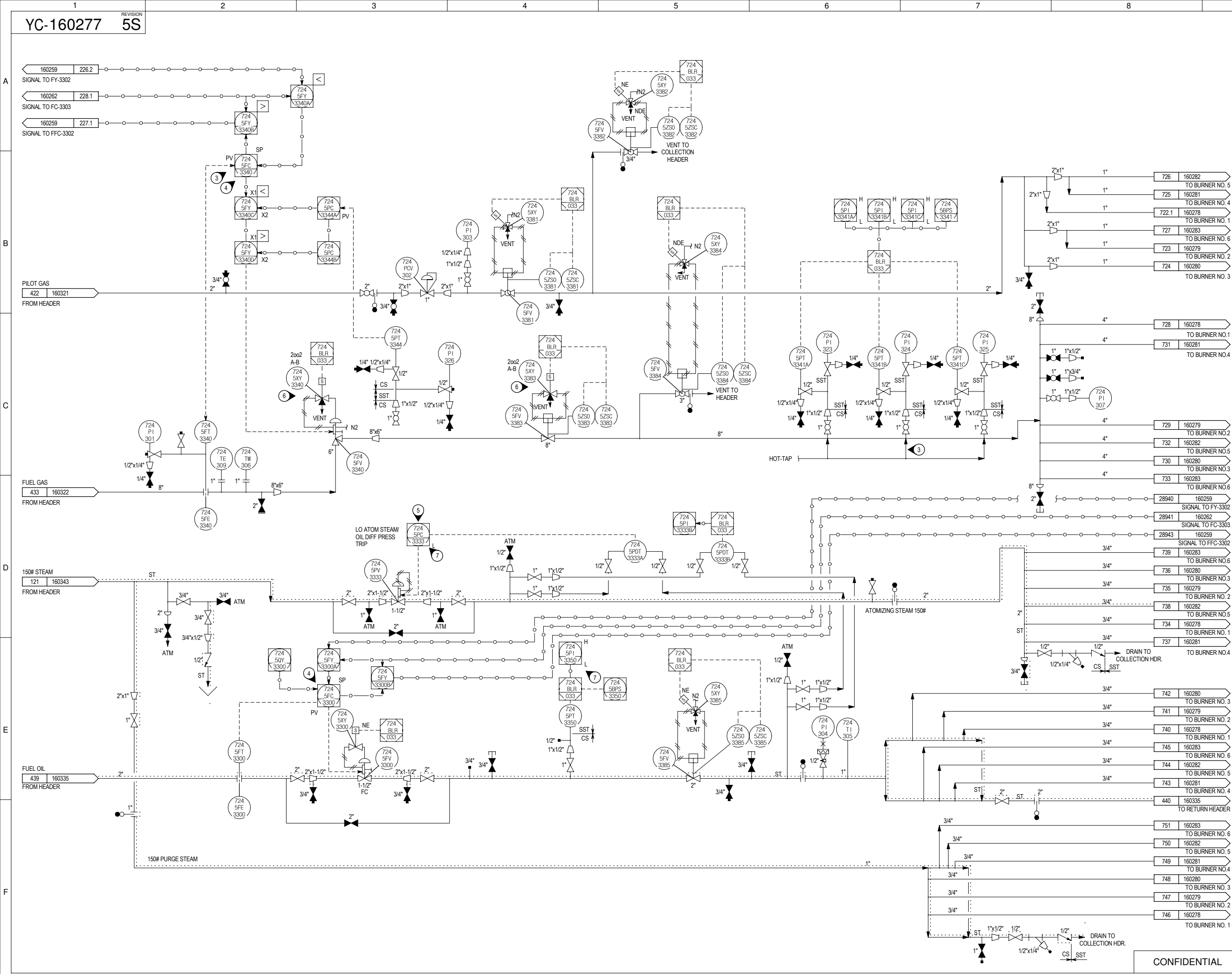
PIPING & INSTRUMENTATION DIAGRAM

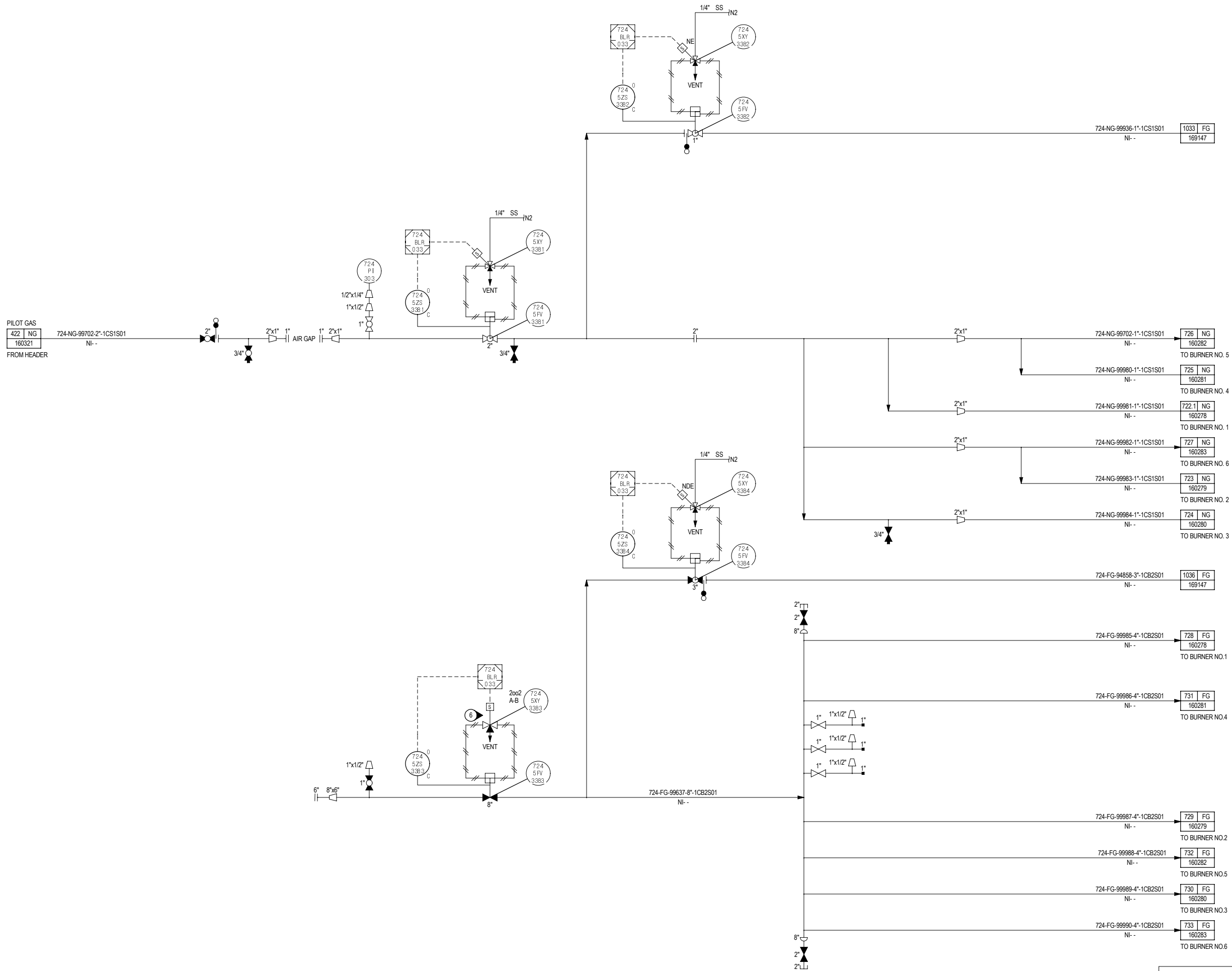
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DRAWN BY: JRC	1/5/1996	NUMBER ASSIGNED	DATE
DESIGN BY: TJ	1/5/1996	AND TRACING	05/01/96
CHK'D BY: TJ	CHARGE:	RECORD ENTERED	
JOB NO.	YC- 160277		REVISION 5S

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SPPID REV 0





NOTES

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. AUTOMATE LEAK TEST.
3. PRESSURE OVERRIDE FROM PC3244A AND PC3244B.
4. TRIP TO MANUAL.
5. DELETED.
6. REFER TO SPECIFIC VALVE DETAILS ON DWG. YC-181653.
7. DELETED.
8. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
9. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.
10. YC-160277 HAS BEEN SPLIT INTO TWO DRAWINGS. PLEASE REFERENCE YC-160277A AND YC-160277B.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

6	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
5S	08/06/14	SMARTPLANT CONVERSION		
4	03/14/08	AS-BUILT	ARS	4372
3	10/05/06	ISSUE FOR BOILER 32-33 INSTR IFC	ARS	4372
2	08/14/00	REVISED EMS SYSTEM	EM	3925
1	11/01/96	GENERAL REVISION	RLS	3742
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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P3ST
724BLR33
BURNER CONTROLS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	1/5/1996	NUMBER ASSIGNED	DATE
DESIGN BY: TJ	1/5/1996	AND TRACING	05/01/96
CHKD BY: TJ	CHARGE:	RECORD ENTERED	REVISION
JOB NO.	YC- 160277A		6



1. AUTOMATE LEAK TEST.
2. TRIP TO MANUAL.
3. TRIP TO NORMAL AFTER ONE MINUTE.
4. ALARMS ACTIVE IF FUEL OIL PUMP RUNNING (XS-3065 OR COMMANDED STATE STATION XV-3110).
5. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-1999989 TO YC-1999999.
6. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
7. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.
8. YC-160277 HAS BEEN SPLIT INTO TWO DRAWINGS. PLEASE REFERENCE YC-160277A AND YC-160277B.

00401706.XLSX LINE LIST

NOTE

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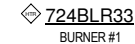
P3ST

724BLR33

BURNER CONTROLS

PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY:		NUMBER ASSIGNED	DATE
DESIGN BY:		AND TRACING	
CHK'D BY:	CHARGE:	RECORD ENTERED	
JOB NO.		YC- 160277B	0



CHKD BY: TJ	CHARGE: 3742	YC- 160278	REVISION 4
JOB NO.			

- ## SAFETY---A LIFETIME COMMITMENT

00401706.XLSX LINE LIST

00401706.XLSX LINE LIST

NOTE	
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References

References

P3ST

724BLR33

BURNER #1 CONTROLS

BIRING 3 INSTRUMENTATION DIAGRAM

CHKD BY: T.J.	CHARGE: 3742	REVISION
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CHKD BY: T.J.	CHARGE: 3742	REVISION
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10	0742	YC-160278	4
JOB NO.			

JOB NO.	10-100-100
FIFTY - A LIFETIME COMMITMENT	SPRID REV 0

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

00401706.XLSX LINE LIST

4	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
3S	08/13/14	SMARTPLANT CONVERSION		
2	03/14/08	AS-BUILT	ARS	4372
1	10/05/06	ISSUE FOR BOILER 32-33 INSTR IFC	ARS	4372
				4372
REV	DATE	DESCRIPTION	BY	CHARGE

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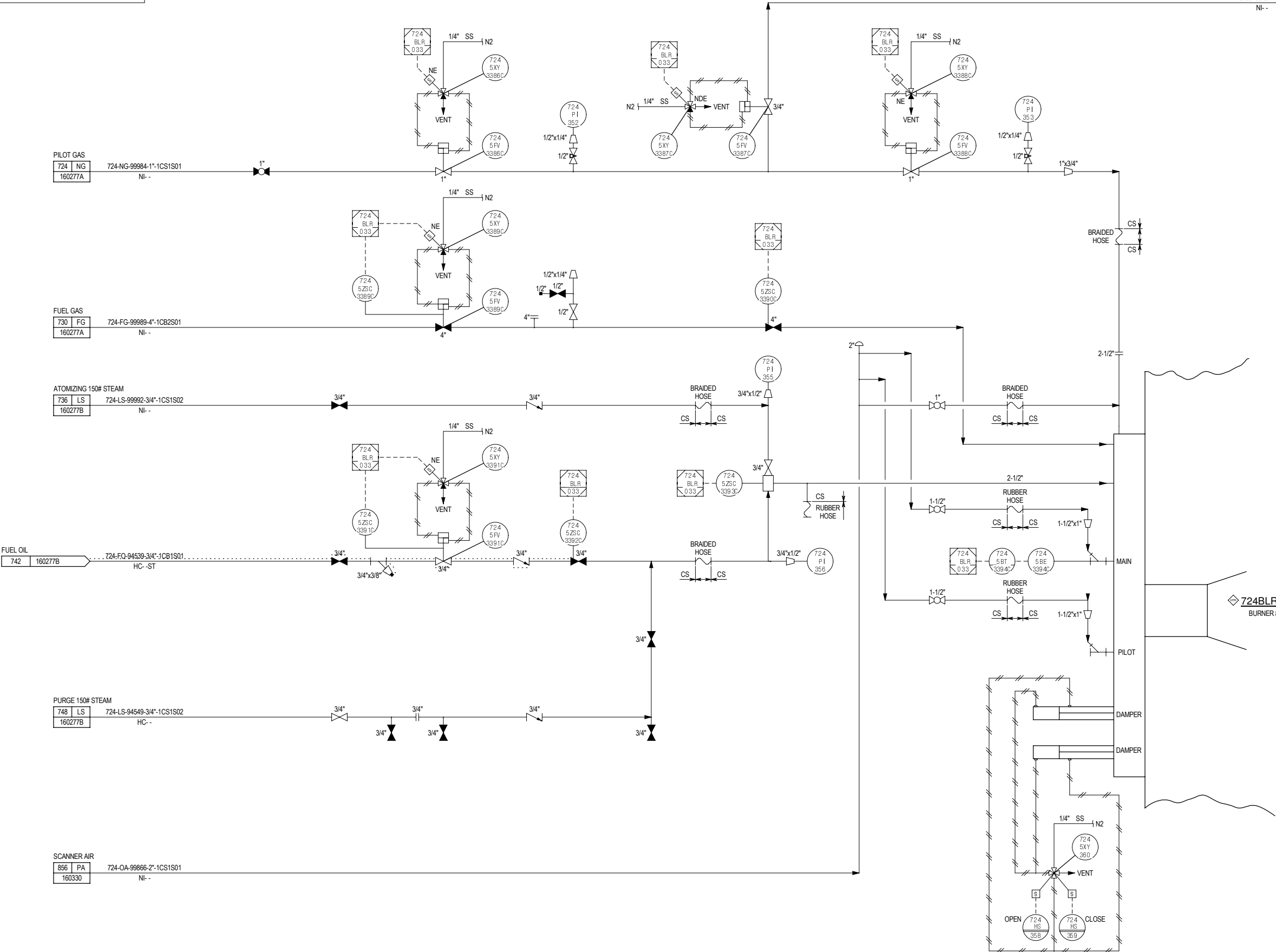
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED AND TRACING	DATE
DESIGN BY: TJ	5/1/1996	RECORD ENTERED	05/01/96
CHKD BY: TJ		CHARGE:	REVISION
JOB NO.		YC- 160280	4

YC- 160280

4

SPPID REV 0



SAFETY---A LIFETIME COMMITMENT

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

5	04/01/20	REVISED PER P3ST WALK DOWN		WCT	MOC 19-1361
4S	08/13/14	SMARTPLANT CONVERSION			
3	03/14/08	AS-BUILT		ARS	4372
2	10/05/06	ISSUE FOR BOILER 32-33 INSTR IFC		ARS	4372
1	11/27/96	GENERAL REVISION		ARS	3742
REV	DATE	DESCRIPTION		BY	CHARGE

NOTE

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P3ST
724BLR33
BURNER #4 CONTROLS
PIPING & INSTRUMENTATION DIAGRAM

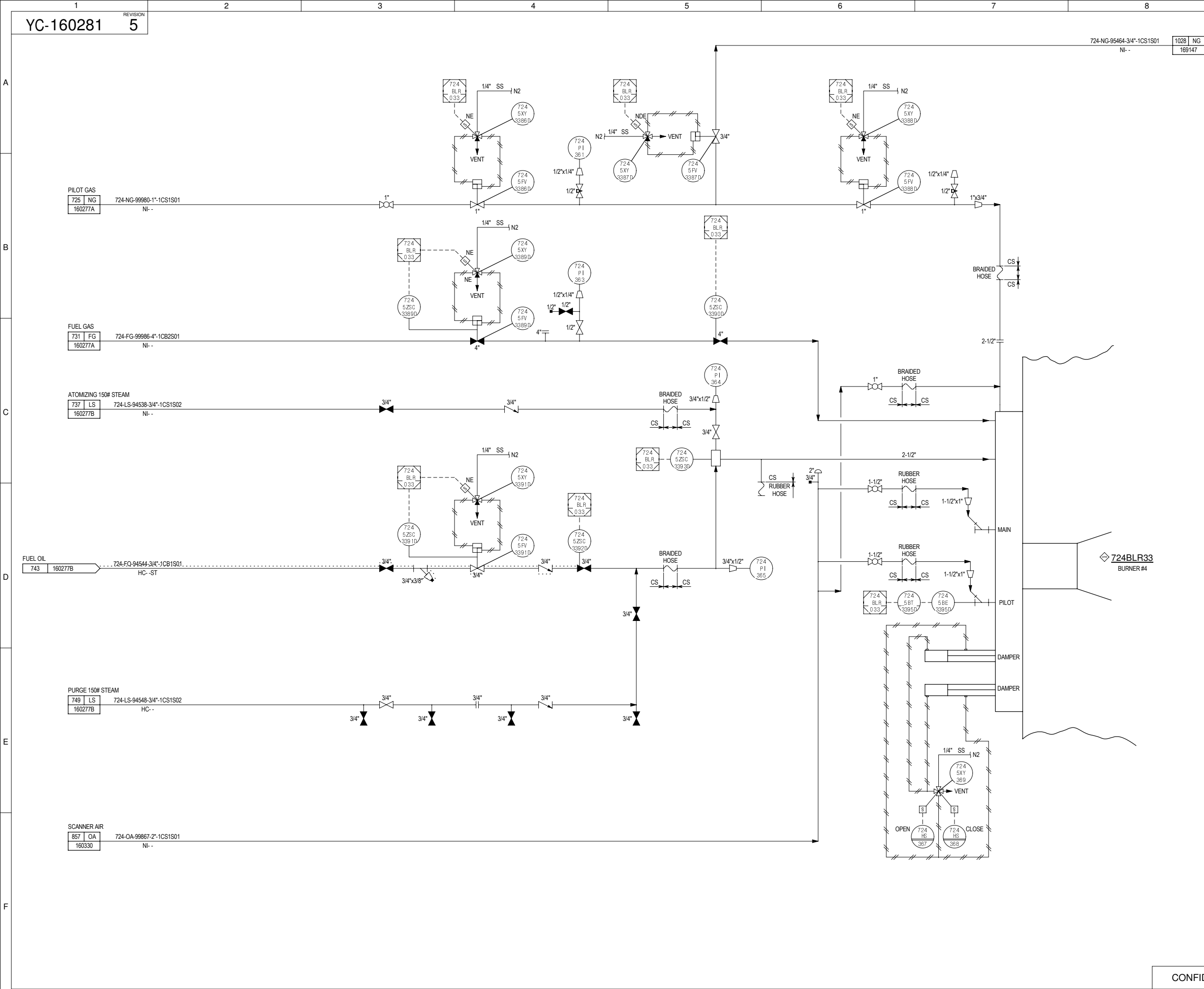
	DATES	SCALE: NONE
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED DATE
DESIGN BY: TJ	5/1/1996	AND TRACING RECORD ENTERED 05/01/96

YC- 160281	REVISION 5
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SAFETY---A LIFETIME COMMITMENT

SPPID REV 0



1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

4	04/01/20	REVISED PER P3ST WALK DOWN		WCT	MOC 19-1361
3S	08/07/14	SMARTPLANT CONVERSION			
2	03/14/08	AS-BUILT		ARS	4372
1	10/05/06	ISSUE FOR BOILER 32-33 INSTR FC		ARS	4372
					3742
REV	DATE	DESCRIPTION		BY	CHARGE

NOTE

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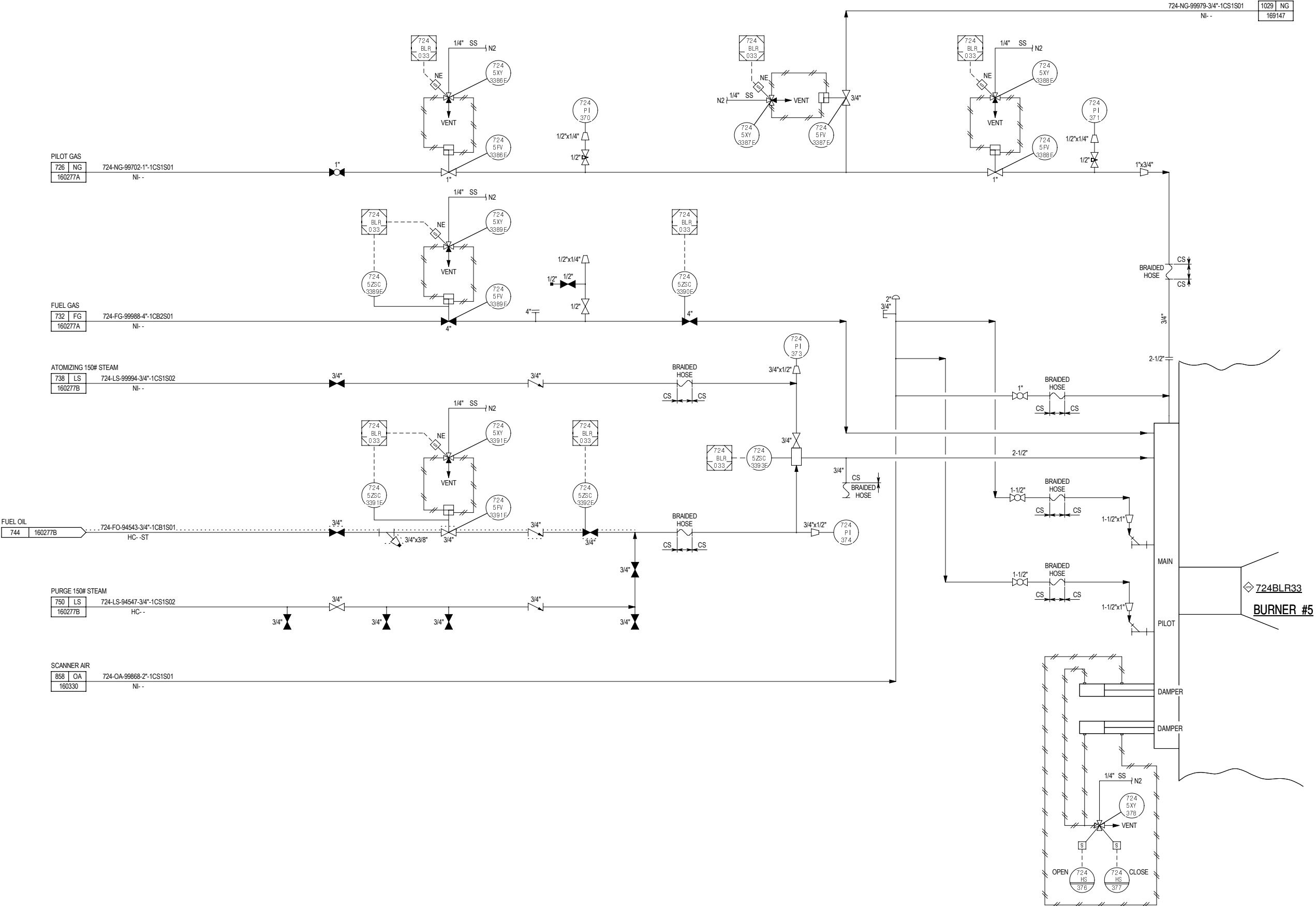


P3ST
724BLR33
BURNER #5 CONTROLS
PIPING & INSTRUMENTATION DIAGRAM

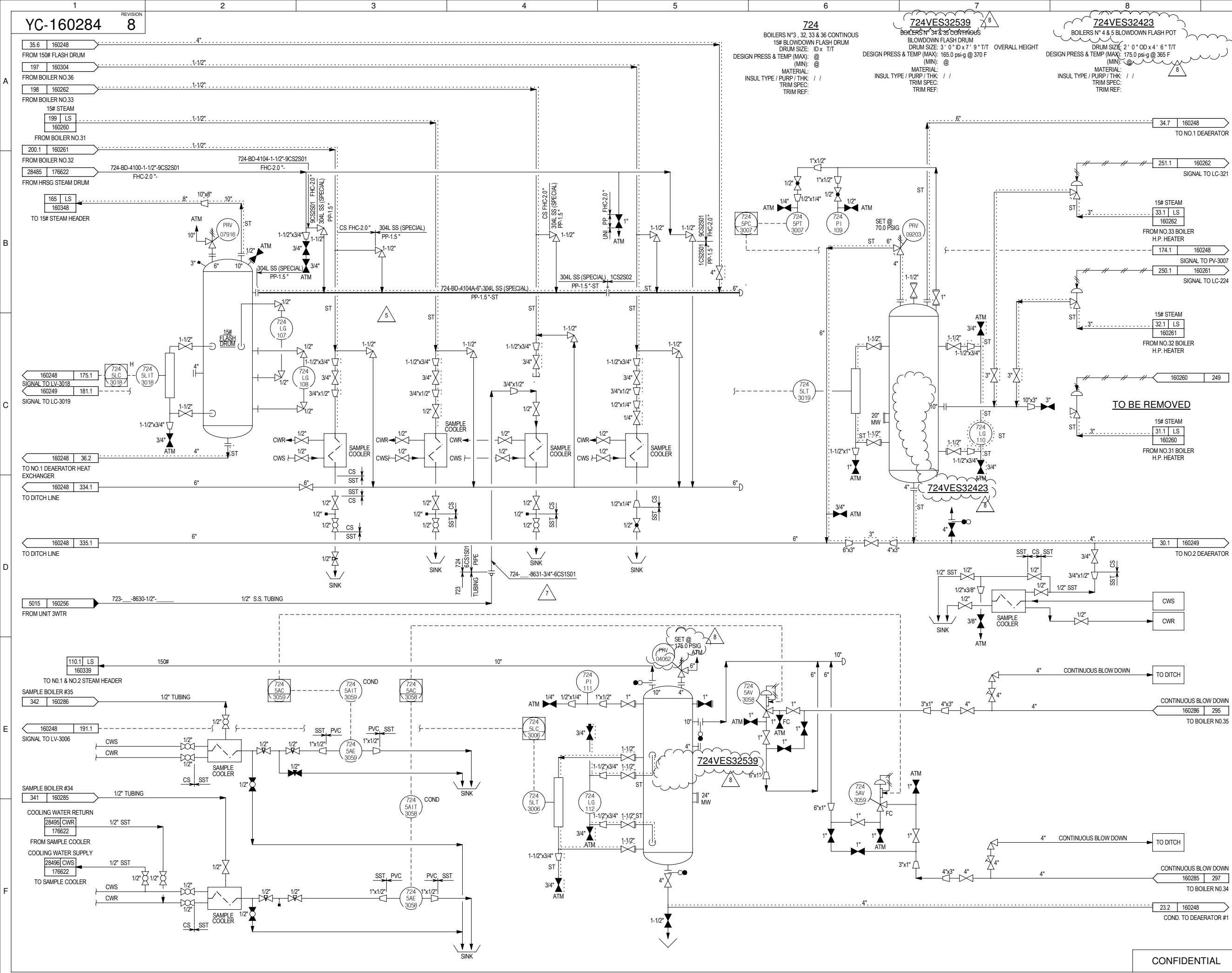
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DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED AND TRACING	DATE 05/01/96
DESIGN BY: TJ	5/1/1996	RECORD ENTERED	

CHCKED BY: TJ	CHARGE: 3742	YC- 160282	REVISION
JOB NO.			4

CONFIDENTIAL



SAFETY--A LIFETIME COMMITMENT



NOTES

1. ALL INSTRUMENTS IN DCS LOOPS ARE PREFIXED WITH "5". NON DCS (LOCAL) INSTRUMENTS ARE PREFIXED WITH "XXX".
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.

REFERENCE DRAWINGS

7	07/23/15	GENERAL REVISION	WRW	6000997184
6S	08/01/14	SMARTPLANT CONVERSION		
5	05/11/10	6"-WBB-4104A-3CS1502-PP	ELP	6000798765
4	07/23/07	GENERAL REVISION	ARS	4372
3	05/10/07	AS-BUILT	ARS	4372
2	07/20/06	ISSUE FOR WTP3 INSTRUMENTATION	ARS	4372
1	05/23/05	MOC REVISION M2005682-001	CHI	
8	10/20/16	MOC REVISION	JBB	M20162669
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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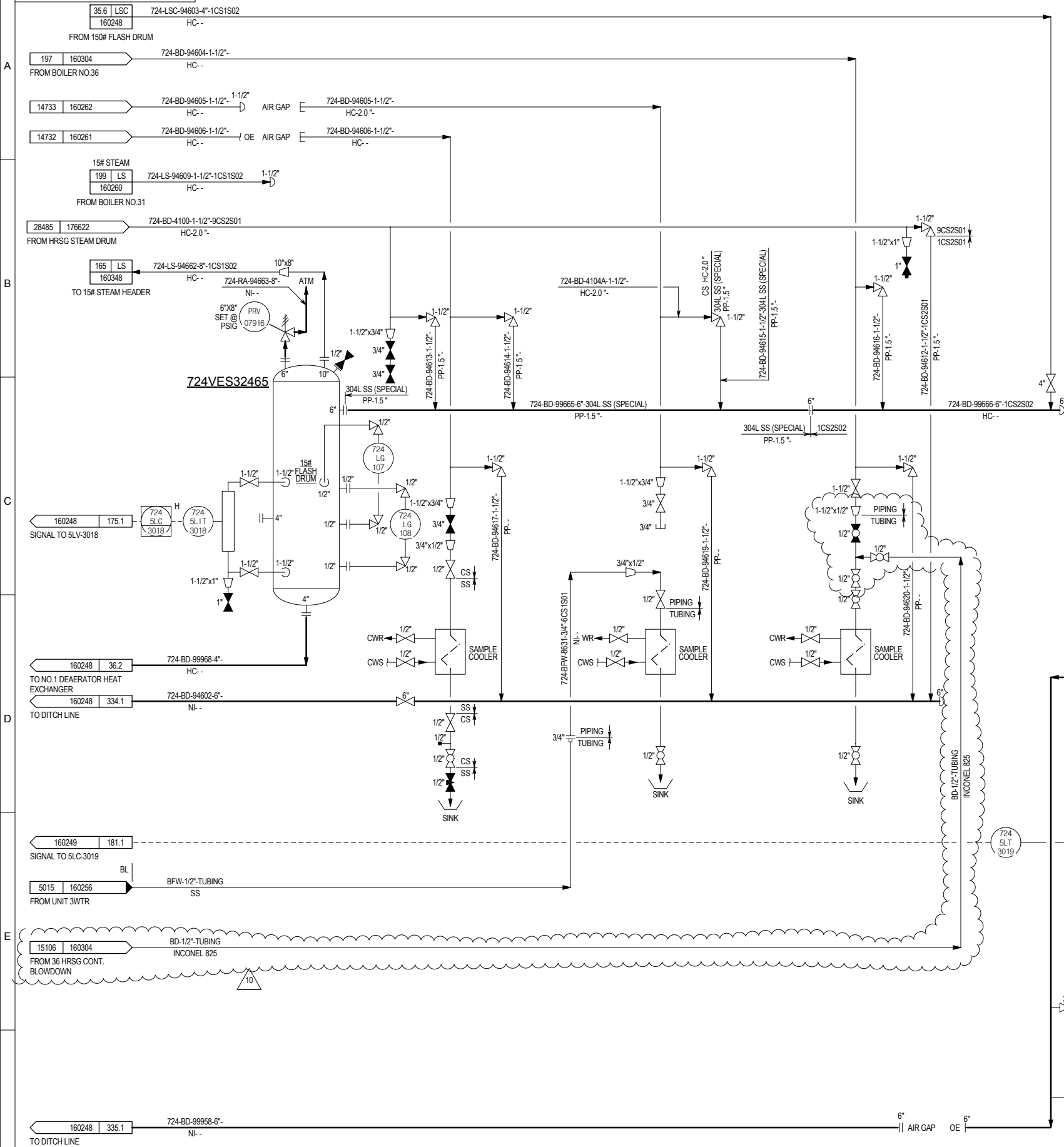
MOTIVA
ENTERPRISES LLC

PORT ARTHUR
REFINERY

P3ST
CONTINUOUS BLOWDOWN FLASH DRUMS
CONTINUOUS BLOWDOWN
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED AND TRACING	DATE 05/01/96
DESIGN BY: TJ	5/1/1996	RECORD ENTERED	
CHKD BY: TJ	CHARGE:		REVISION
JOB NO.		YC- 160284	8

YC-160284A 10

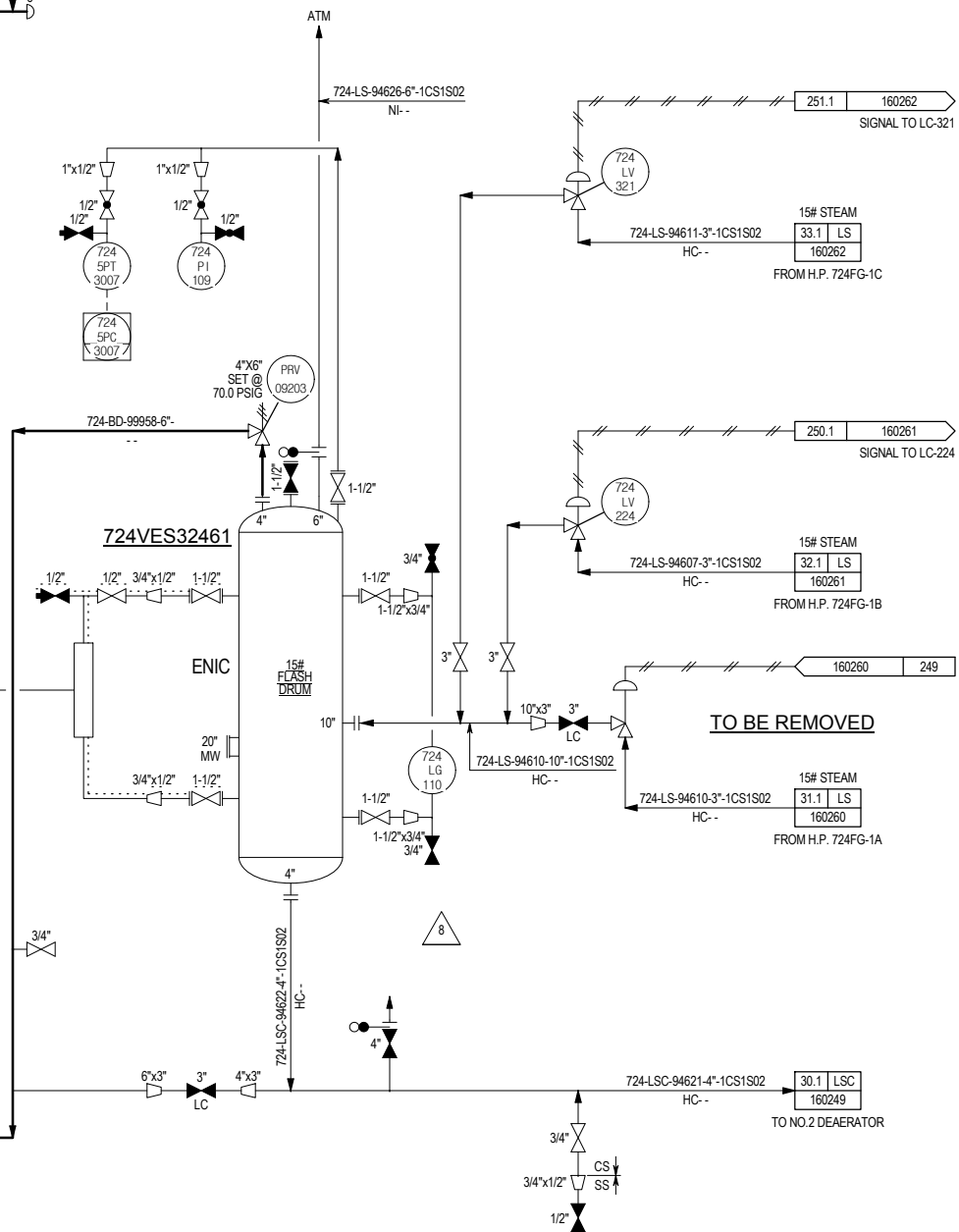


724VES32465
 BOILERS N°3 , 32, 33 & 36 CONTINUOUS
 15# BLOWDOWN FLASH DRUM
 DRUM SIZE: ID x T/T
 DESIGN PRESS & TEMP (MAX): @
 (MIN): @
 MATERIAL:
 INSUL TYPE / PURP / THK: / /
 TRIM SPEC:
 TRIM REF:

OVERALL HEIGHT

724VES32461

DRUM SIZE: 2' 0" OD x 4' 6"
DESIGN PRESS & TEMP (MAX): 175.0 psi-g @ 365 F
(MIN): @
MATERIAL:
INSUL TYPE / PURP / THK: / /
TRIM SPEC:
TRIM REF:



NOTES

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.
4. YC-160284 HAS BEEN SPLIT INTO TWO DRAWINGS. PLEASE REFERENCE YC-160284A AND YC-160284B.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

10	04/03/23	MOC REVISION	JMC	MOC 21-1171
9	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
8	10/20/16	MOC REVISION	JBB	M20162669
7	07/23/15	GENERAL REVISION	WRW	6000997184
6S	08/01/14	SMARTPLANT CONVERSION		
5	05/11/10	6"-WBB-4104A-3CS1S02-PP	ELP	6000798765
4	07/23/07	GENERAL REVISION	ARS	4372
3	05/10/07	AS-BUILT	ARS	4372
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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P3ST
724VES32461, 724VES32465
CONTINUOUS BLOWDOWN
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED AND TRACING RECORD ENTERED	DATE
DESIGN BY: TJ	5/1/1996		05/01/96

YC- 160284A 10

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SAFETY---A LIFETIME COMMITMENT

SPPID REV 0

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

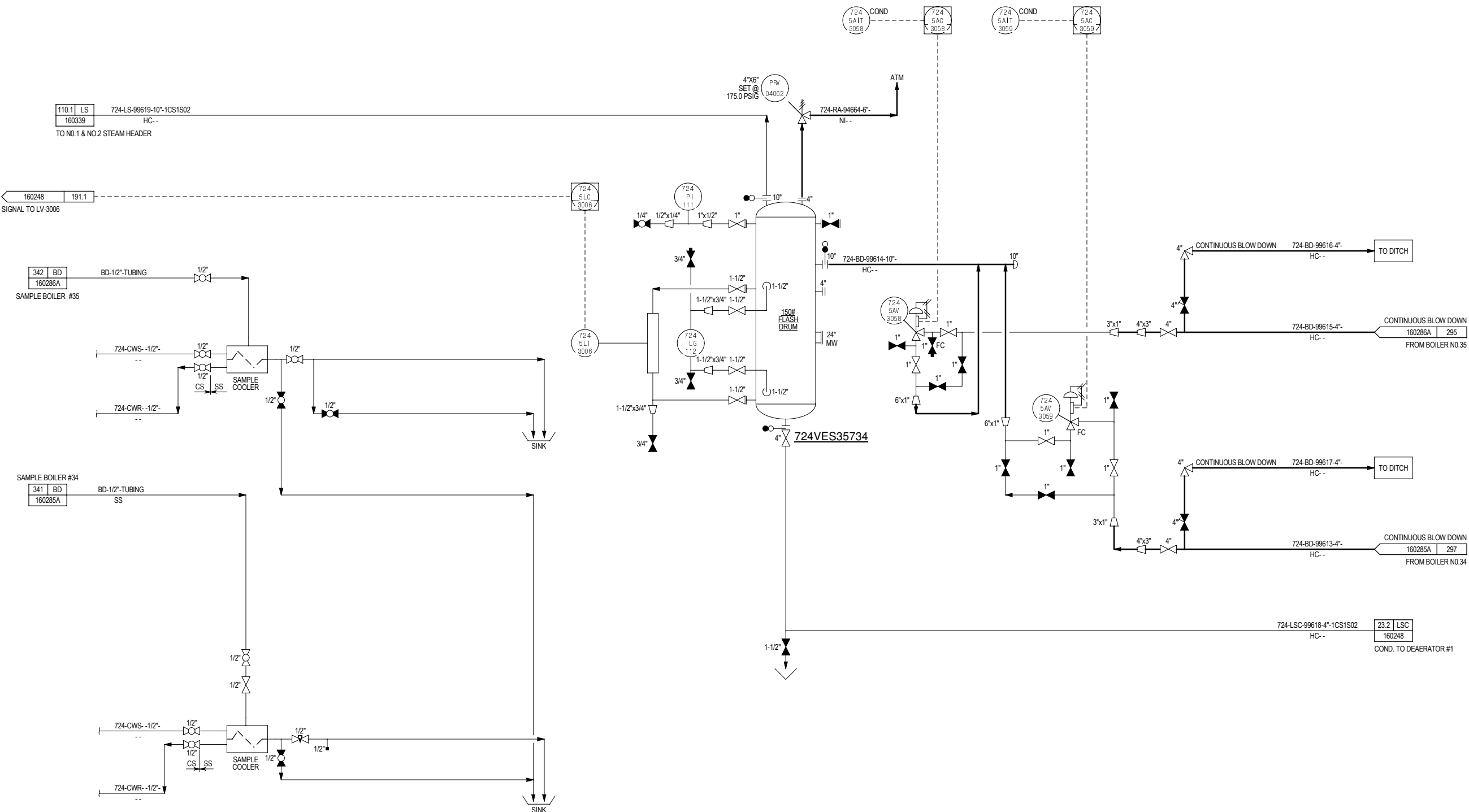
SAFETY---A LIFETIME COMMITMENT

724VES35734

BOILERS N° 34 & 35 CONTINUOUS
150# BLOWDOWN FLASH DRUM
DRUM SIZE: 3' 0" ID x 7' 9" T/T
DESIGN PRESS & TEMP (MAX): 165.0 psi-g @ 370 F
(MIN): @
MATERIAL:
INSUL TYPE / PURP / THK: / /
TRIM SPEC:
TRIM REF:

NOTES

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.
4. YC-160284 HAS BEEN SPLIT INTO TWO DRAWINGS. PLEASE REFERENCE YC-160284A AND YC-160284B.



REFERENCE DRAWINGS

00401706.XLSX LINE LIST

0	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361	
REV	DATE	DESCRIPTION	BY	CHARGE	

NOTE

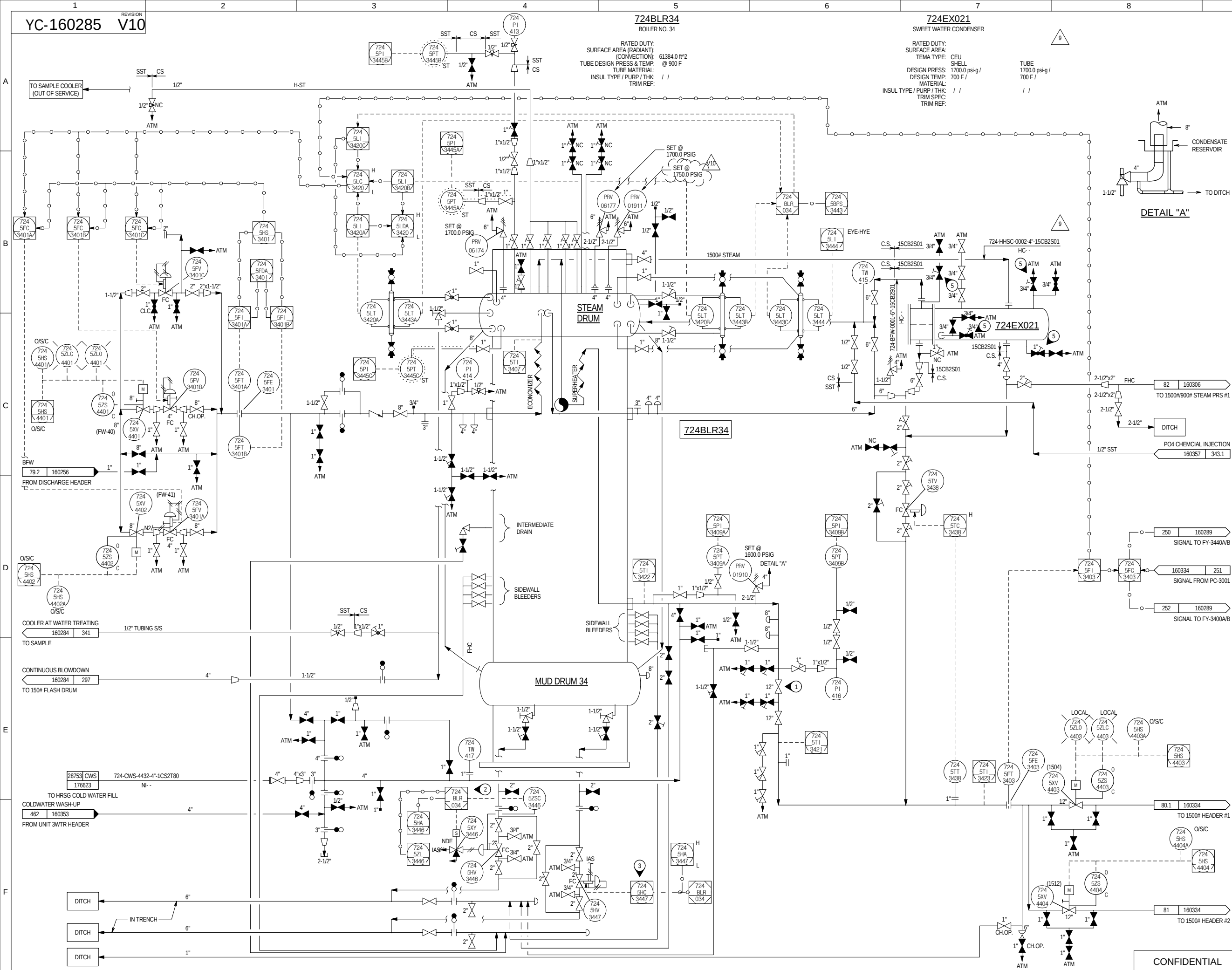
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P3ST
724VES35734
CONTINUOUS BLOWDOWN
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY:		NUMBER ASSIGNED	DATE
DESIGN BY:		AND TRACING	
CHKD BY:	CHARGE:	RECORD ENTERED	
JOB NO.		YC- 160284B	REVISION 0

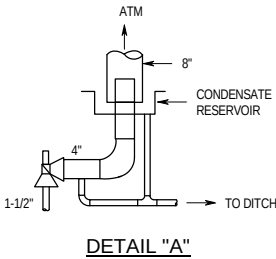
CONFIDENTIAL



NOTES

1. ALL INSTRUMENTS IN DCS LOOPS ARE PREFIXED WITH "5". NON DCS (LOCAL) INSTRUMENTS ARE PREFIXED WITH "XXX".
2. BLOWDOWN VALVES AUTOMATICALLY OPEN ON HIGH LEVEL.
3. LOGIC OVERRIDE AUTOMATICALLY OPENS VALVE ON HIGH LEVEL AND CLOSES VALVE ON LOW LEVEL.
4. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
5. ANGLE BLOW-OFF VALVE V-85033 PER PIPE SPEC J911A.
6. DUE TO CONCURRENT REVISIONS THE FOLLOWING REVISION REQUIRED MERGING:

REV.	DATE	DESCRIPTION	BY	APP	CHARGE
7A	11/14/13	REPLACE SWEET WATER CONDENSER	CH		5384



REFERENCE DRAWINGS

YC-181633 723BLR34 BOILER FEED WATER
SUPPLY & 1500# STM. OUTLET
PAR IPF OVERVIEW

9S	09/23/14	MERGER REVISION		JBB
8S	08/04/14	SMARTPLANT CONVERSION		
7	04/26/11	ADDED STEAM DRUM PRESS TRANSMITTER	BBG	6000904901
6	03/14/08	AS-BUILT	ARS	4372
5	06/21/07	GENERAL REVISION	ARS	4372
4	05/09/07	ISSUE FOR GENERAL MOV REV	ARS	4372
3	12/06/06	ISSUE FOR BOILER 34/35 INSTR IFC	ARS	4372
2	07/26/04	REVISED PER PS3 HRSG NO. 37 PRJ	JN	4164
1	08/14/00	REVISED EMS SYSTEM	EM	3925
V10	03/05/19	MOC REVISION	JBB	M2019122-001
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

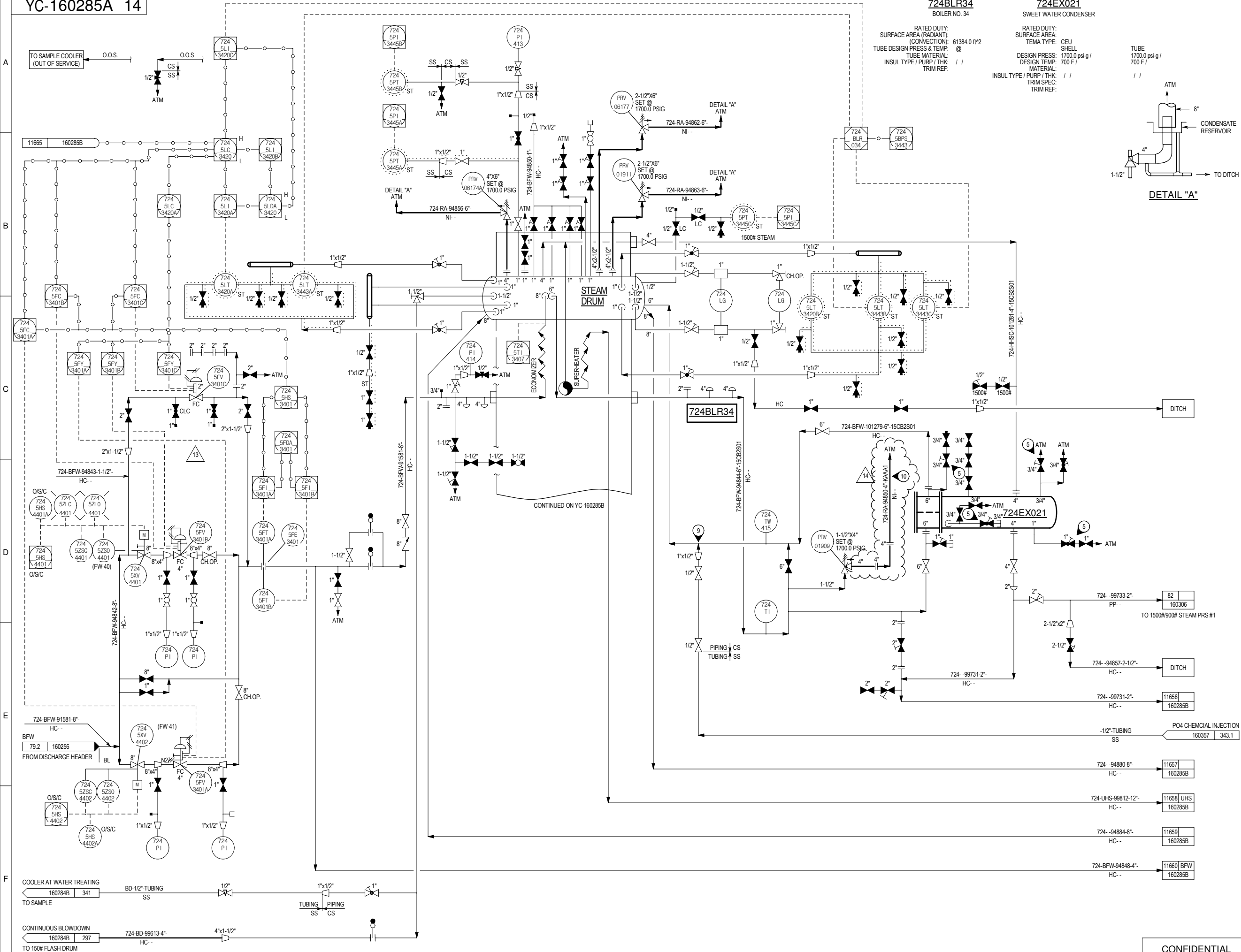
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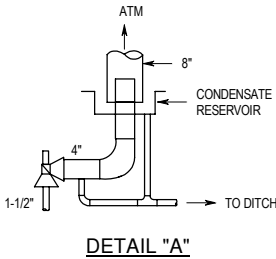
P3ST
724BLR34
BOILER FEED WATER SUPPLY & 1500# STEAM OUTLET
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: SJC	3/4/1996	NUMBER ASSIGNED	DATE
DESIGN BY: T.J	5/1/1996	AND TRACING	05/01/96
CHKD BY: T.J	CHARGE:	RECORD ENTERED	REVISION
JOB NO.		YC- 160285	V10



1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "75", UNLESS OTHERWISE NOTED.
2. BLOWDOWN VALVES AUTOMATICALLY OPEN ON HIGH LEVEL.
3. LOGIC OVERRIDE AUTOMATICALLY OPENS VALVE ON HIGH LEVEL AND CLOSURES VALVE ON LOW LEVEL.
4. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199998 TO YC-199999.
5. ANGLE BLOW-OFF VALVE V-850J3 PER PIPE SPEC J911A.
6. DUE TO CONCURRENT REVISIONS THE FOLLOWING REVISION REQUIRED MERGING:

REV.	DATE	DESCRIPTION	BY	APP	CHARGE
7A	11/14/13	REPLACE SWEET WATER CONDENSER	GH		5384
7. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.
8. YC-160285 HAS BEEN SPLIT INTO TWO DRAWINGS. PLEASE REFERENCE YC-160285A AND YC-160285B.
9. STAINLESS INJECTION LINE PROJECTS 6" INTO THE 6" BFW PIPING AT INJECTION TIE IN POINT. SEE MOC-22-1203
10. REROUTE OF 34 BOILER PRV-01909 DISCHARGE TO SAFE LOCATION



REFERENCE DRAWINGS

00401706.XLSX LINE LIST

12	09/21/22	MOC REVISION	JBB	MOC 22-1203
11	04/01/20	REVISED PER P3ST WALK DOWN	WCOT	MOC 19-1361
V10	03/05/19	MOC REVISION	JBB	M2019122-001
9	09/23/14	MERGER REVISION	JBB	
8S	08/04/14	SMARTPLANT CONVERSION		
7	04/26/11	ADDED STEAM DRUM PRESS TRANSMITTER	BBG	6000904901
14	02/11/25	MOC REVISION	JTF	MOC 23-1603
13	11/18/24	MOC REVISION	JTF	MOC 24-1220
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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PORT ARTHUR
REFINERY

P3ST
724BLR34, 724EX021
BOILER FEED WATER SUPPLY & 1500# STEAM OUTLET
PIPING & INSTRUMENTATION DIAGRAM

	DATES DRAWN BY: SJC 3/4/1996 DESIGN BY: T.J. 5/1/1996 CHKED BY: TJ CHARGE:	SCALE: NONE NUMBER ASSIGNED AND TRACING RECORD ENTERED	DATE 05/01/96	REVISION
JOB NO.	YC- 160285A			14

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. BLOWDOWN VALVES AUTOMATICALLY OPEN ON HIGH LEVEL.
3. LOGIC OVERRIDE AUTOMATICALLY OPENS VALVE ON HIGH LEVEL AND CLOSES VALVE ON LOW LEVEL.
4. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
5. ANGLE BLOW-OFF VALVE V-850J3 PER PIPE SPEC J911A.
6. DUE TO CONCURRENT REVISIONS THE FOLLOWING REVISION REQUIRED MERGING:
REV. DATE DESCRIPTION
7A 11/14/13 REPLACE SWEET WATER CONDENSER
7. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.
8. YC-160285 HAS BEEN SPLIT INTO TWO DRAWINGS. PLEASE REFERENCE YC-160285A AND YC-160285B.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

NOTE

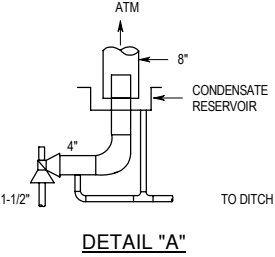
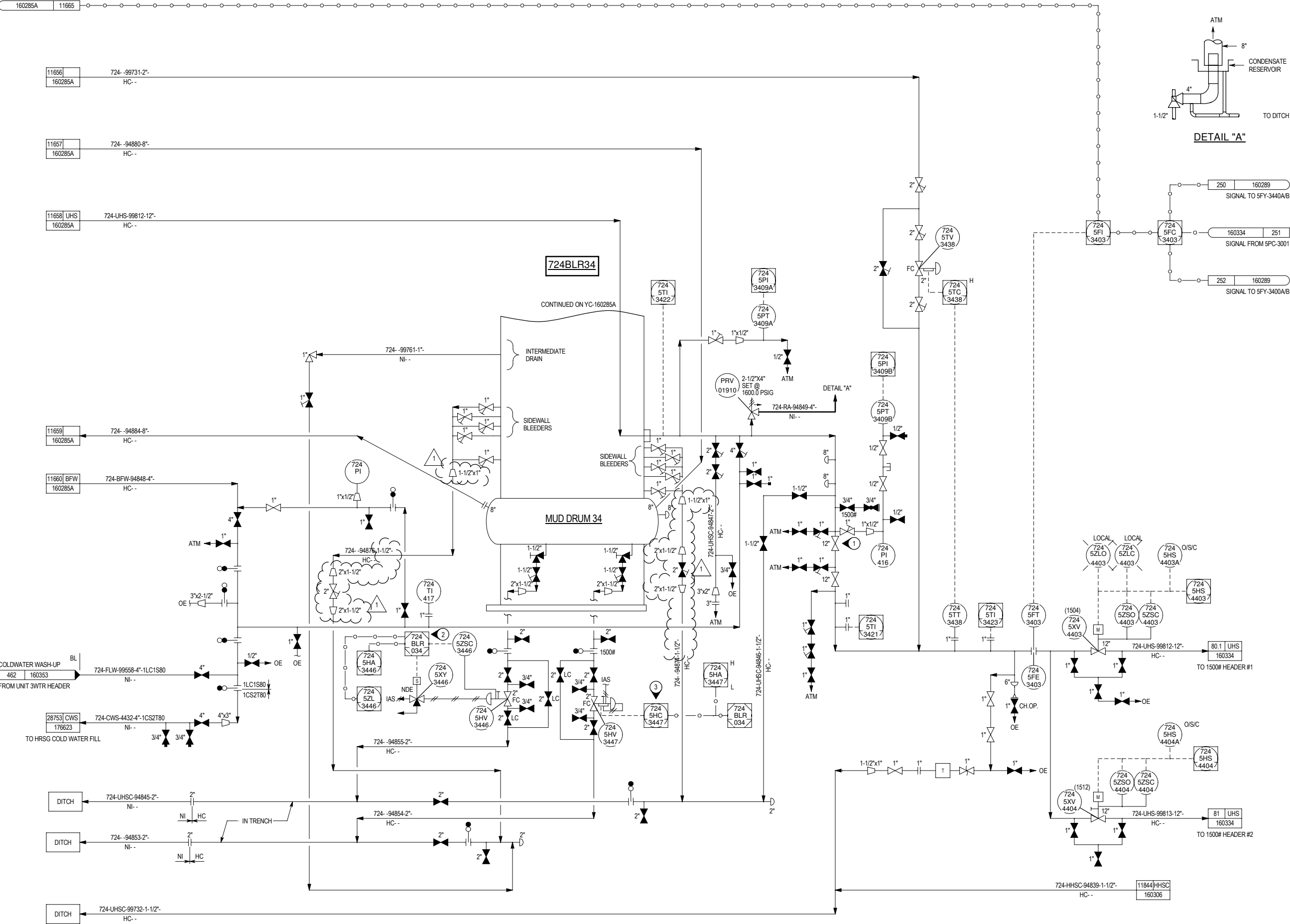
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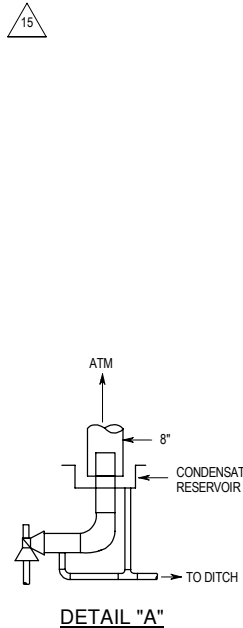
P3ST
724BLR34
BOILER FEED WATER SUPPLY & 1500# STEAM OUTLET
PIPING & INSTRUMENTATION DIAGRAM

DATES		SCALE: NONE	
DRAWN BY:		NUMBER ASSIGNED	DATE
DESIGN BY:		AND TRACING	
CHKD BY:	CHARGE:	RECORD ENTERED	
JOB NO.		YC- 160285B	
		1	

CONFIDENTIAL



1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5". UNLESS OTHERWISE NOTED.
2. BLOWDOWN VALVES AUTOMATICALLY OPEN ON HIGH LEVEL.
3. LOGIC OVERRIDE AUTOMATICALLY OPENS VALVE ON HIGH LEVEL AND CLOSSES VALVE ON LOW LEVEL.
4. FLOW CONTROL FC-3503 IS PRESSURE (P-3509A) AND TEMPERATURE (T-3510) COMPENSATED.
5. YARWAY/TYCO WELBOND Y-PATTERN GLOBE VA. F22 BODY, 1700" SW PROCESS CONN. STELLITE SEATS AND DISC MODEL 5617.
6. 1500" GATE VA., SW 1-1/4CR-1/2MO BODY, GA15CJ300.
7. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-1999999 TO YC-1999999.
8. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.
9. YC-160286 HAS BEEN SPLIT INTO TWO DRAWINGS. PLEASE REFERENCE YC-160286A AND YC-160286B.
10. STAINLESS INJECTION LINE PROJECTS "6" INTO THE 6" BFW PIPING AT INJECTION TIE-IN POINT. SEE MOC-22-1189.



REFERENCE DRAWINGS

00401706.XLSX LINE LIST

15	08/22/22	MOC REVISION	JBB	MOC 22-1189
14	07/20/22	MOC REVISION	CLF	MOC 22-999
13	02/11/21	MOC REVISION	JTF	MOC 20-210
12	09/30/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
V11	03/05/19	MOC REVISION	JBB	M2019122-001
V10	09/18/18	MOC REVISION	JBB	M20143230
95	08/18/14	SMARTPLANT CONVERSION		
6	11/18/24	MOC REVISION	JTF	MOC 24-1220
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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MOTIVA
ENTERPRISES LLC

PORT ARTHUR
REFINERY

P3ST
723BLR35
BOILER FEED WATER SUPPLY & 1500# STM OUTLET
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	3/4/1996	NUMBER ASSIGNED	DATE
DESIGN BY: TJ	5/1/1996	AND TRACING	
CHKD BY: TJ	CHARGE:	RECORD ENTERED	REVISION
JOB NO.		YC- 160286A	16

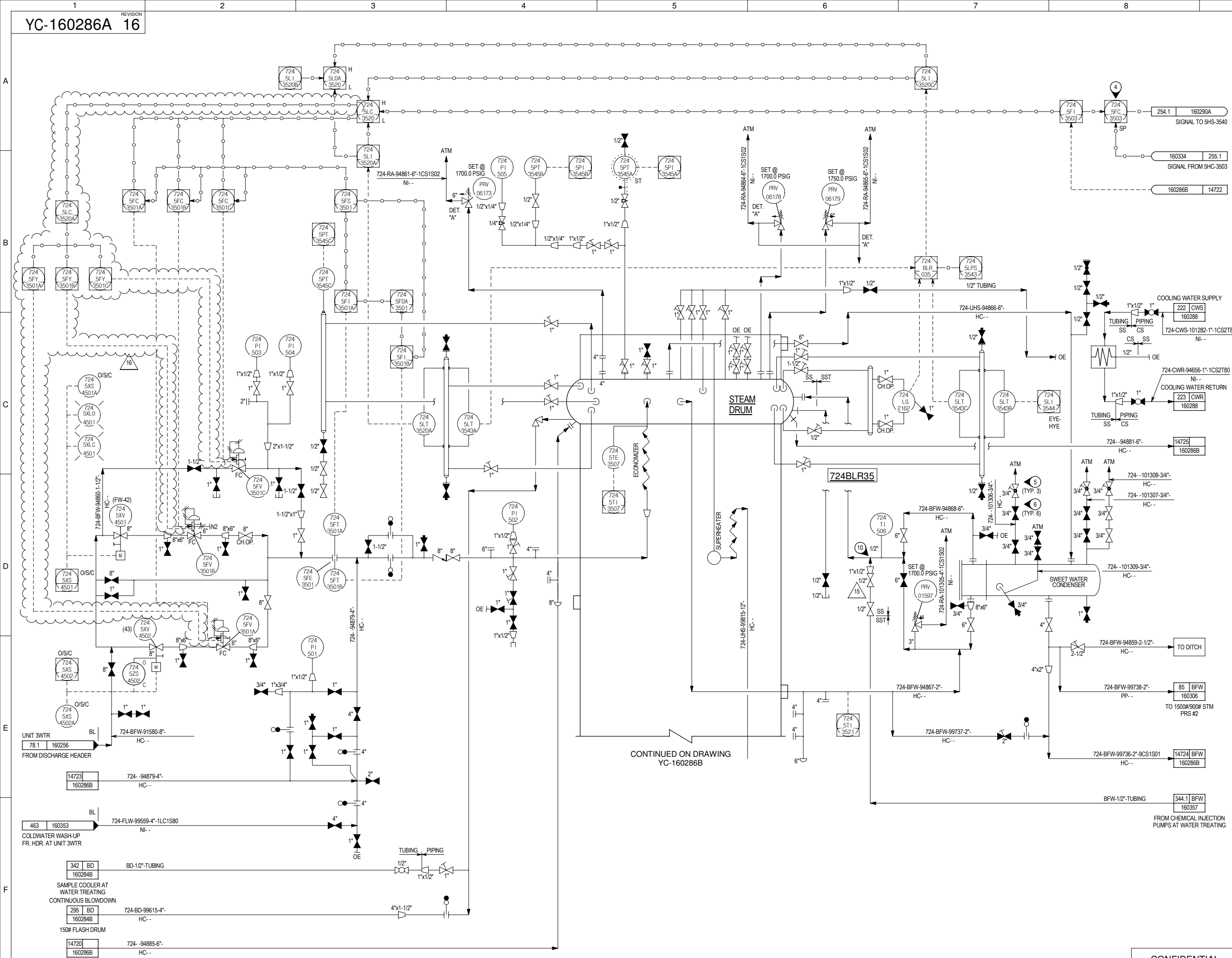
CONFIDENTIAL

SAFETY---A LIFETIME COMMITMENT

SPPID REV 0

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT





-

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

NOTE

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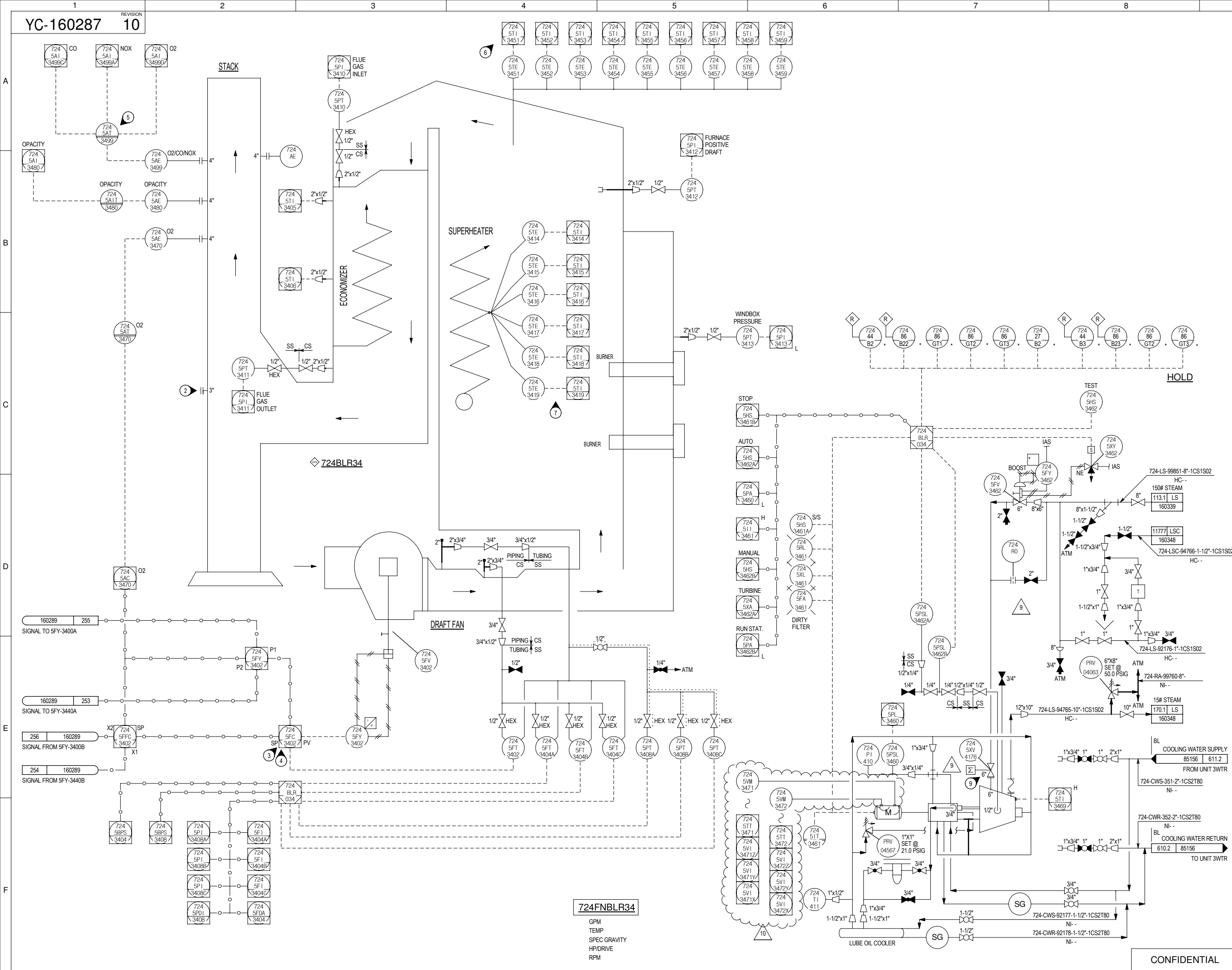


P3ST
723BLR35
BOILER FEED WATER SUPPLY & 1500# STM OUTLET
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE
DRAWN BY:		NUMBER ASSIGNED
DESIGN BY:		DATE
		AND TRACING
		RECORD ENTERED

CONFIDENTIAL

YC- 160286B 2



NOTES

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. TEST NOZZLES (4 REQUIRED).
3. BMS SYSTEM DRIVES OUTPUT TO APPROXIMATELY 65% FOR PURGE.
4. BMS SYSTEM DRIVES OUTPUT TO APPROXIMATELY 10% FOR INITIAL LIGHT OFF.
5. SAT-3499 LOCATED IN ANALYZER SHELTER AS-0929. REFER TO P&ID YC-176618.
6. LOCATION TO BE DETERMINED BY FIELD.
7. TWELVE (12) THERMOCOUPLES, SIX (6) SPARES FOR BURNOUTS.
8. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
9. SEE DWG. YC-197885 FOR TURBINE SPEED AND TRIP CONTROL DETAILS.
10. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE RETIRED MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

8	04/10/20	REVISED PER PS3T WALK DOWN	WCT	MOC 19-1361
7	10/10/18	MOC REVISION (PER USMF/3155)	JTF	19-703
6S	08/11/14	SMARTPLANT CONVERSION		
5	06/13/07	AS-BUILT	ARS	4372
4	12/06/06	ISSUE FOR BOILER 34-35 INSTR IFC	ARS	4372
3	07/26/04	REVISED PER PS3 HRSG NO. 37 PROJECT	JN	4164
10	12/11/24	MOC REVISION	JTF	MOC 24-1628
9	12/04/24	MOC REVISION	JTF	MOC 24-1659
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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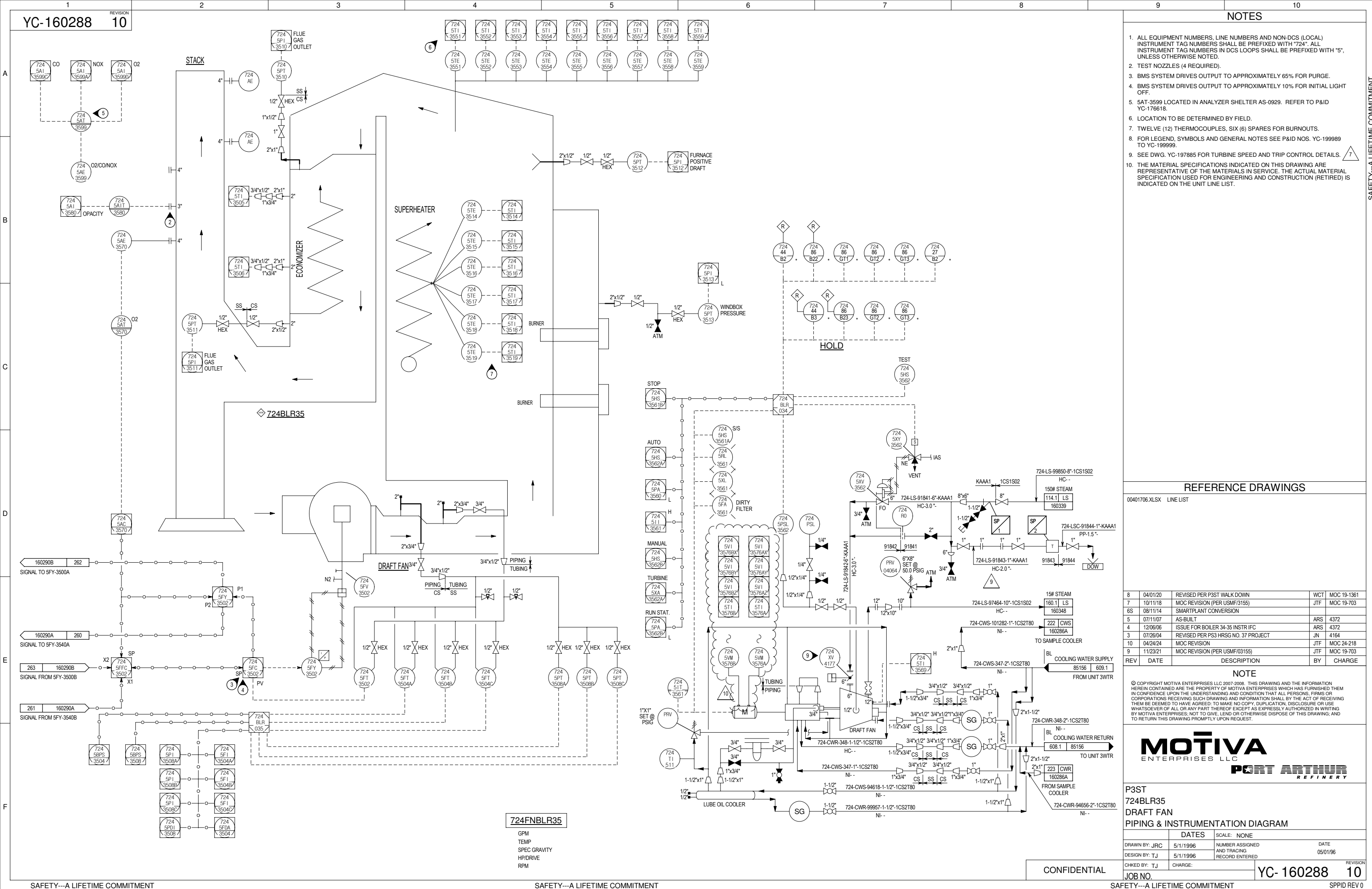
MOTIVA
ENTERPRISES LLC

PORT ARTHUR
REFINERY

P3ST
724BLR34
DRAFT FAN
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED	DATE
DESIGN BY: JT	5/1/1996	AND TRACING	05/01/96
CHKD BY: TJ	CHARGE:	RECORD ENTERED	
JOB NO.	YC- 160287		REVISION 10

SAFETY--A LIFETIME COMMITMENT



NOTES

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. TEST NOZZLES (4 REQUIRED).
3. BMS SYSTEM DRIVES OUTPUT TO APPROXIMATELY 65% FOR PURGE.
4. BMS SYSTEM DRIVES OUTPUT TO APPROXIMATELY 10% FOR INITIAL LIGHT OFF.
5. SAT-3599 LOCATED IN ANALYZER SHELTER AS-0929. REFER TO P&ID YC-176618.
6. LOCATION TO BE DETERMINED BY FIELD.
7. TWELVE (12) THERMOCOUPLES, SIX (6) SPARES FOR BURNOUTS.
8. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
9. SEE DWG. YC-197885 FOR TURBINE SPEED AND TRIP CONTROL DETAILS.
10. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

8	04/01/20	REVISED PER P33T WALK DOWN	WCT	MOC 19-1361
7	10/11/18	MOC REVISION (PER USMF/3155)	JTF	MOC 19-703
6S	08/11/14	SMARTPLANT CONVERSION		
5	07/11/07	AS-BUILT	ARS	4372
4	12/06/06	ISSUE FOR BOILER 34-35 INSTR IFC	ARS	4372
3	07/26/04	REVISED PER PS3 HRSG NO. 37 PROJECT	JN	4164
10	04/24/24	MOC REVISION	JTF	MOC 24-218
9	11/23/21	MOC REVISION (PER USMF/03155)	JTF	MOC 19-703
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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MOTIVA
ENTERPRISES LLC

PORT ARTHUR
REFINERY

P3ST
724BLR35
DRAFT FAN
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED	DATE
DESIGN BY: TJ	5/1/1996	AND TRACING	05/01/96
CHKD BY: TJ	CHARGE:	RECORD ENTERED	
JOB NO.	YC- 160288		REVISION 10

SAFETY---A LIFETIME COMMITMENT

00401706.XLSX LINE LIST

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P3ST

724BLR34

BURNER CONTROLS

PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED	DATE
DESIGN BY: TJ	5/1/1996	AND TRACING	05/01/96
		RECORD ENTERED	

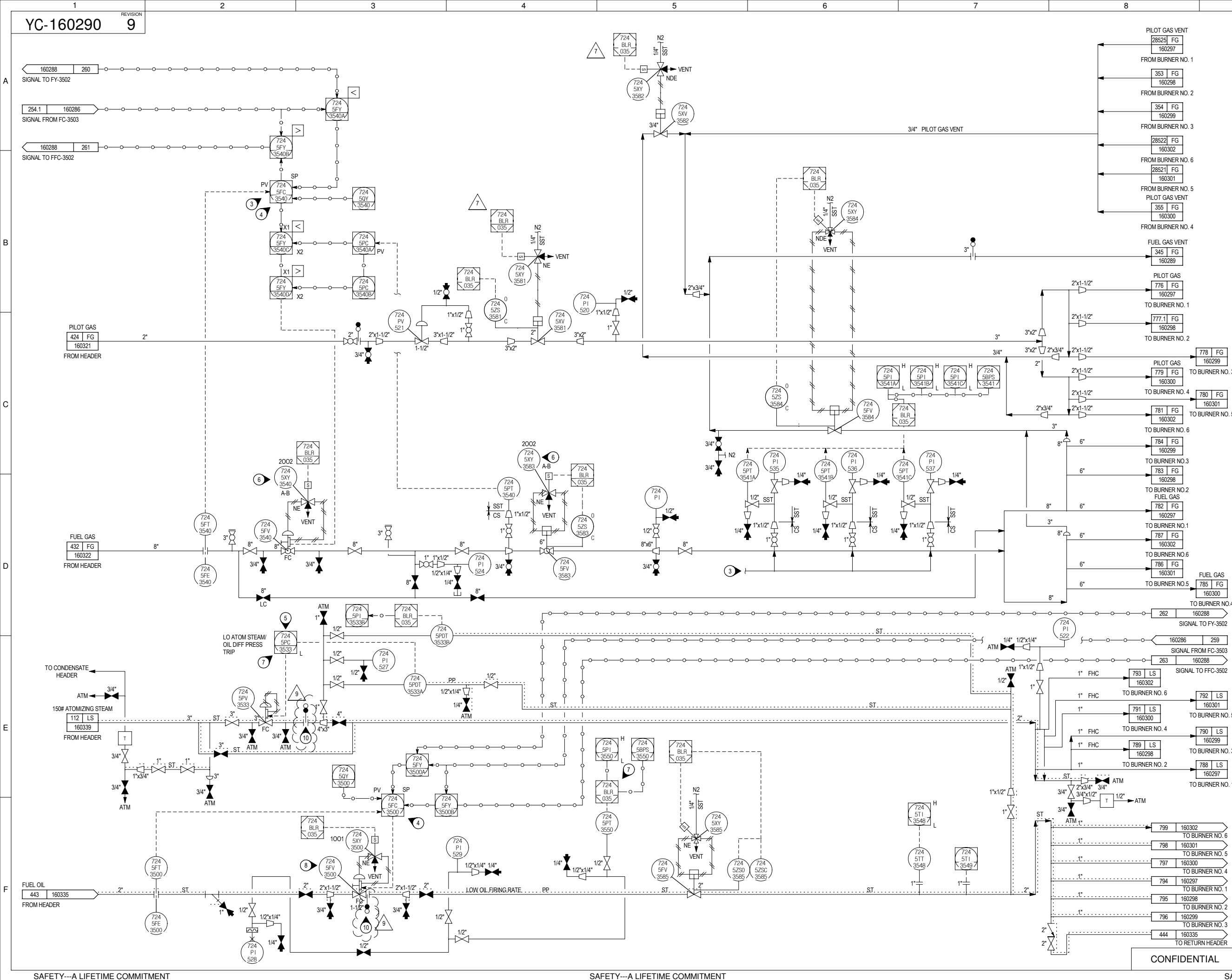
CHKD BY:

YC- 160289

10

SAFETY---A LIFETIME COMMITMENT

SPPID REV 0



1. ALL INSTRUMENTS IN DCS LOOPS ARE PREFIXED WITH "5". NON DCS (LOCAL) INSTRUMENTS ARE PREFIXED WITH "XXX".
2. AUTOMATE LEAK TEST.
3. PRESSURE OVERRIDE FROM MID OF 3 PT-3541A/B/C.
4. TRIP TO MANUAL.
5. TRIP TO NORMAL AFTER ONE MINUTE.
6. REFER TO SPECIFIC VALVE DETAILS ON DWG. YC-181653.
7. ALARMS ACTIVE IF FUEL OIL PUMP RUNNING (XS-3065 OR COMMANDED STATE XV-3110).
8. REFER TO DETAIL 5, DWG. YC-184535.
9. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199889 TO YC-199899.
10. 150# ATOMIZING STEAM AND FUEL OIL HEADERS ARE BLINDED AND OUT OF SERVICE.

YC-181638	723BLR35 BURNER CONTROLS PAR IPF OVERVIEW
YC-181653	2002 VALVES & RCS PANELS
YC-184535	MISCELLANEOUS DETAILS

8S	08/04/14	SMARTPLANT CONVERSION		
7	07/11/07	AS BUILT	ARS	4372
6	05/17/07	RE-ISSUE FOR BOILER 34-35 INSTR IFC	ARS	4372
5	12/06/06	ISSUE FOR BOILER 34-35 INSTR IFC	ARS	4372
4	01/17/02	ADDED BYPASS AROUND FV-3540	CHI	2001039709
3	12/27/00	REV. FUEL GAS VENT PIPING	HWP	4010
2	08/14/00	REVISED EMS SYSTEM	EM	3925
1	10/20/16	MOC REVISION	JBK	M20162669
REV	DATE	DESCRIPTION	BY	CHARGE

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MOTIVA
ENTERPRISES LLC

PORT ARTHUR
REFINERY

P3ST
724BLR35
BURNER CONTROLS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	1/5/1996	NUMBER ASSIGNED	DATE
DESIGN BY: TJ	1/5/1996	AND TRACING	05/01/96
CHKD BY: TJ	CHARGE:	RECORD ENTERED	
JOB NO.	YC- 160290		REVISION 9

5	04/01/20	REVIEW PER P3ST WALK DOWN	WCT	MOC 19-1361
4S	08/05/14	SMARTPLANT CONVERSION		
3	10/01/13	MOC REVISION	JBB	M20122216
2	03/14/08	AS-BUILT	ARS	4372
1	12/06/06	ISSUE FOR BOILER 34-35 INSTR IFC	ARS	4372
				3742
REV	DATE	DESCRIPTION	BY	CHARGE

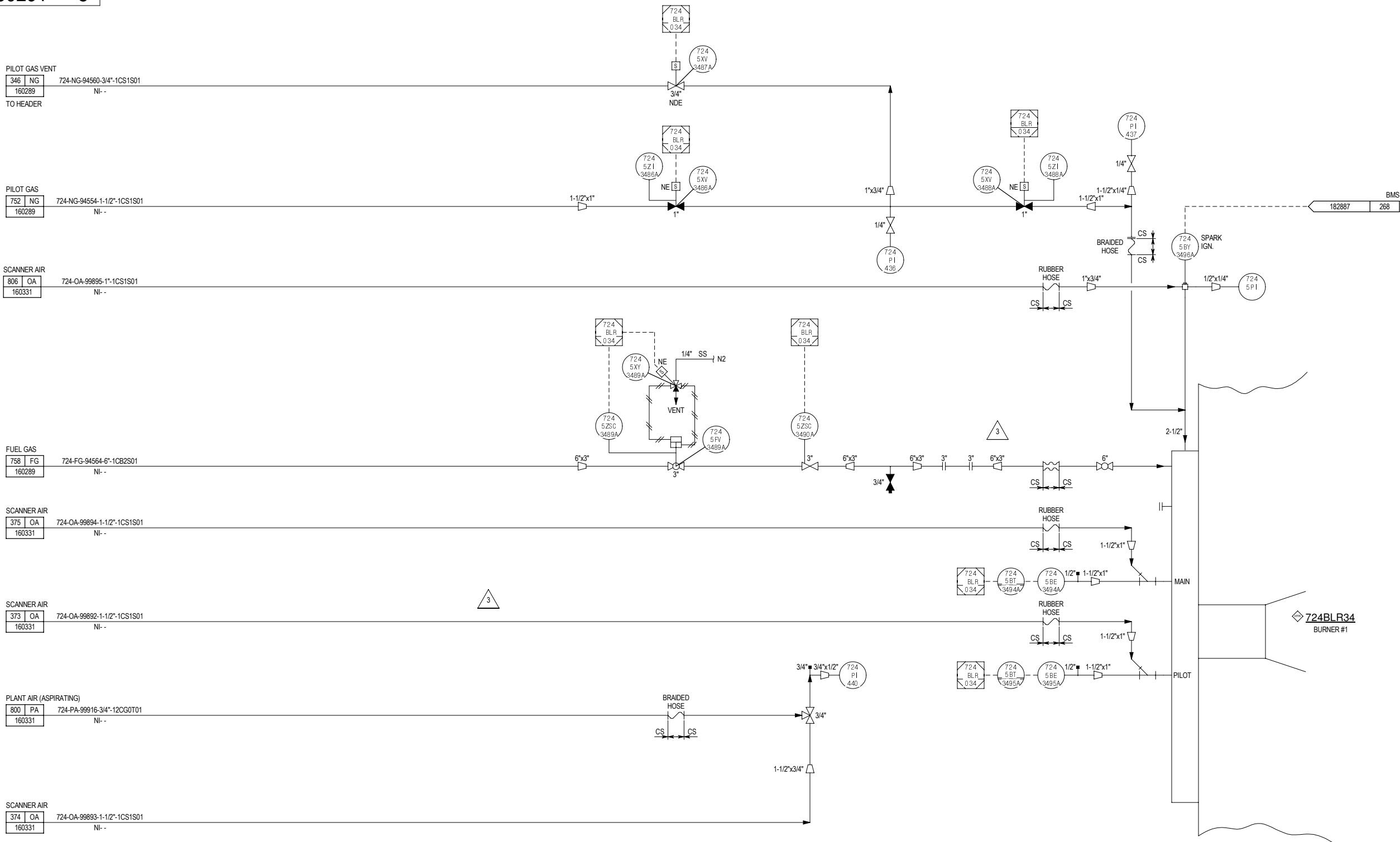
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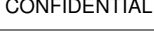
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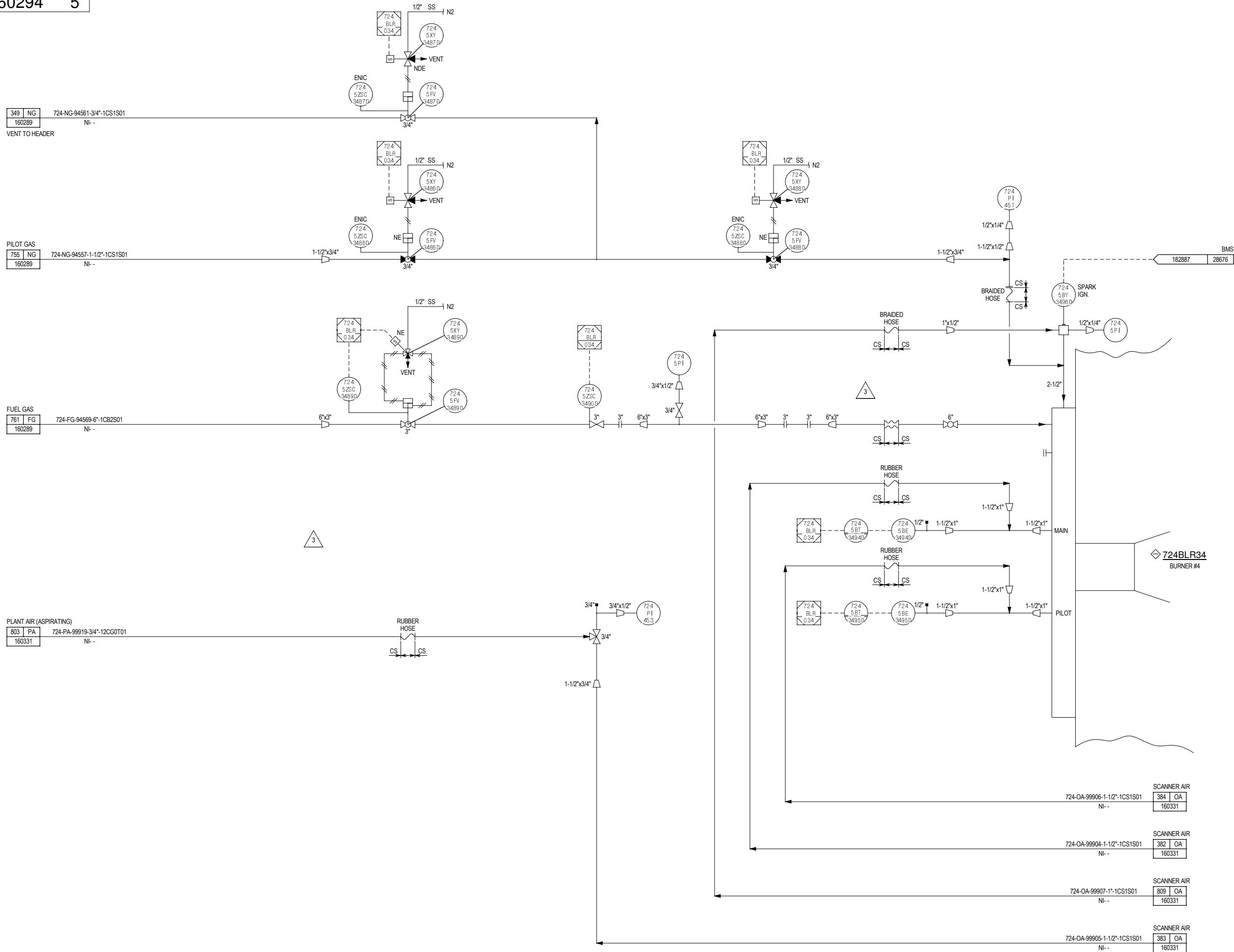
P3ST
724BLR34
BURNER #1 CONTROLS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED	DATE
DESIGN BY: TJ	5/1/1996	AND TRACING	05/01/96
CHECKED BY: TJ	CHARGE: 3742	RECORD ENTERED	
JOB NO.		YC- 160291	REVISION 5





- | | | | |
|---------------|------------|-----------------|------------------|
| | DATES | SCALE: NONE | |
| DRAWN BY: JRC | 5/1/1996 | NUMBER ASSIGNED | DATE
05/01/96 |
| DESIGN BY: TJ | 5/1/1996 | AND TRACING | |
| CHKD BY: TJ | CHARGE: | RECORD ENTERED | |
| JOB NO. | YC- 160293 | | REVISION
5 |



NOTES

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

5	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
4S	08/04/14	SMARTPLANT CONVERSION		
3	10/02/13	MOC REVISION	JBB	M20122216
2	03/14/08	AS-BUILT	ARS	4372
1	12/06/06	ISSUE FOR BOILER 34-35 INSTR IFC	ARS	4372
				3742
REV	DATE	DESCRIPTION	BY	CHARGE

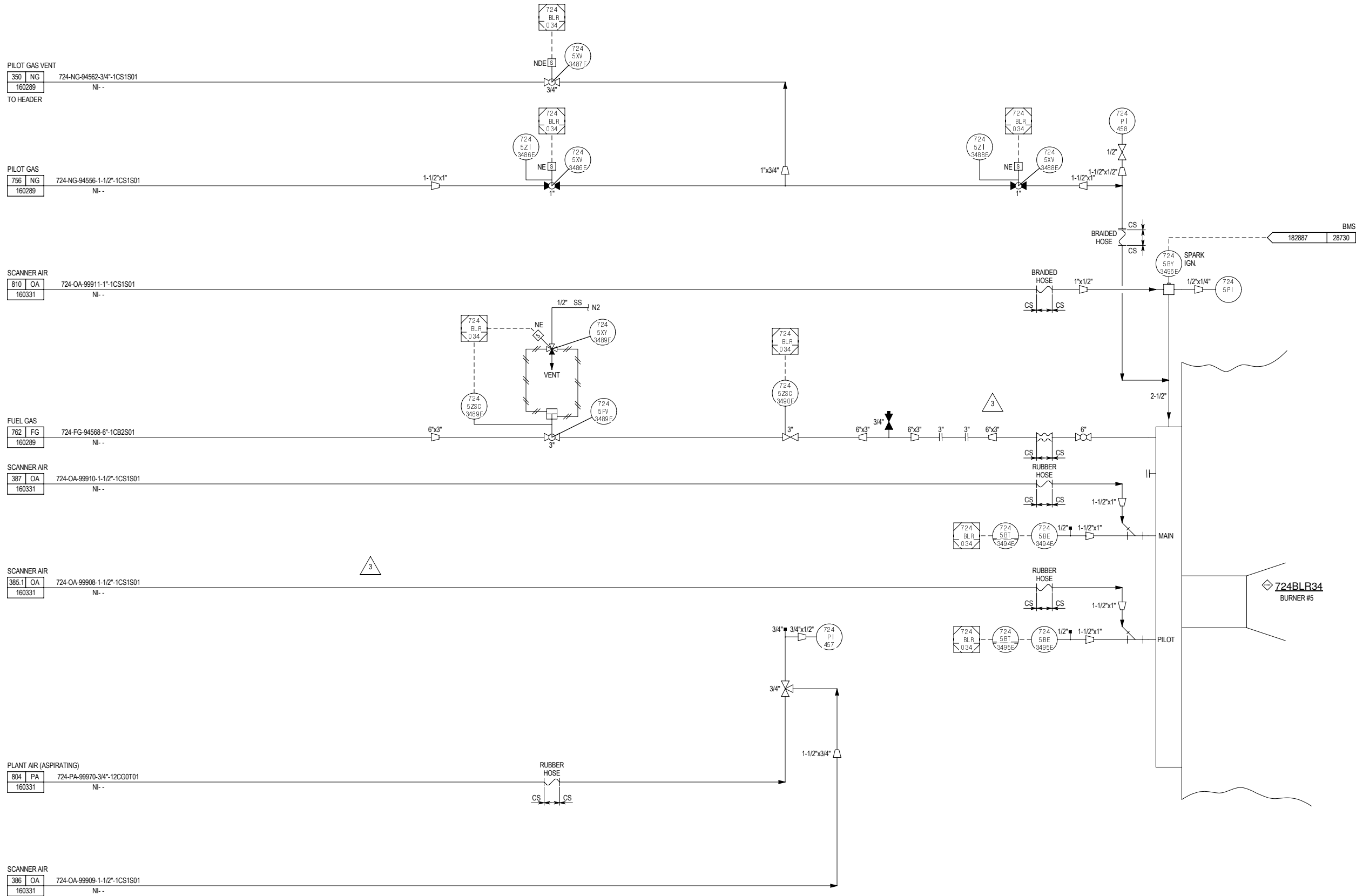
NOTE

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P3ST
724BLR34
BURNER #4 CONTROLS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED	DATE 05/01/96
DESIGN BY: TJ	5/1/1996	AND TRACING	
CHKD BY: TJ	CHARGE:	RECORD ENTERED	
JOB NO.		YC- 160294	REVISION 5



1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

5	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
4S	08/04/14	SMARTPLANT CONVERSION		
3	10/02/13	MOC REVISION	JBB	M20122216
2	03/14/08	AS-BUILT	ARS	4372
1	12/06/06	ISSUE FOR BOILER 34-35 INSTR IFC	ARS	4372
				3742
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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P3ST
724BLR34
BURNER #5 CONTROLS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED AND TRACING	DATE 05/01/96
DESIGN BY: TJ	5/1/1996	RECORD ENTERED	
CHKD BY: TJ	CHARGE: 3742		
JOB NO.		YC- 160295	REVISION 5

CONFIDENTIAL

YC- 160295

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

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SAFETY---A LIFETIME COMMITMENT

SPPID REV 0

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

5	03/09/21	MOC REVISION	JBB	20-209
4	04/01/20	REVISED PER PSTT WALK DOWN	WCT	MOC 19-1361
3S	08/06/14	SMARTPLANT CONVERSION		
2	07/11/07	AS BUILT	ARS	4372
1	12/06/06	ISSUE FOR BOILER 34-35 INSTR IFC	ARS	4372
				3742
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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P3ST
724BLR35
BURNER #1 CONTROLS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED AND TRACING	DATE 05/01/96
DESIGN BY: TJ	5/1/1996	RECORD ENTERED	

CHCKED BY: TJ	CHARGE: 3742	YC- 160297	REVISION
JOB NO.			5

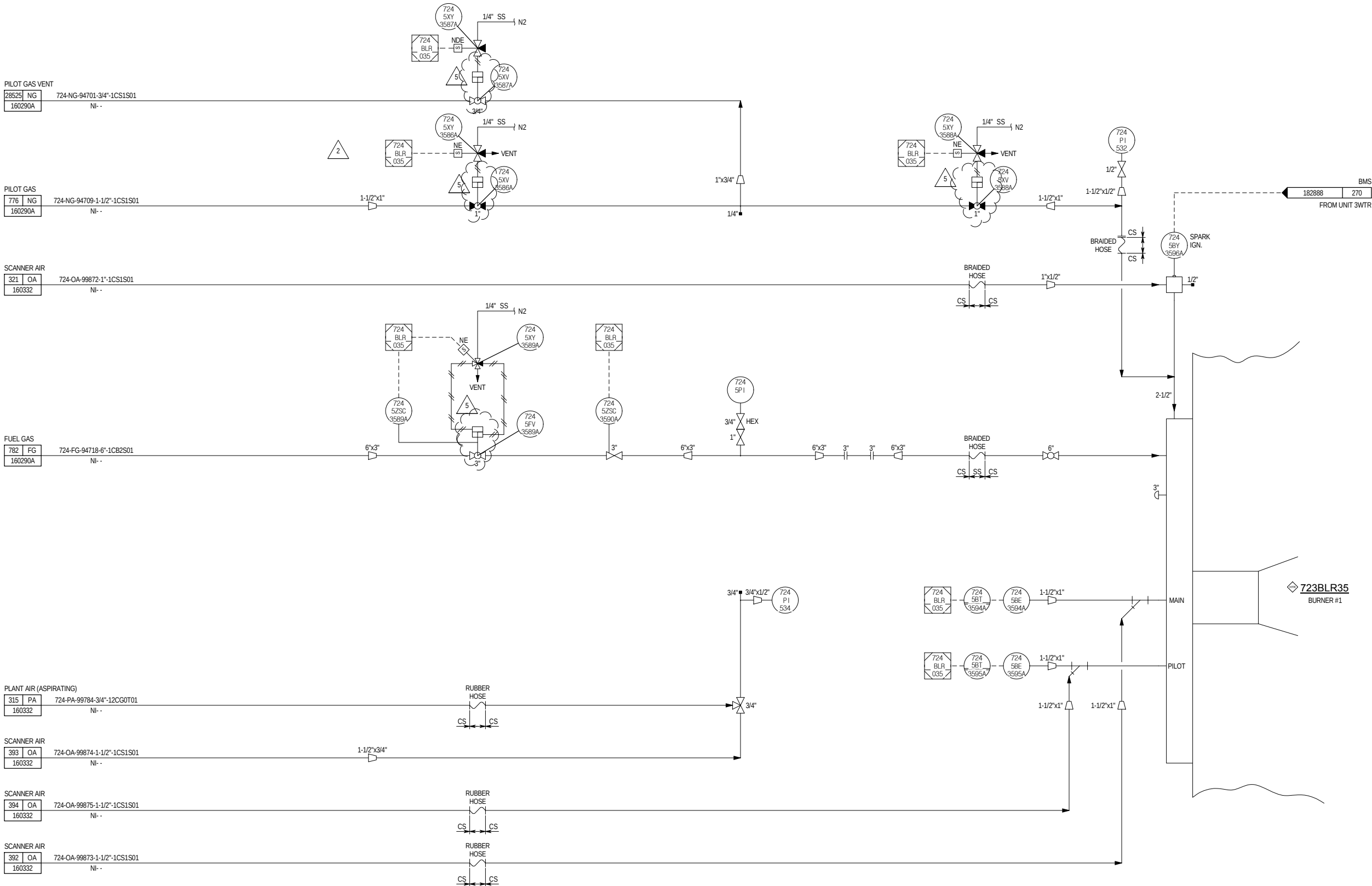
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SAFETY--A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

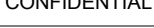
SAFETY---A LIFETIME COMMITMENT

SPPID REV 0





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|----------------|--------------|-----------------|---------------|
| | DATES | SCALE: NONE | |
| DRAWN BY: JRC | 5/1/1996 | NUMBER ASSIGNED | DATE |
| DESIGN BY: TJ | 5/1/1996 | AND TRACING | 05/01/96 |
| CHECKED BY: TJ | CHARGE: 3742 | RECORD ENTERED | |
| JOB NO. | | YC- 160298 | REVISION
5 |



1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

00401706.XLSX LINE LIST

NOTE

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	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED AND TRACING	DATE 05/01/96
DESIGN BY: TJ	5/1/1996	RECORD ENTERED	
CHECKED BY: TJ	CHARGE: 3742		REVISION
JOB NO.		YC- 160299	5

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

5	03/09/21	MOC REVISION	JBB	20-209
4	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
3S	08/13/14	SMARTPLANT CONVERSION		
2	07/11/07	AS-BUILT	ARS	4372
1	12/06/06	ISSUE FOR BOILER 34-35 INSTR IFC	ARS	4372
				3742
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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P3ST

724BLR35

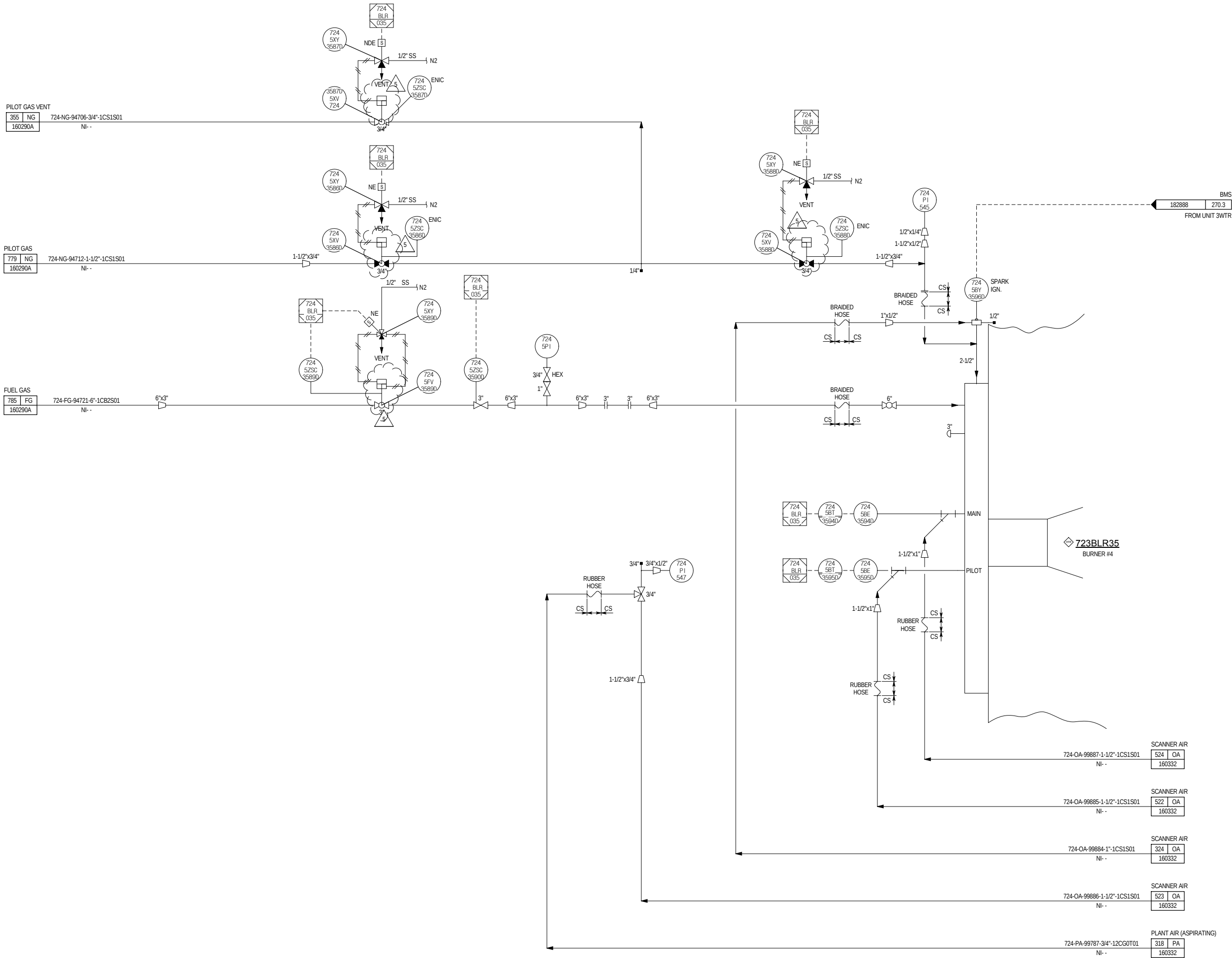
BURNER #4 CONTROLS

PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED AND TRACING	DATE 05/01/96
DESIGN BY: TJ	5/1/1996	RECORD ENTERED	

CHIKED BY:

YC- 160300

REVISION
5

CONFIDENTIAL

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

5	03/09/21	MOC REVISION	JBB	20-209
4	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
3S	08/11/14	SMARTPLANT CONVERSION		
2	07/11/07	AS BUILT	ARS	4372
1	12/06/06	ISSUE FOR BOILER 34-35 INSTR IFRC	ARS	4372
				3742
REV	DATE	DESCRIPTION	BY	CHARGE

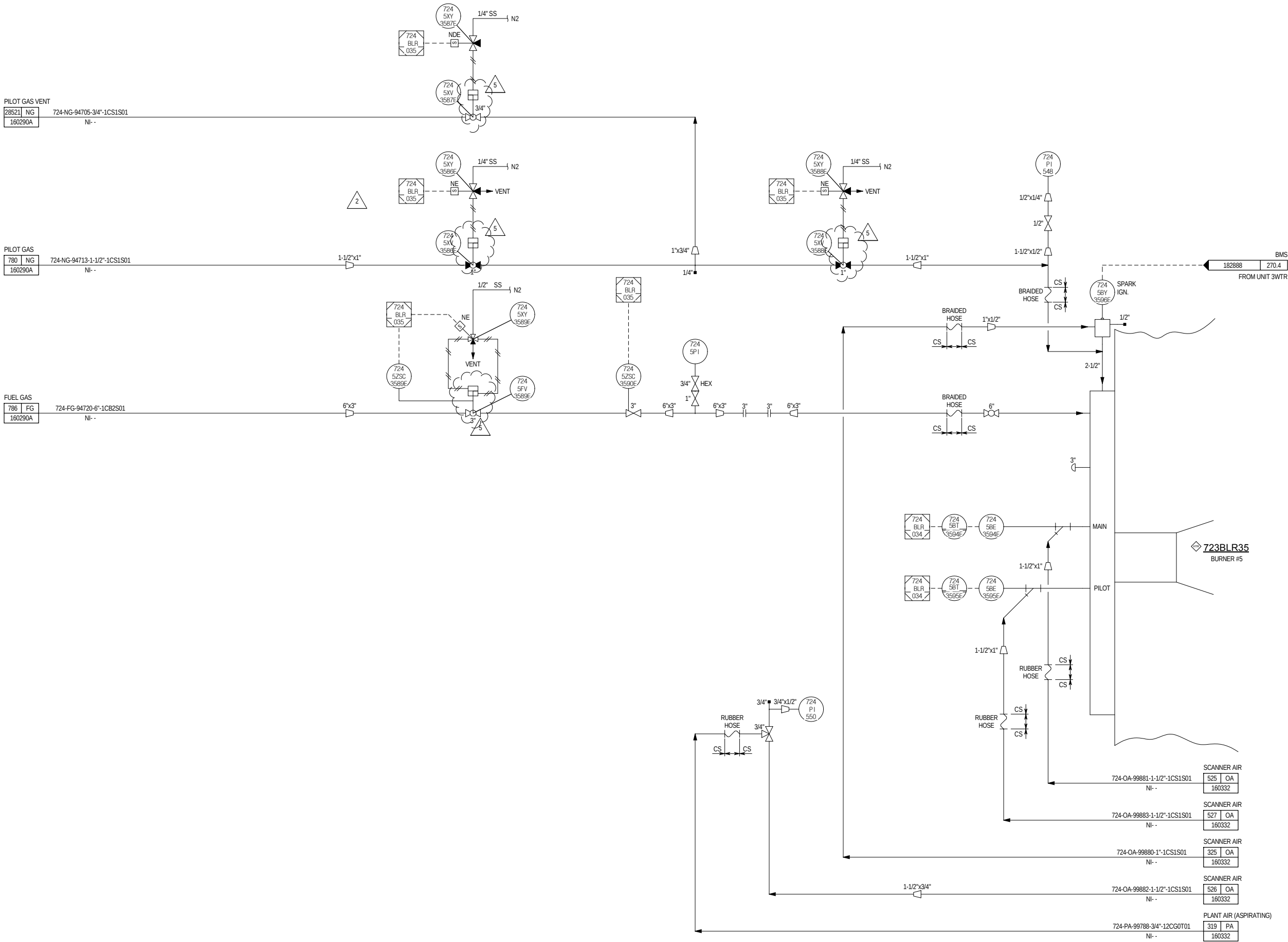
NOTE

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P3ST
724BLR35
BURNER #5 CONTROLS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED AND TRACING	DATE
DESIGN BY: TJ	5/1/1996	RECORD ENTERED	05/01/96

REVISION
5

CONFIDENTIAL

5	03/09/21	MOC REVISION	JBB	20-209
4	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
3S	08/20/14	SMARTPLANT CONVERSION		
2	07/11/07	AS BUILT	ARS	4372
1	12/06/06	ISSUE FOR BOILER 34-35 INSTR IFC	ARS	4372
				3742
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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P3ST

724BLR35

BURNER #6 CONTROLS

PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED AND TRACING	DATE 05/01/96
DESIGN BY: TJ	5/1/1996	RECORD ENTERED	

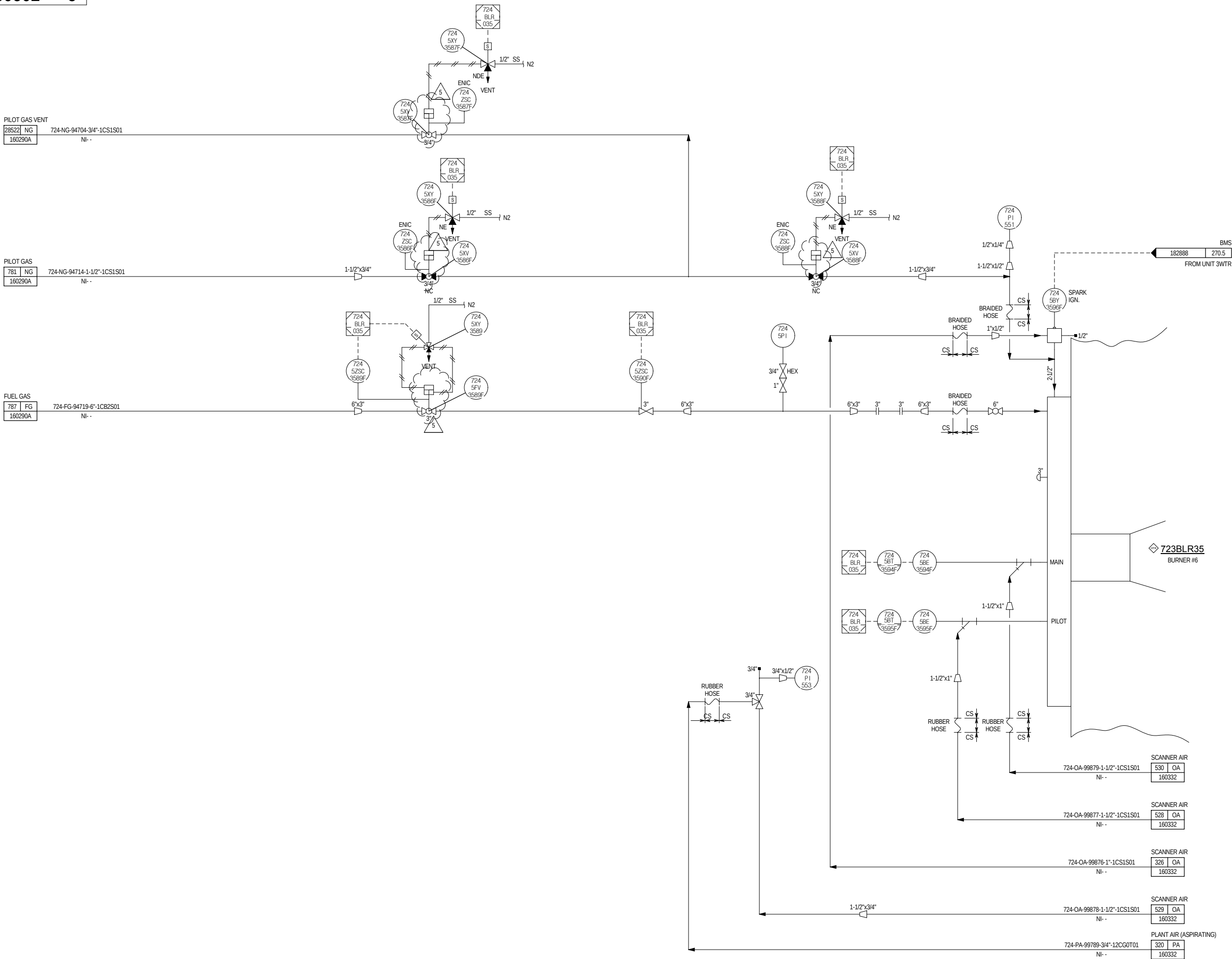
CHKD BY: TJ	CHARGE: 3742	VO 160200	REVISION 5
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YC- 160302

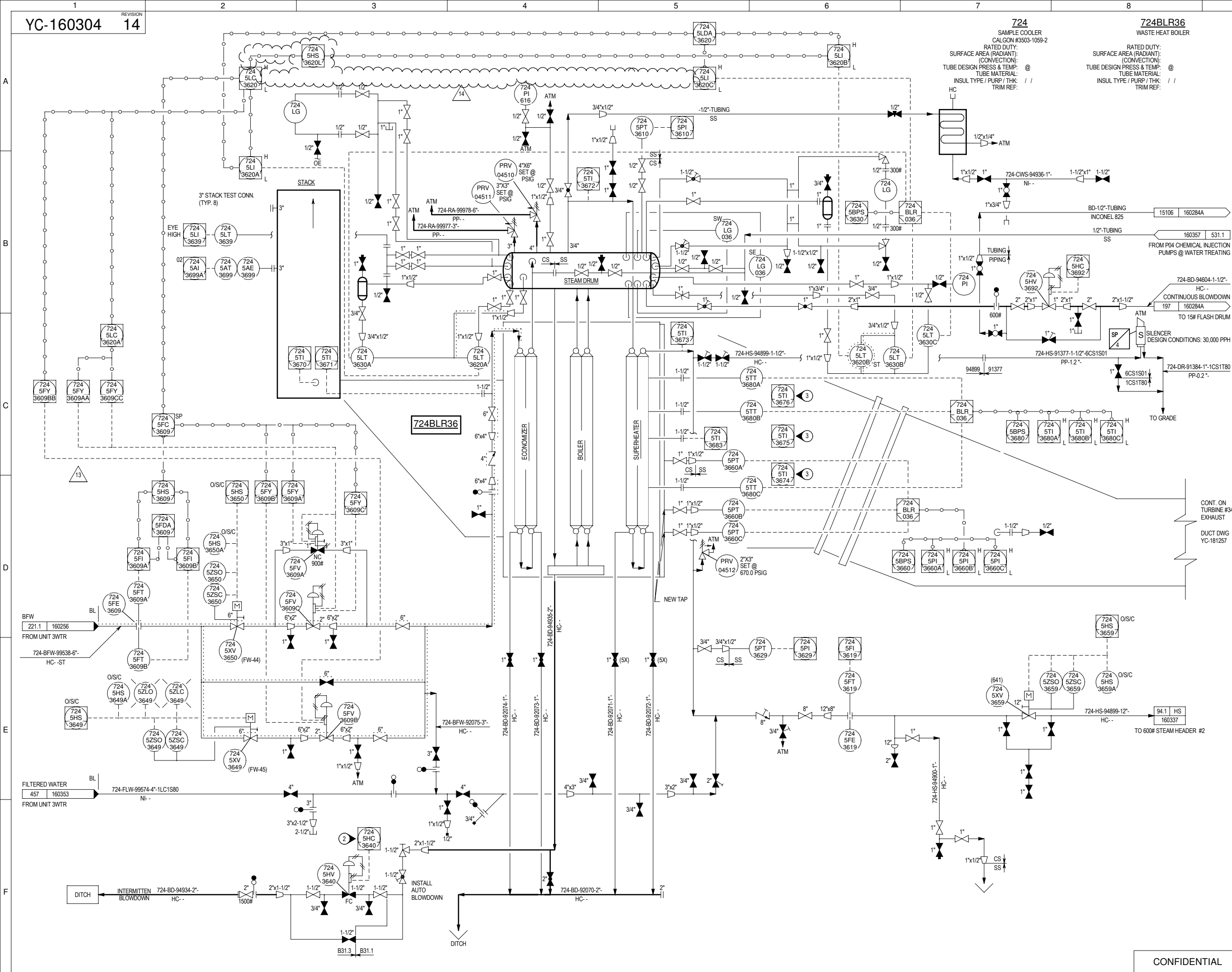
REVISION
5

SAFETY---A LIFETIME COMMITMENT

SPPID REV 0



CONFIDENTIAL



NOTES

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. LOGIC OVERRIDE AUTOMATICALLY OPENS VALVE ON HIGH LEVEL AND CLOSES VALVE ON LOW LEVEL.
3. TEMPERATURE ELEMENT LOCATED ON NORTH SIDE OF BOILER 36 DUCT.
4. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
5. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

8	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC-19-1361
9	09/18/18	MOC REVISION	JBB	M20143230
7S	08/06/14	SMARTPLANT CONVERSION		
14	05/09/25	MOC REVISION	JTF	MOC-25-71
13	11/18/24	MOC REVISION	JTF	MOC-24-1220
12	04/03/23	MOC REVISION	JMC	MOC-21-1171
11	06/10/21	MOC REVISION (PER USMF/1244)	JTF	MOC-21-756
10	05/11/21	MOC REVISION	JTF	MOC-21-631
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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MOTIVA
ENTERPRISES LLC

PORT ARTHUR
REFINERY

P3ST
724BLR36
BOILER FEED WATER SUPPLY AND 600# STM OUTLET
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: TP	05-01-1996	NUMBER ASSIGNED AND TRACING	DATE 05/01/96
DESIGN BY: TJ	05-01-1996	RECORD ENTERED	

CONFIDENTIAL

YC- 160304 14

- ## REFERENCE DRAWINGS

00401706.XLSX LINE LIST

NOTE

MOTIVA
ENTERPRISES LLC

PORT ARTHUR
REFINERY

P3ST

724BLR36

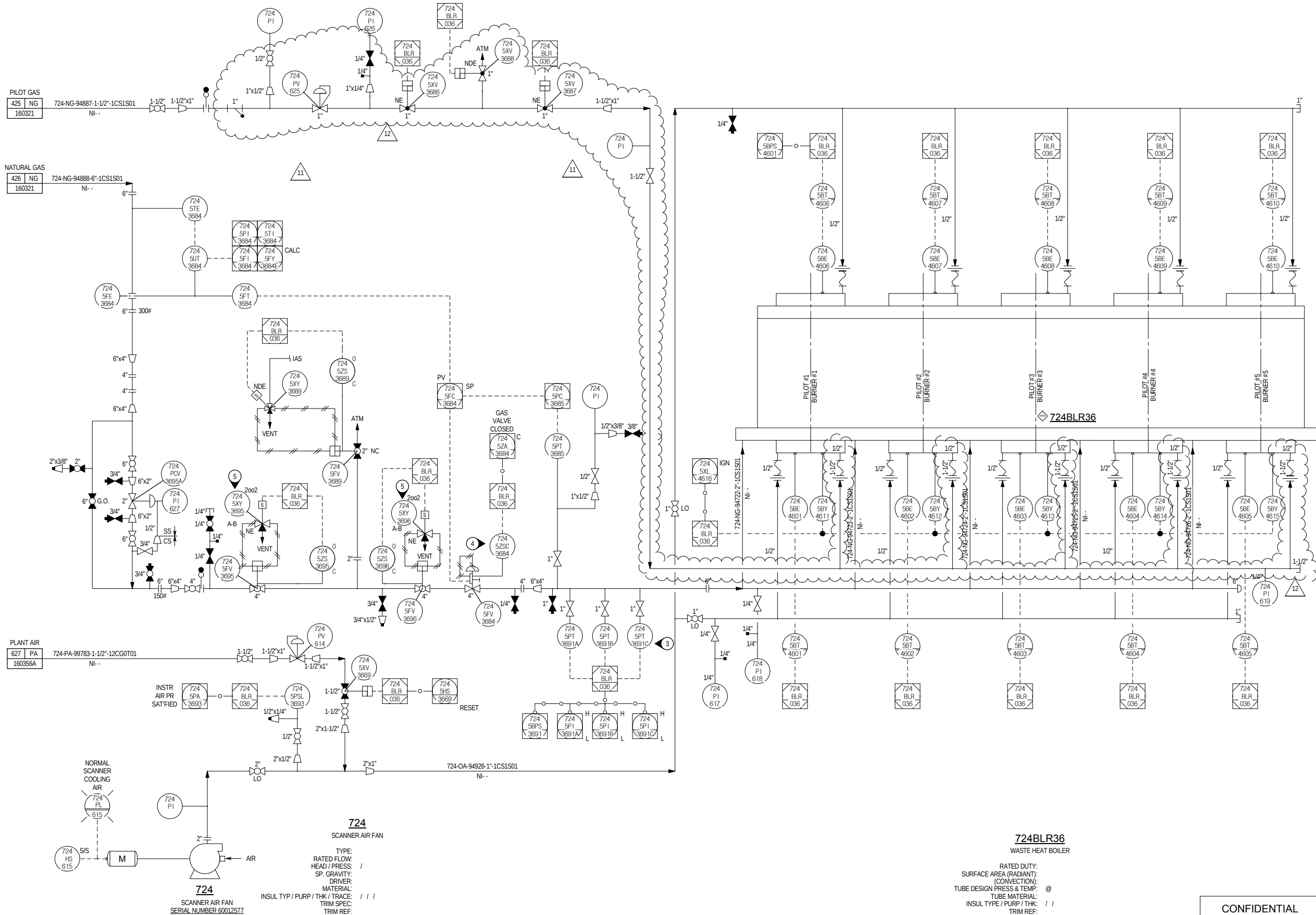
BURNER CONTROLS

PIPING & INSTRUMENTATION DIAGRAM

CONFIDENTIAL

YC- 160305

REV 0



- ## SAFETY--A LIFETIME COMMITMENT



SAFETY--A LIFETIME COMMITMENT

SAFETY--A LIFETIME COMMITMENT

SAFETY--A LIFETIME COMMITMENT

SAFETY--A LIFETIME COMMITMENT

SAFETY--A LIFETIME COMMITMENT

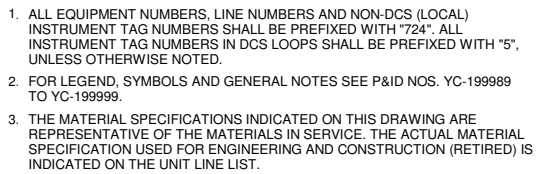
SAFETY--A LIFETIME COMMITMENT

SAFETY--A LIFETIME COMMITMENT

SAFETY--A LIFETIME COMMITMENT

SAFETY--A LIFETIME COMMITMENT

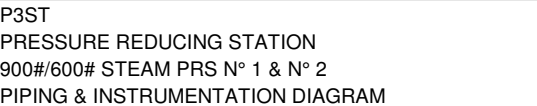
SAFETY--A LIFETIME COMMITMENT



00401706.XLSX LINE LIST

9	10/05/23	MOC REVISION	JMC	MOC 23-1
8	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1
75	08/04/14	SMARTPLANT CONVERSION		
6	03/14/08	AS-BUILT	ARS	4372
5	10/11/07	GENERAL REVISION	ARS	4372
4	05/09/07	ISSUE FOR GENERAL MOV REV	ARS	4372
3	04/20/07	LOW-NOX PROJECT	ARS	4517
2	10/05/06	ISSUE FOR BOILER 32-33 INSTR IFC	ARS	4372
REV	DATE	DESCRIPTION	BY	CHAR#

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CONFIDENTIAL

CHKD BY: TJ	CHARGE: 3742	VC 160207
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- 724

00401706.XLSX LINE LIST

NOTE

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ENTERPRISES LLC

PORT ARTHUR
REFINERY

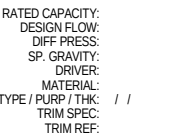
PLANT AIR

PIPING & INSTRUMENTATION DIAGRAM

YC- 160312

REV C





NOTES

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5". UNLESS OTHERWISE NOTED.
2.  REPRESENTS COMPRESSOR 724CP6405 CONTROL PANEL. THE CONTROL PANEL IS A VANTAGE CONTROL SYSTEM THAT CONTROLS, MONITORS AND PROTECTS 724CP6405 OPERATIONS.
3. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
4. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

6	10/12/21	MOC REVISION	JMC	21-1265
5	07/09/21	MOC REVISION	RJB	21-888
4	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
3	10/20/16	MOC REVISION	JBB	M20162669
2S	05/08/14	SMARTPLANT CONVERSION		
1	12/02/05	GENERAL TAGGING REVISION	ARS	4271
0	11/11/05	ISSUED FOR CONSTRUCTION	ARS	4271
				4172
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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MOTIVA
ENTERPRISES LLC

PORT ARTHUR
REFINERY

PLANT AIR PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE
DRAWN BY: GAK	6/9/2005	NUMBER ASSIGNED AND TRACING RECORD ENTERED
DESIGN BY: SM	6/9/2005	DATE 08-15-05

YC- 160313

6

- SPPID REV 0



REFERENCE DRAWINGS

00401706.XLSX LINE LIST

NOTE

MOTIVA
ENTERPRISES LLC

PORT ARTHUR
REFINERY

	DATES	SCALE: NONE
DRAWN BY:		NUMBER ASSIGNED DATE
DESIGN BY:		AND TRACING RECORD ENTERED

CONFIDENTIAL

YC- 160315B 1

SAFETY--A LIFETIME COMMITMENT

SPPID REV 0



SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

00401706.XLSX LINE LIST

NOTE

MOTIVA
ENTERPRISES LLC

PORT ARTHUR
REFINERY

	DATES	SCALE: NONE
DRAWN BY:		NUMBER ASSIGNED DATE
DESIGN BY:		AND TRACING RECORD ENTERED

YC- 160315D 0

SPPID REV 0



SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SPPID REV 0



- 00401706.XLSX LINE LIST

MOTIVA
ENTERPRISES LLC

PORT ARTHUR
REFINERY

YC- 160315E 0



1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

0	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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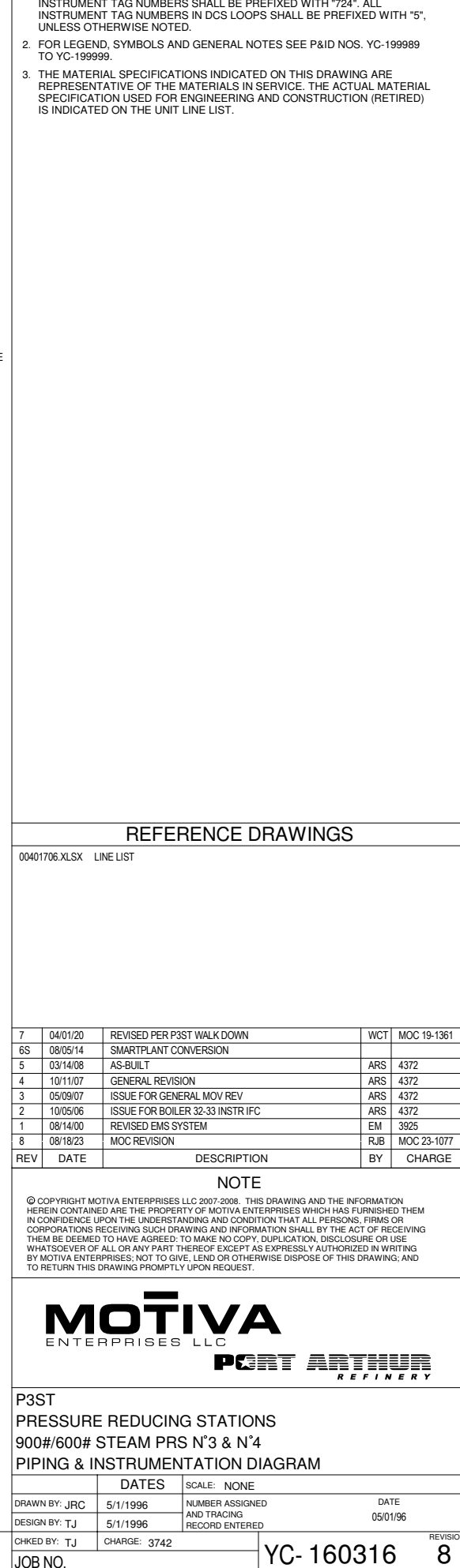


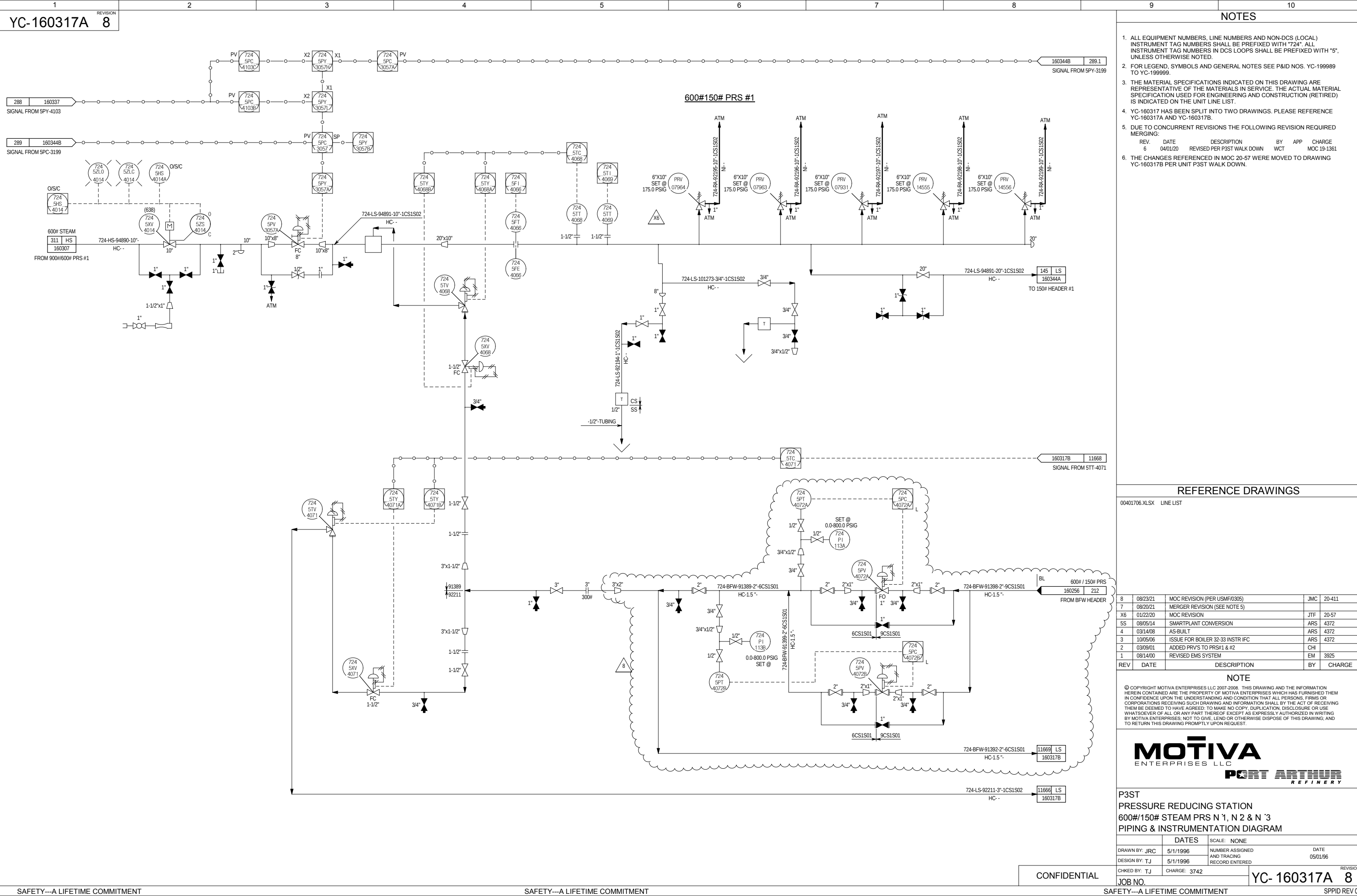
P3ST
INSTRUMENT AIR
DRYERS AND RECEIVING DRUM
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY:		NUMBER ASSIGNED	DATE
DESIGN BY:		AND TRACING	
CHECKED BY:	CHARGE:	RECORD ENTERED	
JOB NO.		YC- 160315F	0

CONFIDENTIAL

YC- 160315F 0





1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

8	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
7S	07/29/14	SMARTPLANT CONVERSION		
6	03/14/08	AS-BUILT	ARS	4372
5	10/08/07	GEN REVISION	ARS	4372
4	10/06/06	ISSUE FOR BOILER 36 INSTR. IFC	ARS	4372
3	07/26/04	REVISED PER PS3 HRSG NO. 37 PROJECT	JM	4164
2	08/14/00	REVISED EMS SYSTEM	EN	3925
1	10/18/96	GENERAL REVISION	RLS	3742
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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P3ST
190# NATURAL GAS
DISTRIBUTION HEADERS
PIPING & INSTRUMENTATION DIAGRAM

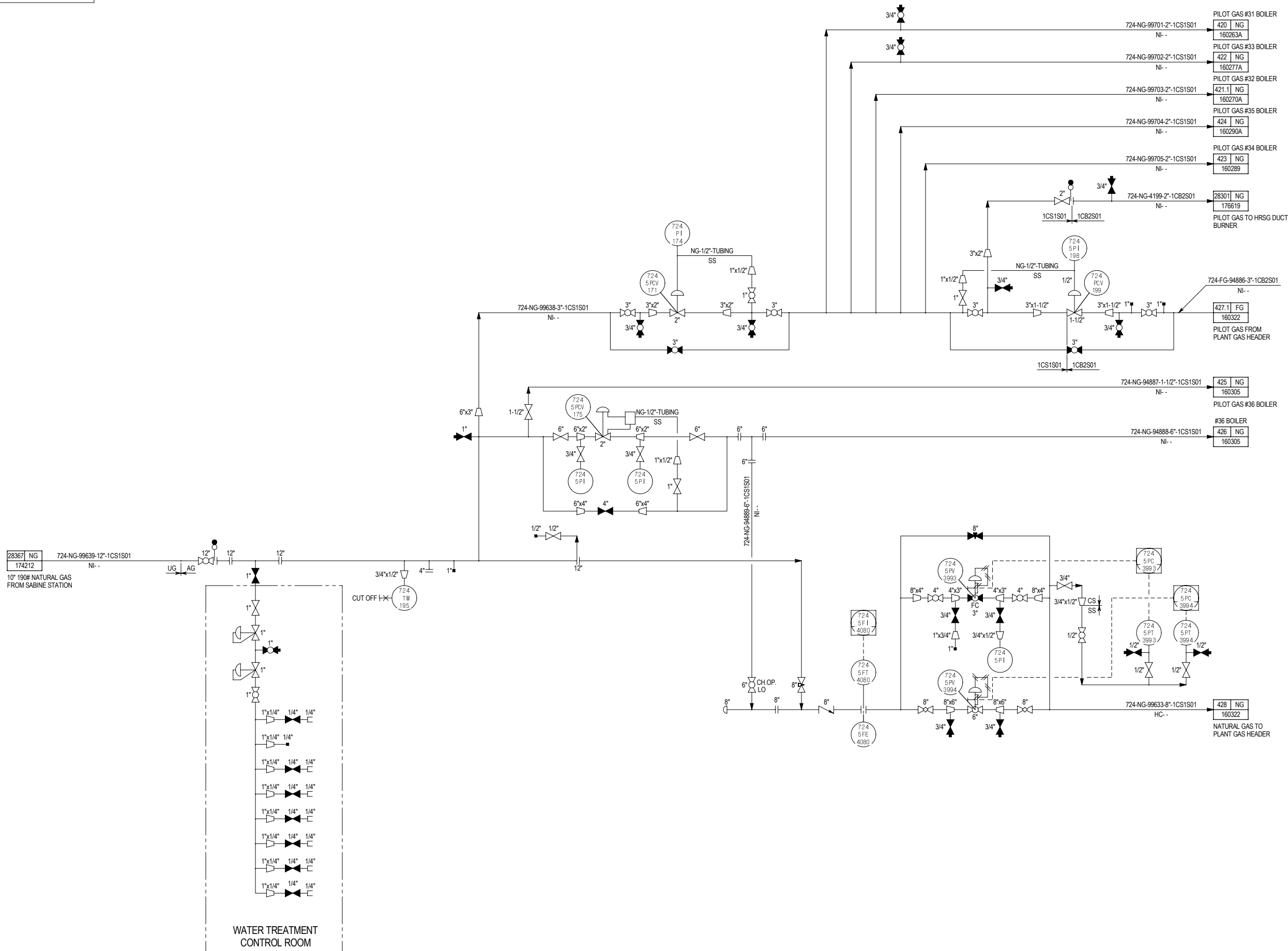
	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED	DATE
DESIGN BY: TJ	5/1/1996	AND TRACING	5-1-96
CHKD BY: TJ	CHARGE: 3742	RECORD ENTERED	
JOB NO.		YC- 160321	REVISION 8

CONFIDENTIAL

YC- 160321

REVISION

SPPID REV 0



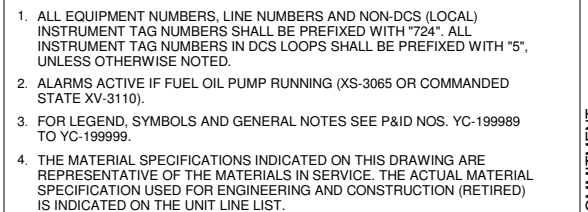
SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SPPID REV 0

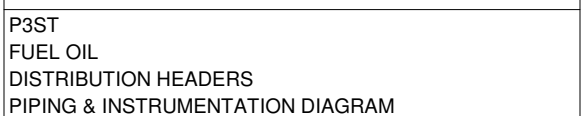
SAFETY---A LIFETIME COMMITMENT



00401706.XLSX LINE LIST

NOTE

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CONFIDENTIAL	CHKD BY: TJ	CHARGE:	YC- 160335	REVISION
	JOB NO.			7

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. CONTROL SIGNAL TO BE HIGHEST OF MID TWO.
3. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
4. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

8	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
7S	08/04/14	SMARTPLAN CONVERSION		
6	03/14/08	AS BUILT	ARS	4372
5	10/11/07	GENERAL REVISION	ARS	4372
4	05/09/07	ISSUE FOR GENERAL MOV REV	ARS	4372
3	10/05/06	ISSUE FOR BOILER 32.33 INSTR IFC	ARS	4372
2	07/26/04	REVISED PER PS3 HRSG NO. 37 PROJECT	JN	4164
1	08/14/00	REVISED EMS SYSTEM	EM	3925
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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P3ST
STEAM HEADERS
900# STEAM HEADER #1 & #2
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JC	3/7/1996	NUMBER ASSIGNED	DATE
DESIGN BY: TJ	5/1/1996	AND TRACING	05/01/96
CHECKED BY: TJ	CHARGE:	RECORD ENTERED	REVISION
JOB NO.		YC- 160336	8

YC- 160337 7



- ## SAFETY--A LIFETIME COMMITMENT

00401706.XLSX LINE LIST

00401706.XLSX LINE LIST

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P3ST
STEAM HEADERS
150# STEAM NORTH LOOP HEADERS NO.1 & NO.2
PIPING & INSTRUMENTATION DIAGRAM

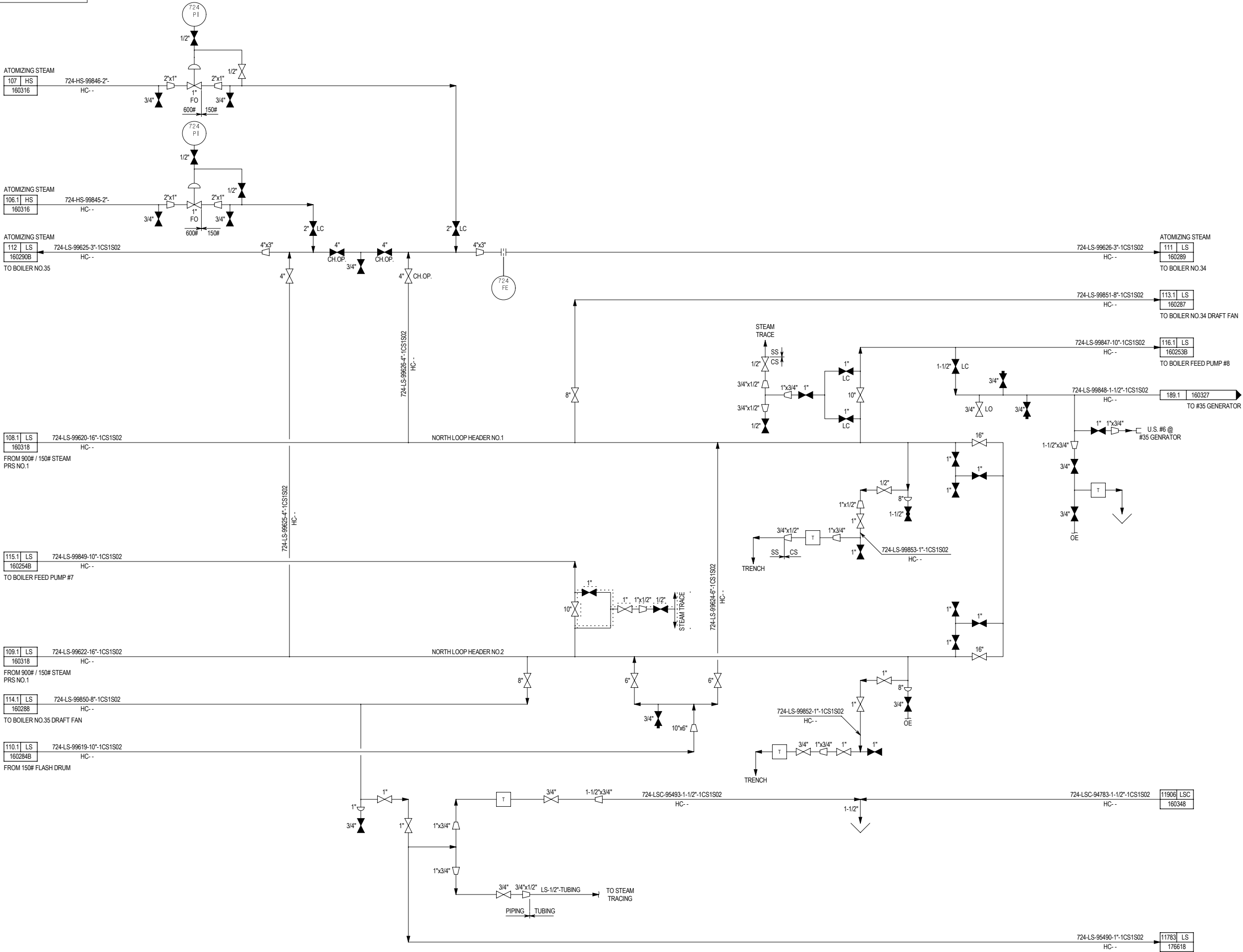
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DRAWN BY: JRC	4/1/1996	NUMBER ASSIGNED AND TRACING	DATE
DESIGN BY: TJ	5/1/1996	RECORD ENTERED	5-1-96

YC- 160339

CONFIDENTIAL

SAFETY---A LIFETIME COMMITMENT

SPPID REV 0

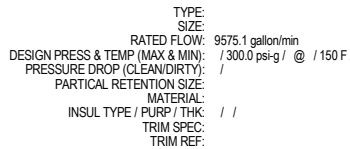


SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SPPID REV 0



00188623.XLS LINE LIST

NOTE

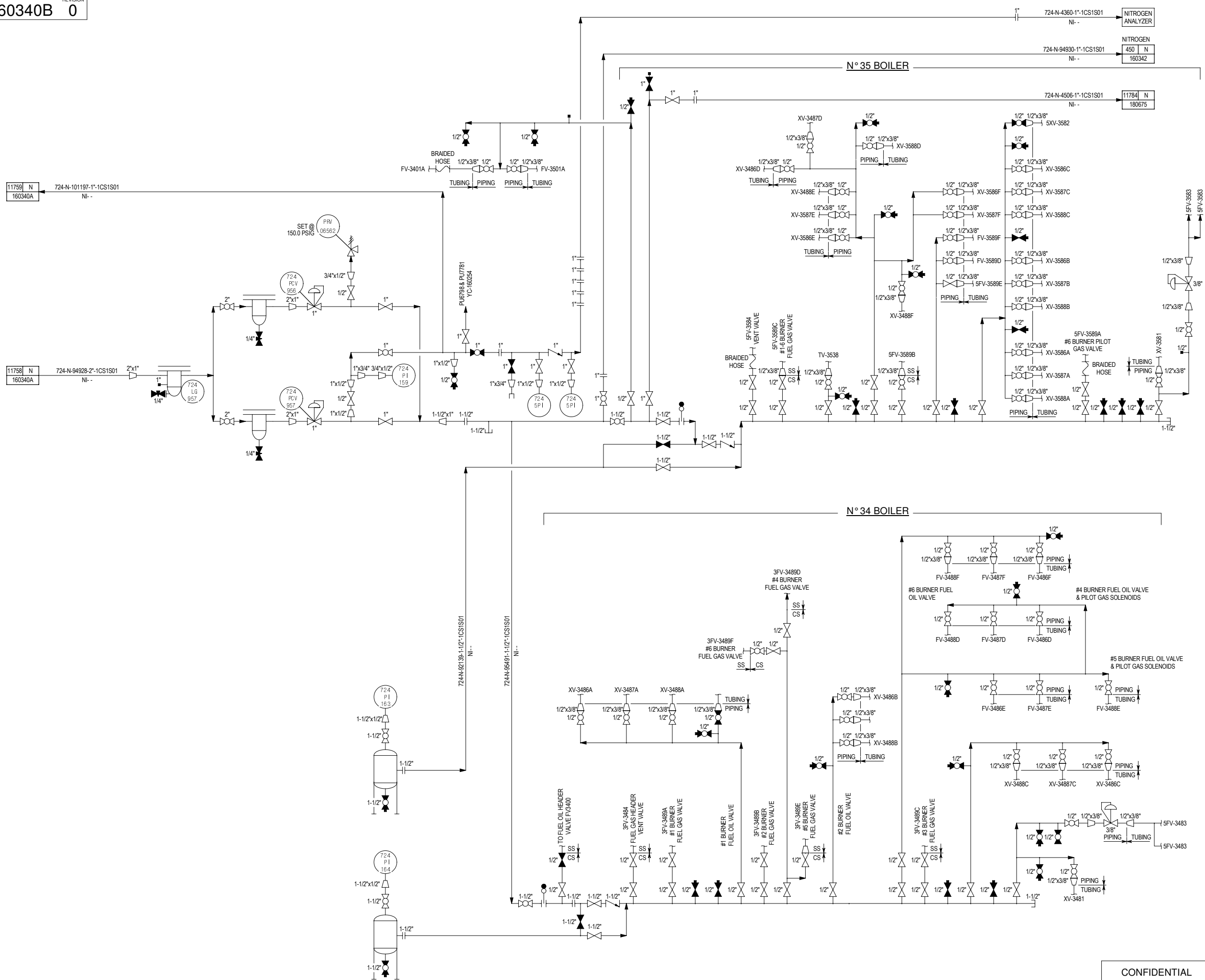
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MOTIVA
ENTERPRISES LLC

PORT ARTHUR
REFINERY

	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED	DATE
DESIGN BY: TJ	5/1/1996	AND TRACING	05/01/96
CHKD BY: TJ	CHARGE: 3474	RECORD ENTERED	
JOB NO.		YC- 160340	REVISION V9

YC- 160340 V9



NOTES

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.
4. YC-160340 HAS BEEN SPLIT INTO TWO DRAWINGS. PLEASE REFERENCE YC-160340A AND YC-160340B.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

0	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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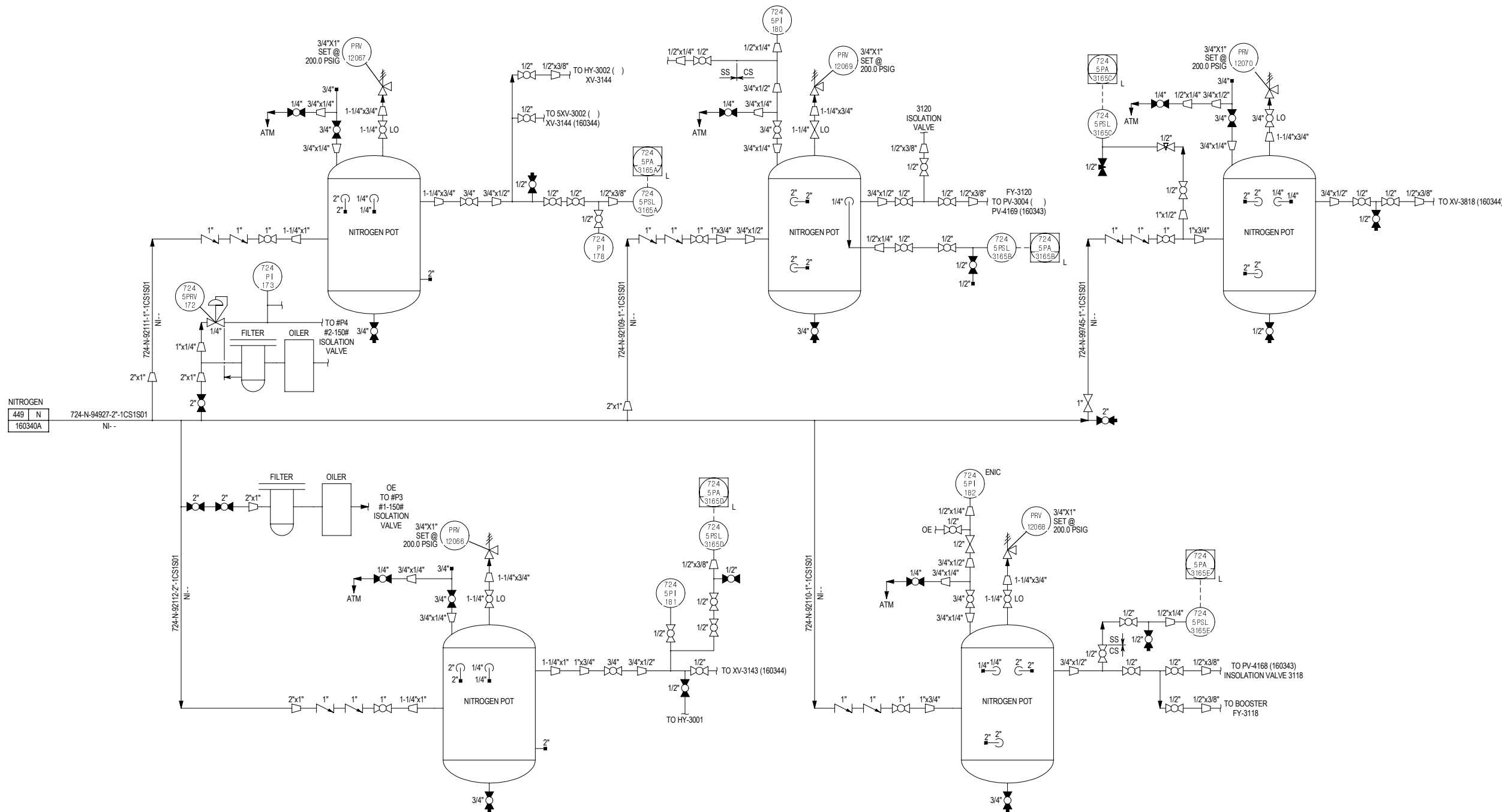
P3ST
NITROGEN
DISTRIBUTION HEADERS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE
DRAWN BY:		NUMBER ASSIGNED DATE
DESIGN BY:		AND TRACING RECORD ENTERED

YC- 160340B 0

CONFIDENTIAL

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.



REFERENCE DRAWINGS

00401706.XLSX LINE LIST

5	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
4S	08/07/14	SMARTPLANT CONVERSION		
3	03/14/08	AS-BUILT	ARS	4372
2	10/06/06	ISSUE FOR BOILER 36 INSTR. IFC	ARS	4372
1	12/01/04	MVH PS MODERNIZATION	ARS	4372
				3742
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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MOTIVA
ENTERPRISES LLC

PORT ARTHUR
REFINERY

P3ST
NITROGEN
SUPPLY TO ISOLATION VALVES
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRS	5/1/1996	NUMBER ASSIGNED AND TRACING	DATE 05/01/96
DESIGN BY: TJ	5/1/1996	RECORD ENTERED	

YC- 160341 5

CONFIDENTIAL

SAFETY---A LIFETIME COMMITMENT

SPPID REV 0

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

6	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
5	11/15/18	MOC REVISION	RJB	M2018823
4S	08/04/14	SMARTPLANT CONVERSION		
3	03/14/08	AS-BUILT	ARS	4372
2	04/15/07	GENERAL REVISION	ARS	4372
1	12/01/04	MVH PS MODERNIZATION	ARS	4372
				3742
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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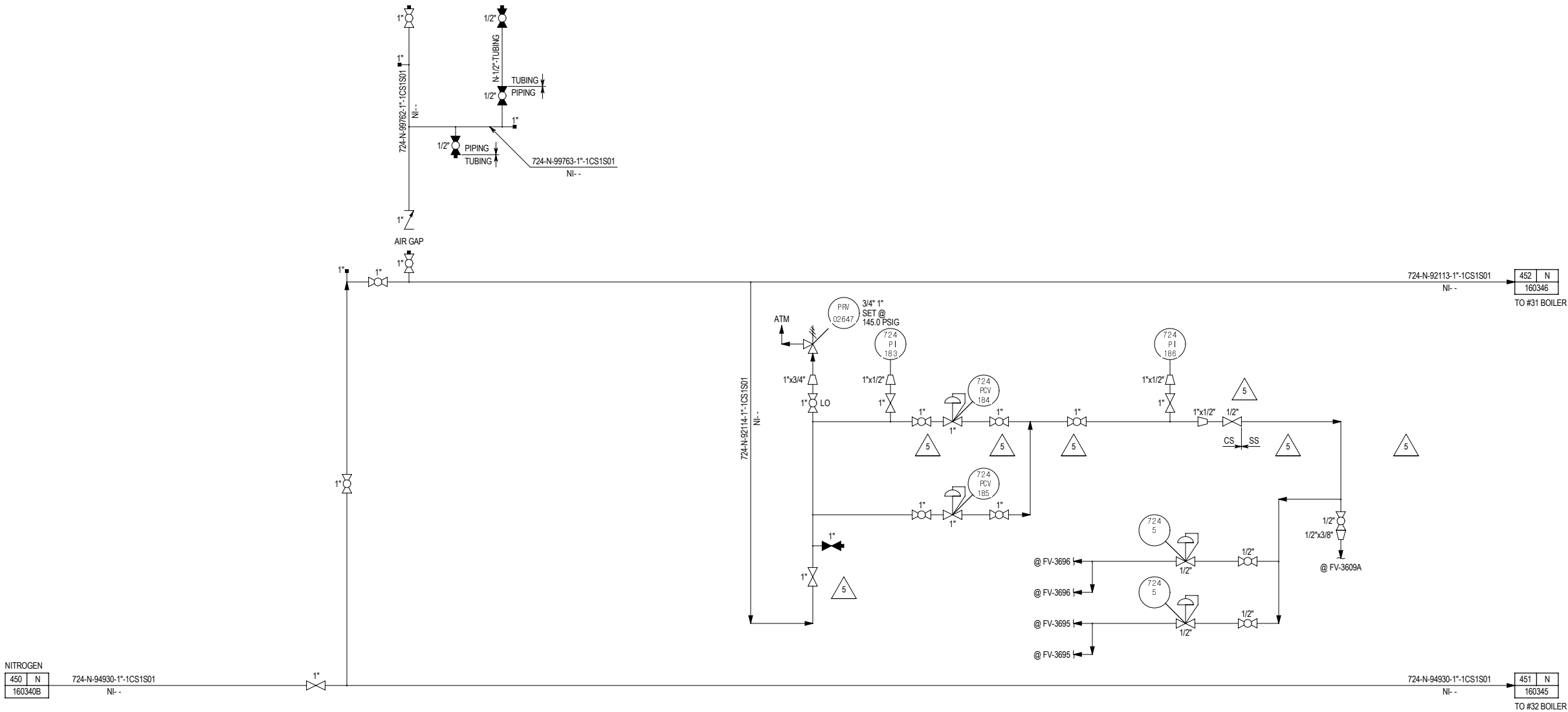
P3ST
NITROGEN
#33 BOILER
PIPING & INSTRUMENTATION DIAGRAM

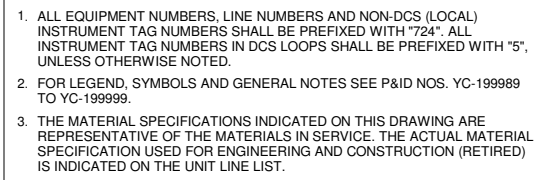
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DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED AND TRACING RECORD ENTERED
DESIGN BY: TJ	5/1/1996	DATE 05/01/96

CONFIDENTIAL

YC- 160342

SAFETY---A LIFETIME COMMITMENT

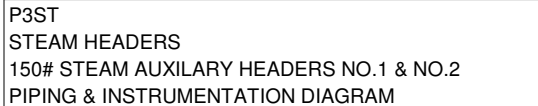




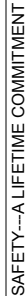
00401706.XLSX LINE LIST

NOTE

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	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED AND TRACING	DATE 05/01/96
DESIGN BY: TJ	5/1/1996	RECORD ENTERED	
CHKD BY: TJ	CHARGE: 3742		
JOB NO.		YC- 160343	



- ## SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT



SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT



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YC-160346

REVISION
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O.O.S.

FUEL VALVES

O.O.S.

FUEL VALVES

O.O.S.

FUEL VALVES

NOTES

1. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.

2. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.

3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

3	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
2S	08/05/14	SMARTPLANT CONVERSION		
1	12/01/04	MVH PS MODERNIZATION	ARS	4372
				3742
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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MOTIVA

ENTERPRISES LLC

PORT ARTHUR

REFINERY

P3ST

NITROGEN

#31 BOILER

PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED AND TRACING
DESIGN BY: TJ	5/1/1996	DATE 05/01/96
CHKD BY: TJ	CHARGE: 3742	RECORD ENTERED

CONFIDENTIAL

JOB NO.

YC- 160346

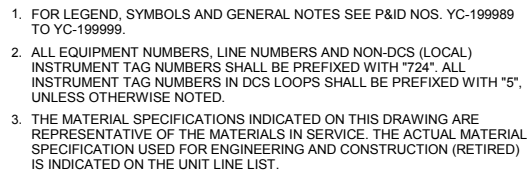
REVISION
3

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SPPID REV 0



00401706.XLSX LINE LIST

NOTE

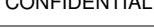
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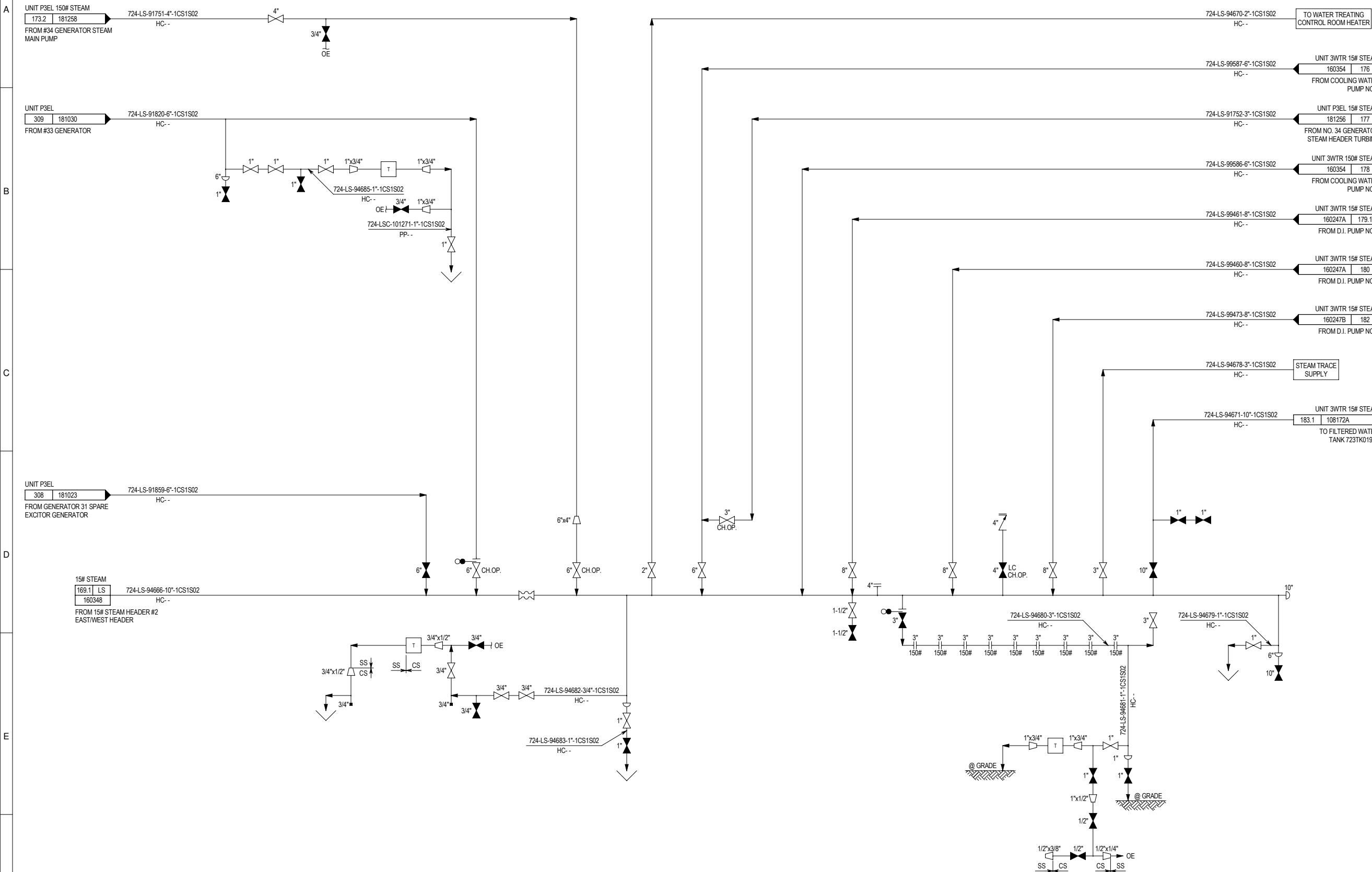


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DESIGN BY: TJ	5/1/1996	AND TRACING	
CHKD BY: TJ	CHARGE:	RECORD ENTERED	
JOB NO.		YC- 160347	REVISION 4

YC- 160347

- | | | | |
|---------------|----------|--------------------------------|------------------|
| | DATES | SCALE: NONE | |
| DRAWN BY: JRC | 5/1/1996 | NUMBER ASSIGNED
AND TRACING | DATE
05/01/96 |
| DESIGN BY: TJ | 5/1/1996 | RECORD ENTERED | |
| CHK'D BY: TJ | CHARGE: | YC- 160348 | REVISION |
| JOB NO. | | | 5 |





NOTES

1. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
2. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

3	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
2S	08/06/14	SMARTPLANT CONVERSION		
1	01/14/05	ISSUED AS-BUILT	ARS	4372
				3742
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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P3ST
STEAM HEADERS
15# EAST/WEST STEAM HEADER
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NTS	
DRAWN BY: JRC	3/1/1996	NUMBER ASSIGNED	DATE 05/01/96
DESIGN BY: TJ	3/1/1996	AND TRACING RECORD ENTERED	
CHKD BY: TJ	CHARGE: 3742	YC- 160349	REVISION
JOB NO.			3

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

4	11/30/21	MOC REVISION (PER USMF01769)	JMC	19-655
3	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
2S	08/05/14	SMARTPLANT CONVERSION		
1	01/14/05	ISSUED AS-BUILT	ARS	4372
				3742
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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P3ST
STEAM HEADERS
15# STEAM HEADER NO.3
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	3/1/1996	NUMBER ASSIGNED AND TRACING	DATE
DESIGN BY: TJ	5/1/1996	RECORD ENTERED	05/01/96

CHKD BY: TJ	CHARGE:	YC- 160350	REVISION
JOB NO.			4

CONFIDENTIAL

SAFETY--A LIFETIME COMMITMENT

SPPID REV 0

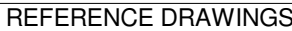
SAFETY--A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY--A LIFETIME COMMITMENT

SPPID REV 0

SAFETY---A LIFETIME COMMITMENT



4	10/07/16	MOC REVISION	JMC	M2015246
3S	08/05/14	SMPLANT PUMP CONVERSION		
3	04/04/08	CHANGED TO 3/4" 1/4-TURN VALVE	ARS	4372
2	05/10/07	AS-BUILT	ARS	4372
1	07/20/06	ISSUE FOR WTP3 INSTRUMENTATION	ARS	4372
				3742
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

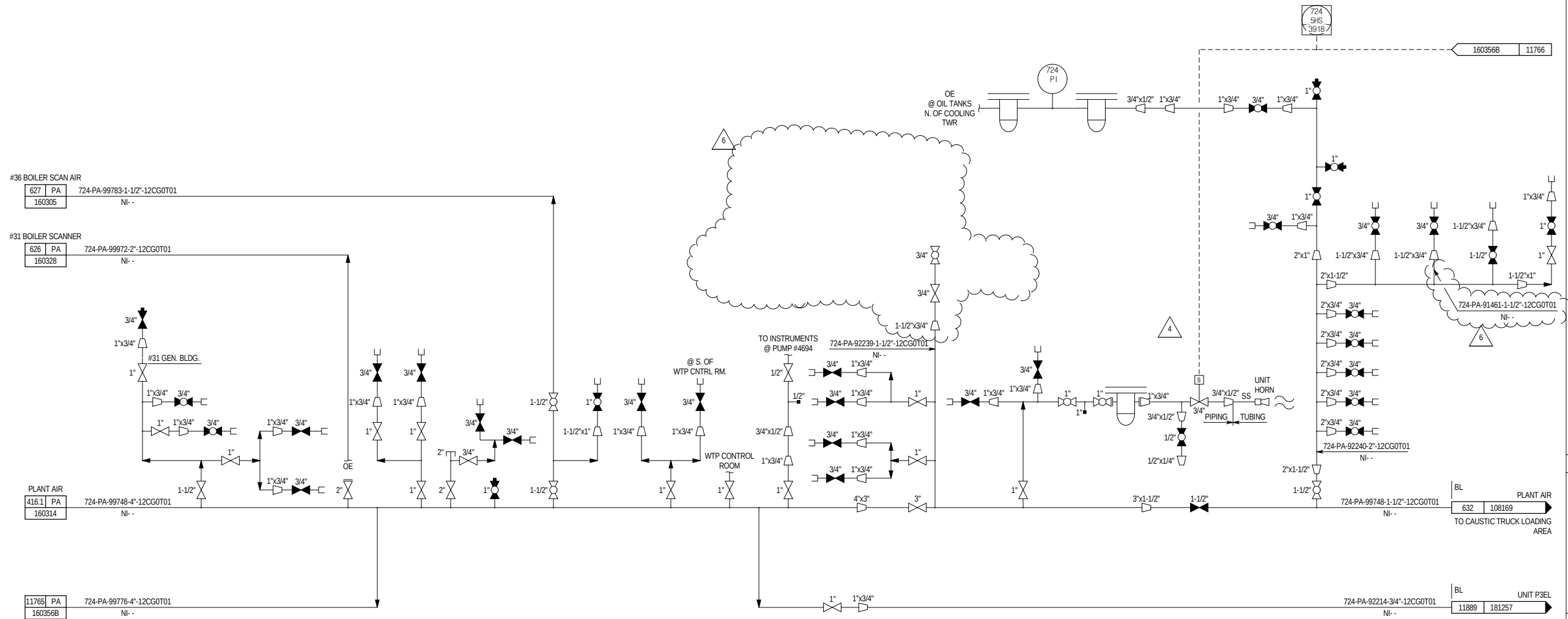
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P3ST
PLANT AIR
DISTRIBUTION HEADER
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED AND TRACING	DATE 05/01/96
DESIGN BY: TJ	5/1/1996	RECORD ENTERED	
CHECKED BY: TJ	CHARGE: 3742	YC- 160356	REVISION
JOB NO.			4

1. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
2. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.



REFERENCE DRAWINGS

00401706.XLSX LINE LIST

LINE LIST

6	11/30/21	MOC REVISION (PER USMF01769)	JMC	19-655
5	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
4	1/00/16	MOC REVISION	JMC	M2015246
3S	08/05/14	SMARTPLANT CONVERSION		
3	04/04/08	CHANGED TO 3/4" 14-TURN VALVE	ARS	4372
2	05/10/07	AS-BUILT	ARS	4372
1	07/20/06	ISSUE FOR WTP3 INSTRUMENTATION	ARS	4372
				3742
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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P3ST
PLANT AIR
DISTRIBUTION HEADER
PIPING & INSTRUMENTATION DIAGRAM

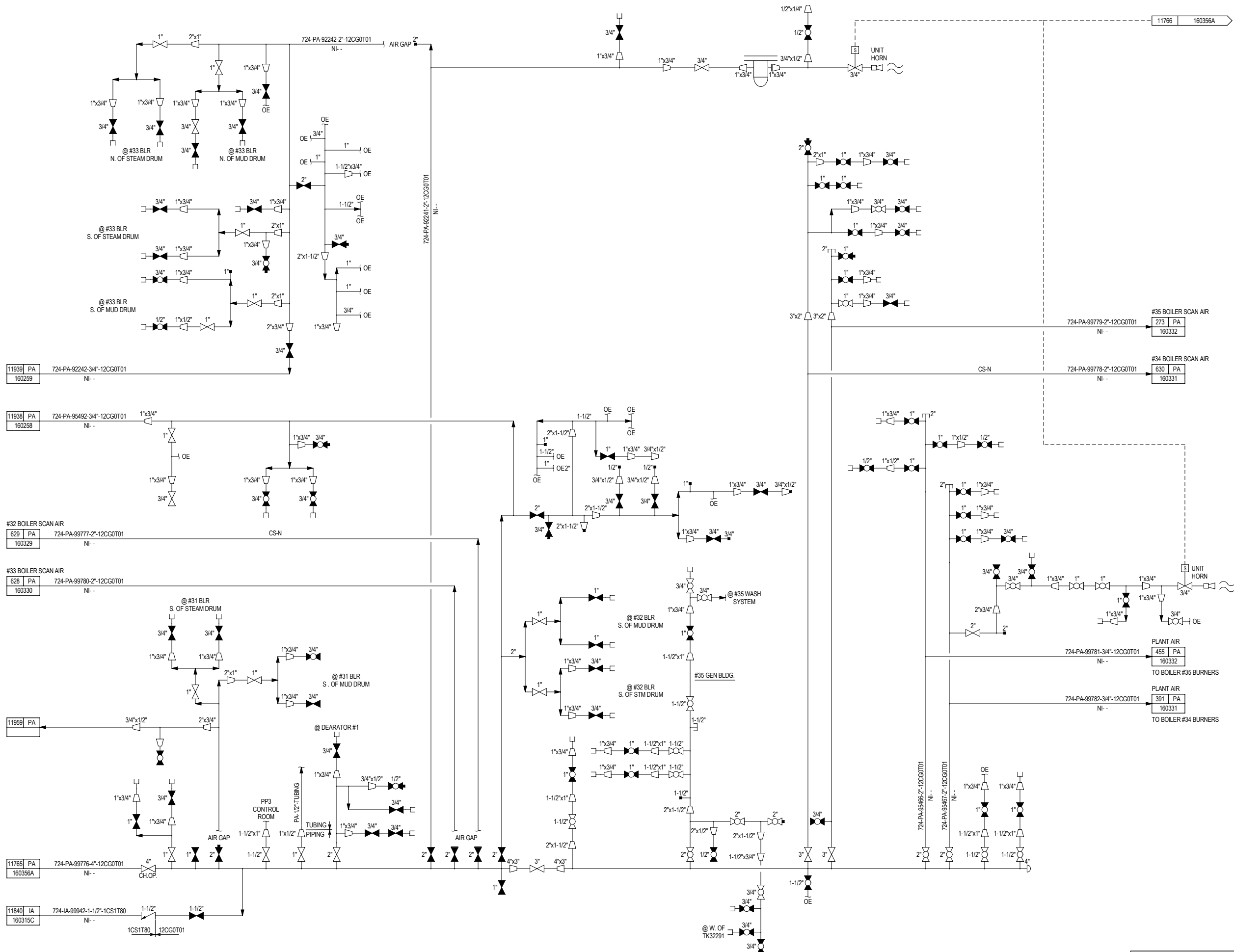
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DRAWN BY: JRC	5/1/1996	NUMBER ASSIGNED AND TRACING	DATE 05/01/96
DESIGN BY: TJ	5/1/1996	RECORD ENTERED	

CONFIDENTIAL

JOB NO.

YC- 160356A 6

1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
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REFERENCE DRAWINGS

00401706.XLSX LINE LIST

0	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
REV	DATE	DESCRIPTION	BY	CHARGE

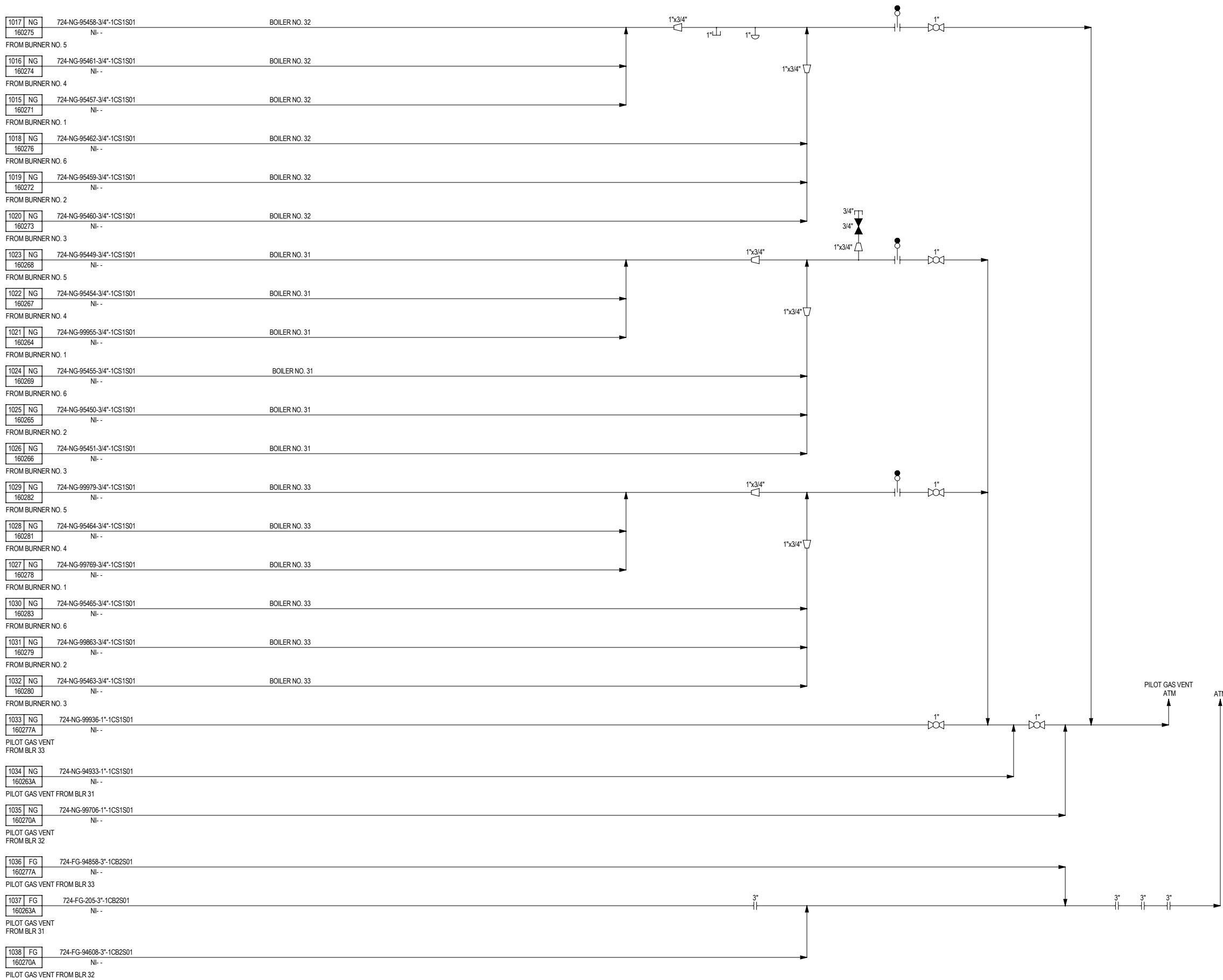
NOTE

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P3ST
PLANT AIR
DISTRIBUTION HEADER
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY:		NUMBER ASSIGNED	DATE
DESIGN BY:		AND TRACING	
CHKD BY:	CHARGE:	RECORD ENTERED	
JOB NO.		YC- 160356B	REVISION 0



NOTES

1. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
2. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
3. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

2	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
1S	08/05/14	SMARTPLANT CONVERSION		
A	12/27/00	3 PS BOILER FRONT VENT PIPING	HWP	
		MOC REVIEW		4010
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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P3ST
724BLR31, BLR32, BLR33, BLR34, BLR35
BURNER CONTROL VENTS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE
DRAWN BY: HWP	11/6/2000	NUMBER ASSIGNED AND TRACING
DESIGN BY: HWP	11/6/2000	DATE

CONFIDENTIAL

JOB NO. _____
SAFETY---A LIFETIME COMMITMENT

YC- 169147 REVISION 2

SAFETY...A LIFETIME COMMITMENT



1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. USE BALL VALVES IN LIEU OF GATE VALVES.
3. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
4. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.
5. PRV-09271, 09272 SHOULD ALTERNATE LO AND LC ON 4" GATE VALVE.
6. PRV-15012, 15013 SHOULD ALTERNATE LO AND LC ON 6" BALL VALVE.

00401706.XLSX LINE LIST

NOTE

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	DATES	SCALE: NONE	
DRAWN BY: KJH	10/14/2002	NUMBER ASSIGNED	DATE
DESIGN BY: KJH	10/14/2002	AND TRACING RECORD ENTERED	
CHECKED BY:	CHARGE: 419322	YC- 174212	REVISION
JOB NO.			10

- ## REFERENCE DRAWINGS

00401706.XLSX LINE LIST

5	02/08/22	MOC REVISION	JTF	21-1286
4	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
3	04/02/18	MOC REVISION	JBB	M2018302
2S	08/08/14	SMARTPLANT CONVERSION		
1	09/22/06	AS-BUILT - CONVERTED TO CAD	ARS	4372 4164
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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MOTIVA
ENTERPRISES LLC

PORT ARTHUR
REFINERY

P3ST
724BLR37, 724BRN002
EXHAUST GAS SYSTEM
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: MGB	7/30/2003	NUMBER ASSIGNED	DATE
DESIGN BY:		AND TRACING	
CHKD BY:	CHARGE:	RECORD ENTERED	
JOB NO.		YC- 176618	REVISION 5



SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SPPID REV 0

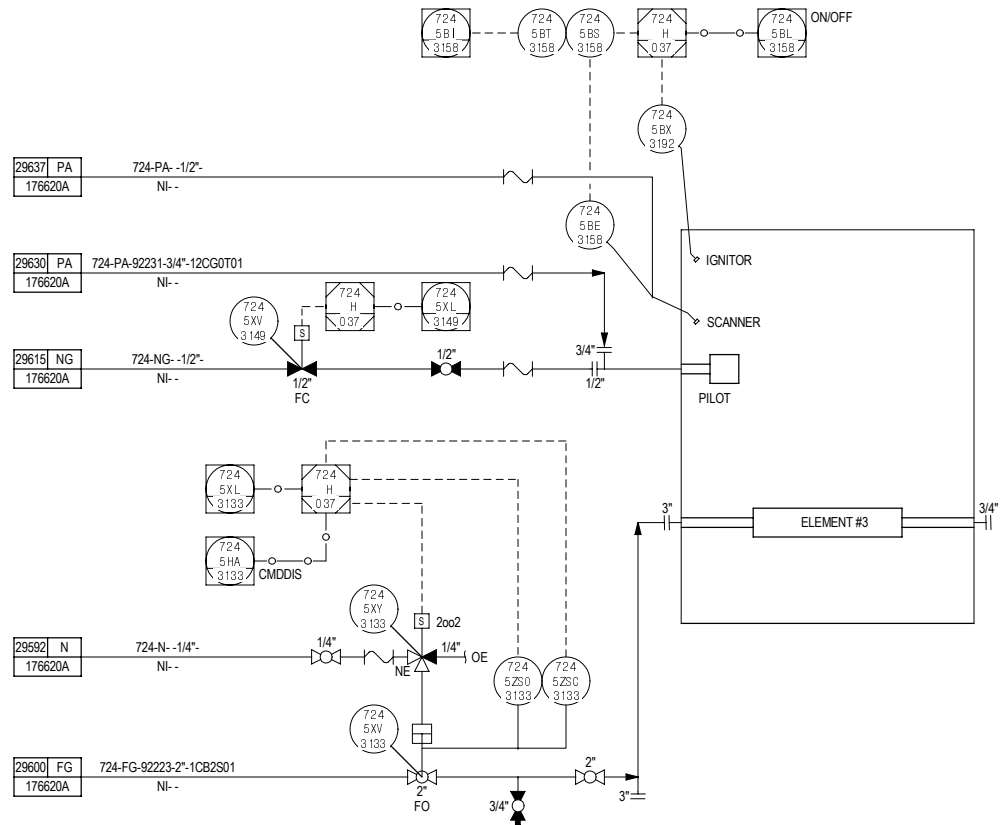
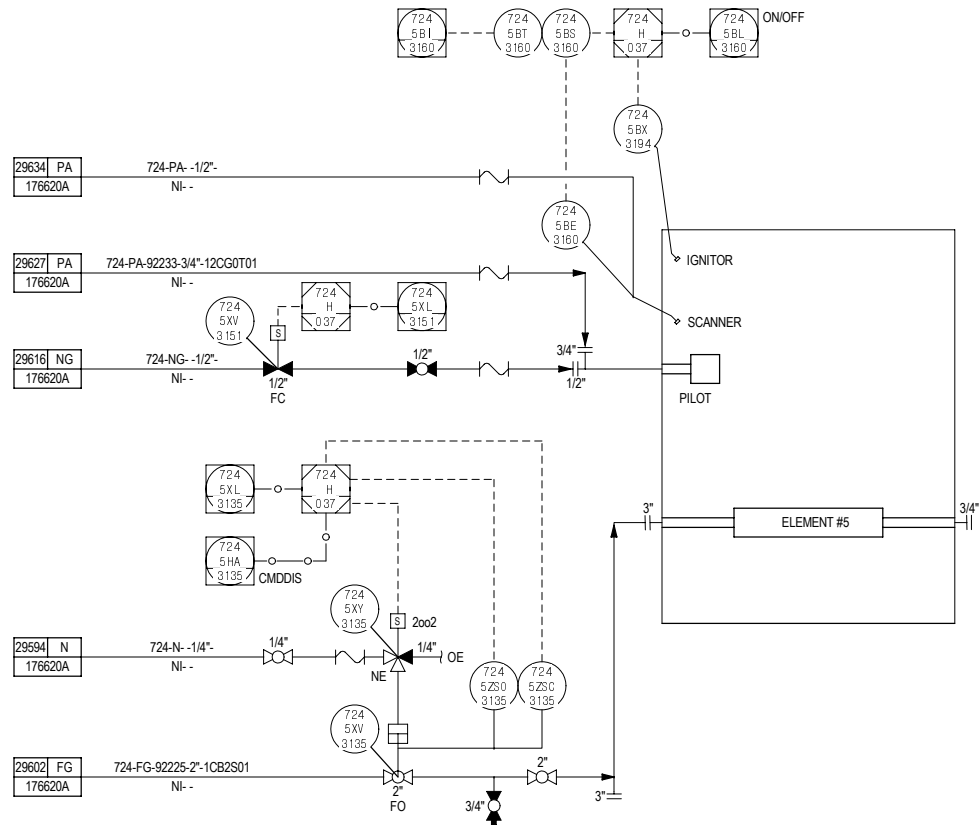
1. SEE DWG. YC-143764, YC-143765, YC-143766 & YC-143767 FOR GENERAL NOTES AND SYMBOLS.
2. SIGHT PORTS (3).
3. DELETED.
4. DELETED.
5. VENT TO SAFE LOCATION. DO NOT MANIFOLD.
6. SEE DWG. YC-176624 FOR SOLENOID VALVE DETAILS. THESE VALVES ARE 2002 DETT (DE-ENERGIZE TO TRIP).
7. RO-129 TO BE MULTI-PLATE (4) DESIGN.
8. THE PILOT SYSTEM VALVES ARE DETT AND ARE TRIPPED ONCE THE MAIN BURNERS ARE LIT.
9. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199999 TO YC-1999999.
10. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
11. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REV	DATE	DESCRIPTION	BY	CHARGE
3	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
2S	08/18/14	SMARTPLANT CONVERSION		
1	09/22/06	AS-BUILT - CONVERTED TO CAD	ARS	4372 4164

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	DATES	SCALE: NONE	
DRAWN BY: RMG	10/23/2003	NUMBER ASSIGNED AND TRACING RECORD ENTERED	DATE
DESIGN BY:			
CHKD BY:	CHARGE: 4164		REVISION
JOB NO.		YC- 176620A	3



- ## SAFETY---A LIFETIME COMMITMENT

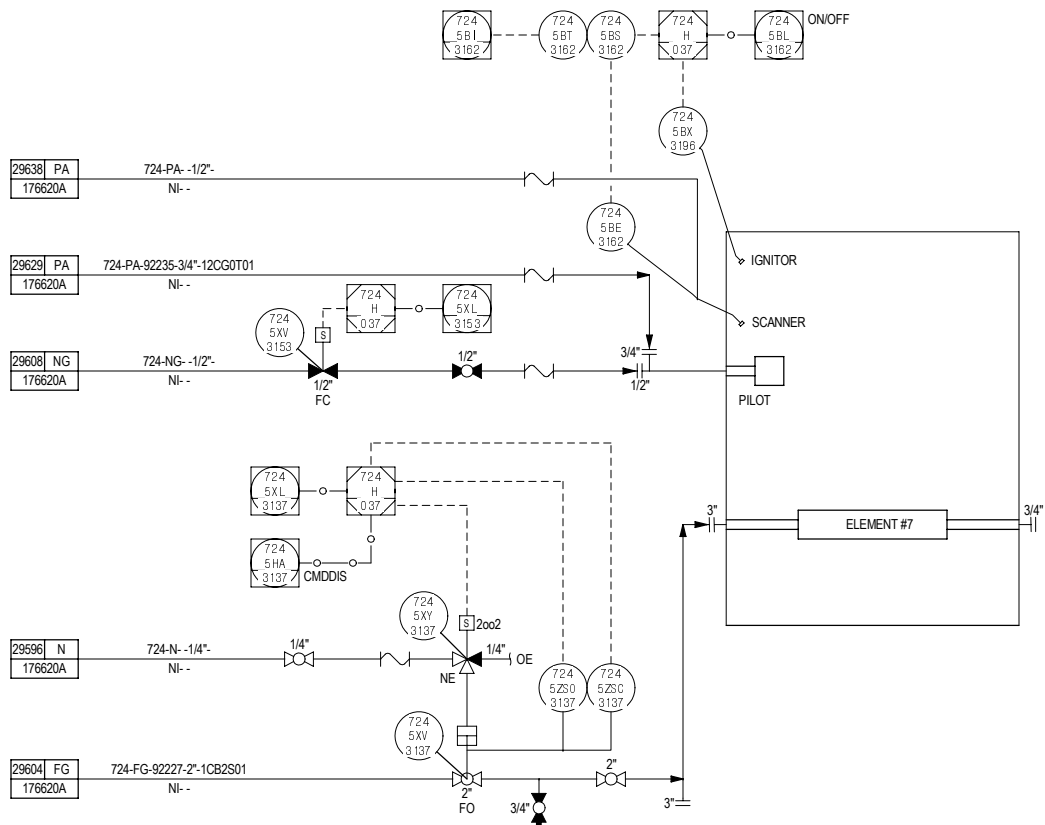
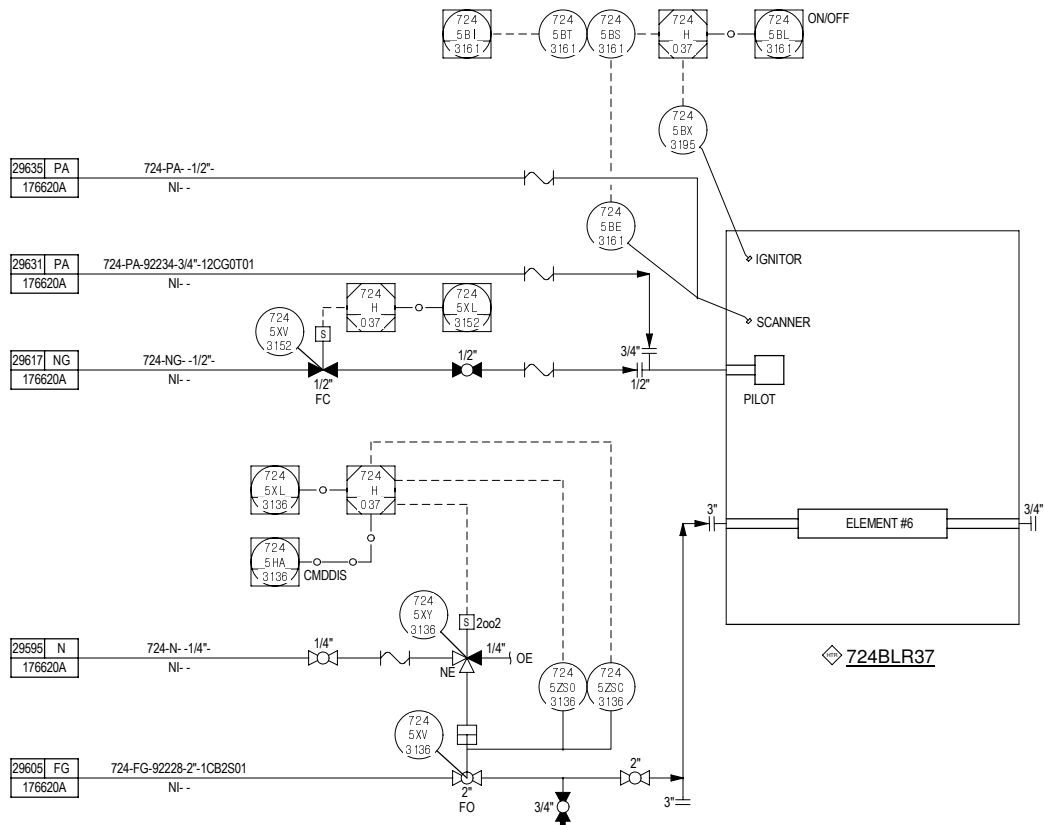
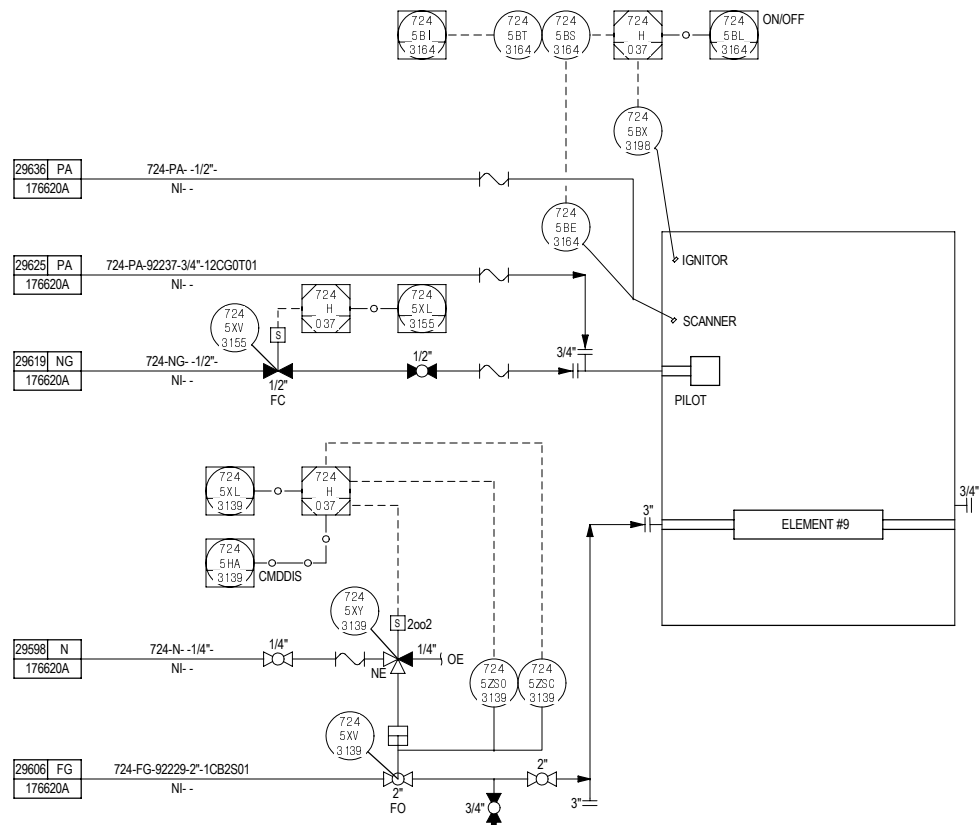
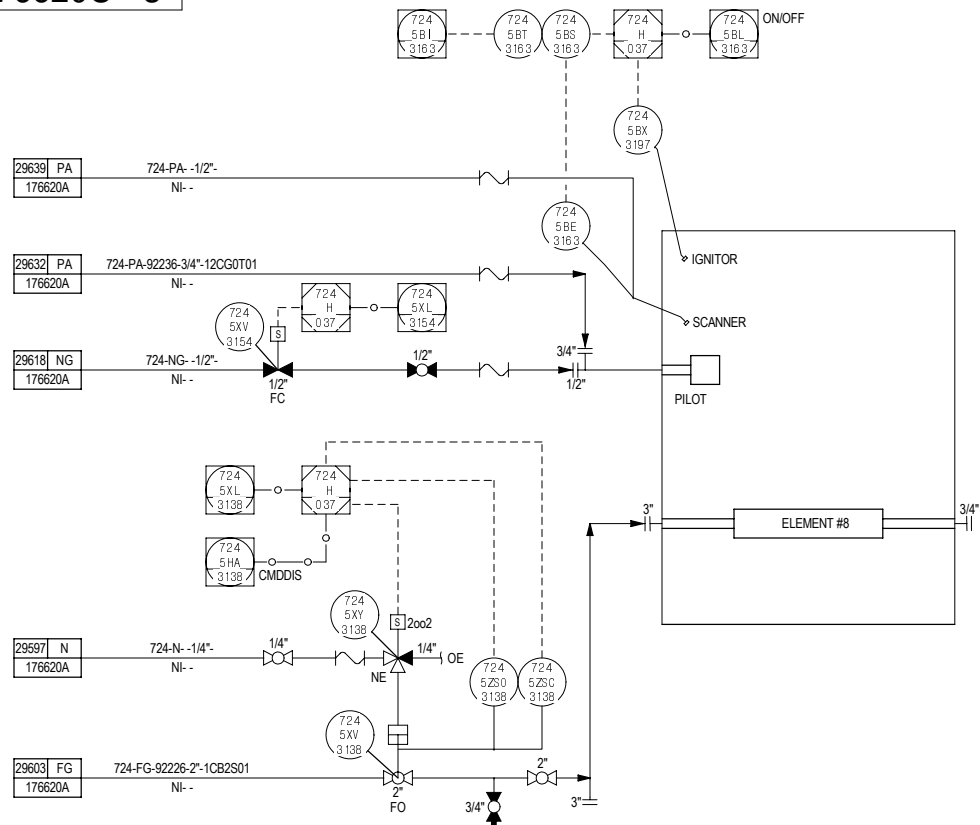
00401706.XLSX LINE LIST

NOTE

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P3ST
724BLR37
DUCT BURNER DETAILS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: RMG	10/23/2003	NUMBER ASSIGNED	DATE
DESIGN BY:		AND TRACING	
CHECKED BY:	CHARGE: 4164	RECORD ENTERED	REVISION
JOB NO.		YC- 176620B	3



- ## NOTES
1. SEE DWG. YC-143764, YC-143765, YC-143766 & YC-143767 FOR GENERAL NOTES AND SYMBOLS.
 2. DELETED.
 3. DELETED.
 4. DELETED.
 5. DELETED.
 6. DELETED.
 7. DELETED.
 8. DELETED.
 9. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
 10. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
 11. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

3	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
2S	08/18/14	SMARTPLANT CONVERSION		
1	09/22/06	AS-BUILT - CONVERTED TO CAD	ARS	4372
				4164
REV	DATE	DESCRIPTION	BY	CHARGE

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MOTIVA
ENTERPRISES LLC

PORT ARTHUR
REFINERY

P3ST
724BLR37
DUCT BURNER DETAILS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: RMG	10/23/2003	NUMBER ASSIGNED	DATE
DESIGN BY:		AND TRACING	
CHECKED BY:	CHARGE: 4164	RECORD ENTERED	
JOB NO.	YC- 176620C		3



INSTRUMENTATION

PIPIN

CONFIDENTIAL

MOTIVA
ENTERPRISES LLC

PORT ARTHUR
REFINERY

P3ST
724BLR37
HP ECONOMIZER SECTION
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: RMGB	8/4/2003	NUMBER ASSIGNED	DATE
DESIGN BY:		AND TRACING	
CHKD BY:	CHARGE:	RECORD ENTERED	
JOB NO.		YC- 176621	REVISION 7

1. SEE DWG. YC-143764, 5, 6 & 7 FOR GENERAL NOTES & SYMBOLS.
2. DELETED.
3. DELETED.
4. SEE DETAIL "A" ON DWG. YC-176624.
5. DELETED.
6. DELETED.
7. DELETED.
8. FOR START UP, LOW FLOW & FILL ONLY.
9. DELETED.
10. ONLY ONE VALVE WILL BE IN USE AT A TIME.
11. SEE VOGT POWER P&ID D-415-042-0 FOR BOILER CODE PIPING SPECIFICATION AND MATERIALS.
12. REFER TO ASME MFC-3M, TABLE 7 WITH DIAMETER RATIO OF 0.45 FOR RECOMMENDED STRAIGHT LENGTHS.
13. SEE DETAIL "A" FOR INSTRUMENT DETAILS.
14. INSTRUMENT TO BE STEAM TRACED FOR WINTERIZATION.
15. CONSTRUCTION TO SUPPLY SWAGE WITH THREADED NIPPLE TO CONNECT SOCKET WELDED VALVE TO INSTRUMENTATION.
16. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199999 TO YC-199999.
17. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
18. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

7	11/18/24	MOC REVISION	JTF	MOC 24-1220
6	03/01/23	MOC REVISION	CLF	MOC 23-106
5	06/27/22	MOC REVISION	JMC	MOC 22-598
4	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
3	10/29/18	MOC REVISION	JTF	M2018907
2S	07/28/14	SMARTPLANT CONVERSION		
1	04/06/05	MOC REVISION M2005478-001	CHI	4164
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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- ## REFERENCE DRAWINGS

00401706.XLSX LINE LIST

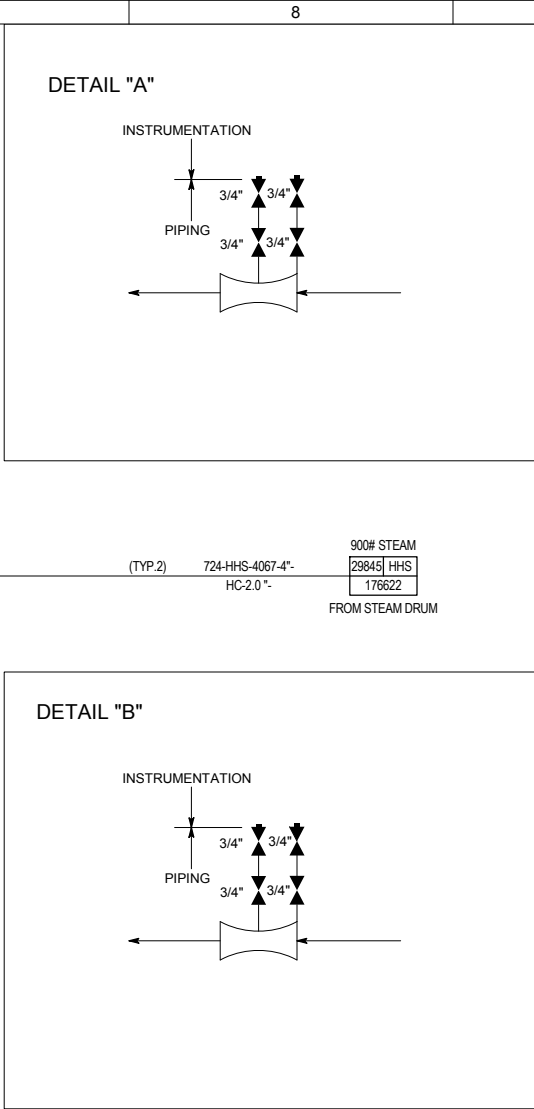
NOTE

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P3ST
724BLR37
STEAM DRUM & HP EVAPORATOR SECTION
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: RMGB	7/30/2003	NUMBER ASSIGNED	DATE
DESIGN BY:		AND TRACING	
CHK'D BY:	CHARGE:	RECORD ENTERED	
JOB NO.		YC- 176622	REVISION 5



- | REFERENCE DRAWINGS | | | | | |
|--------------------|----------|----------------------------|-----|---------|--|
| 040401706.XLSX | | LINE LIST | | | |
| | | | | | |
| | | | | | |
| 4 | 06/27/22 | MOC REVISION | JMC | 22-598 | |
| 3 | 04/01/20 | REVISED PER P3ST WALK DOWN | WCT | MOC 19- | |
| 2 | 12/02/19 | MOC REVISION | JMC | 19-1274 | |
| 1S | 07/28/14 | SMARTPLANT CONVERSION | | | |
| | | | | 4164 | |
| REV | DATE | DESCRIPTION | BY | CHAP | |

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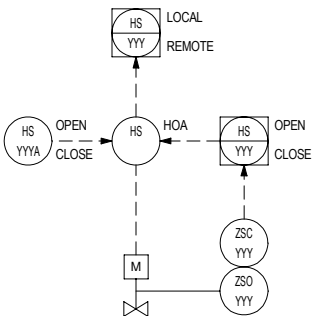
P3ST
724BLR37
HP SUPERHEATER SECTION
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: RMGB	4AUG03	NUMBER ASSIGNED AND TRACING	DATE
DESIGN BY:		RECORD ENTERED	
CHKD BY:	CHARGE: 4164	YC- 176623	
JOB NO			

1. SEE DWG. YC-143764, YC-143765, YC-143766 & YC-143767 FOR GENERAL NOTES AND SYMBOLS.
2. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199985 TO YC-199999.
3. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
4. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

MOTOR OPERATED VALVES

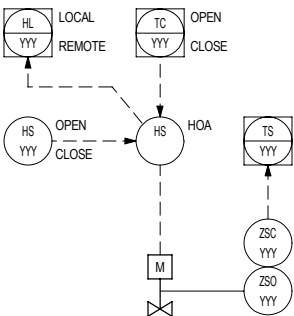
ON-OFF MOV



DETAIL "A"

VALVE NUMBER	P&ID NUMBER
MOV-3746	YC-160336
MOV-3781	YC-160336
MOV-3782	YC-176623
MOV-3789	YC-176621
MOV-3790	YC-176621

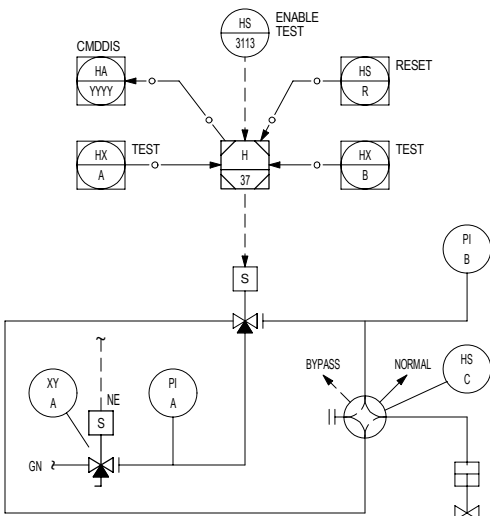
MOV



DETAIL "B"

VALVE NUMBER	P&ID NUMBER
TV-3756	YC-176623

2002 SOLENOIDS TO TRIP VALVE



P&ID NO.	VALVE	VALVE FAIL CONDITION	1ST SOLENOID VALVE	FIRST PI	2ND SOLENOID VALVE	SECOND PI	BYPASS SWITCH	1ST SOLENOID TEST SWITCH	2ND SOLENOID TEST SWITCH
YC-176619	XV-3114	FC	XY-3114A	PI-174A	XY-3114B	PI-174B	HS-3114C	HK-3114A	HK-3114B
YC-176619	XV-3116	FC	XY-3116A	PI-175A	XY-3116B	PI-175B	HS-3116C	HK-3116A	HK-3116B
YC-176619	XV-3115	FO	XY-3115A	PI-176A	XY-3115B	PI-176B	HS-3115C	HK-3115A	HK-3115B
YC-176620	XV-3130	FO	XY-3130A	PI-177A	XY-3130B	PI-177B	HS-3130C	HK-3130A	HK-3130B

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

3	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
2S	08/06/14	SMARTPLANT CONVERSION		
1	09/22/06	AS-BUILT - CONVERTED TO CAD	ARS	4372
				4164
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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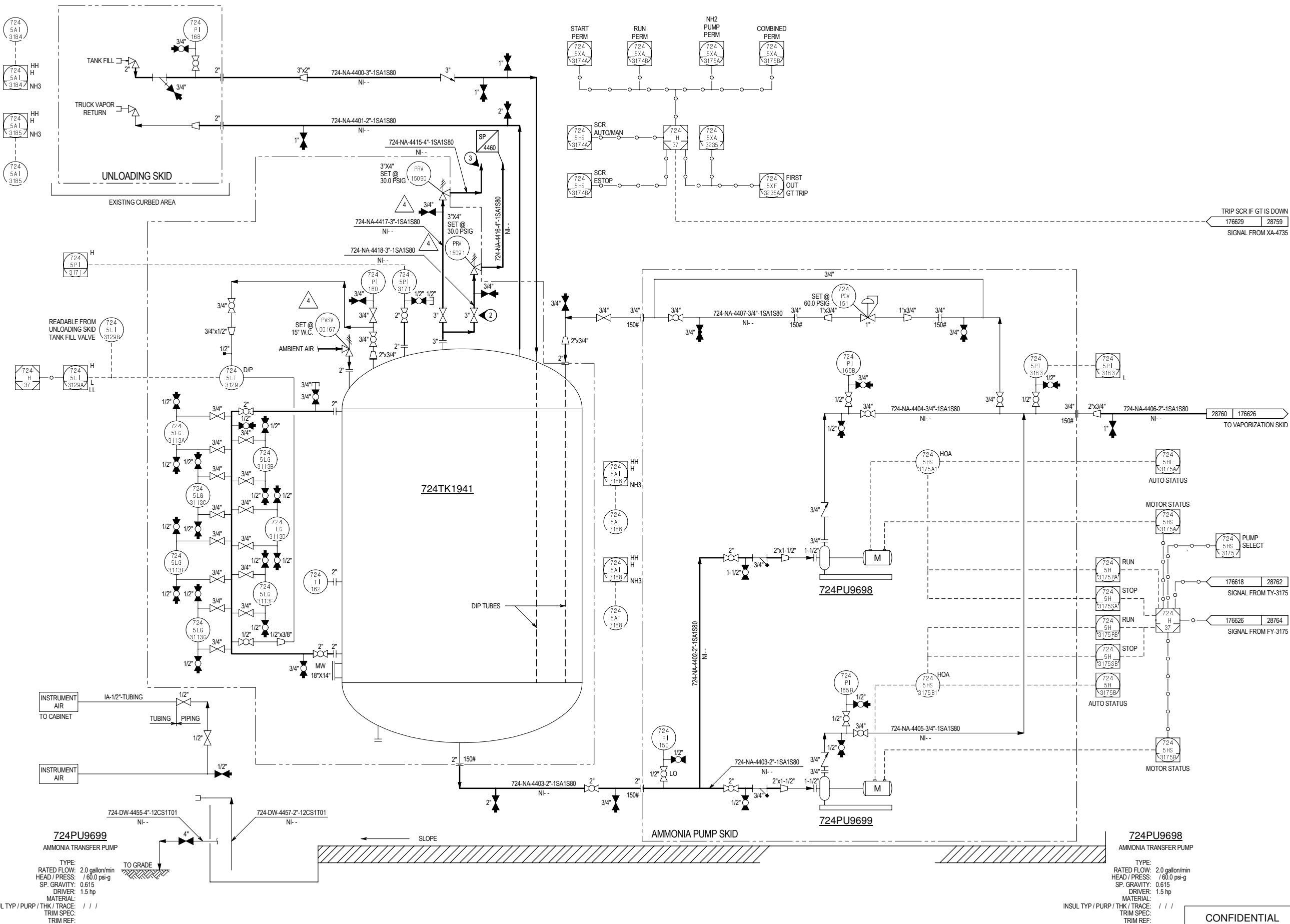
P3ST
HRSG - 724BLR37
GENERAL DETAIL
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: RMB	8/13/2003	NUMBER ASSIGNED AND TRACING RECORD ENTERED	DATE
DESIGN BY:			
CHK'D BY:	CHARGE:		REVISION
JOB NO.		YC- 176624	3

CONFIDENTIAL

YC- 176624

SAFETY...A LIFETIME COMMITMENT



1. SEE DWG. YC-143764, YC-143765, YC-143766, YC-143767 FOR GENERAL NOTES AND SYMBOLS.
2. ONE VALVE WILL BE OPEN AND ONE VALVE WILL BE CLOSED AT ALL TIMES PER ADMINISTRATIVE PROCEDURES.
3. VENT TO ATM AT SAFE LOCATION ADD RAIN HAT, BIRD SCREEN AND WEEP HOLE.
4. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
5. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
6. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

00401706.XLSX LINE LIST

5	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
4	08/16/18	MOC REVISION	JBB	M2018559
3	06/03/16	MOC REVISION	JTF	M20161669
2S	08/05/14	SMARTPLANT CONVERSION		
1	09/22/06	AS-BUILT - CONVERTED TO CAD	ARS	4372 4164
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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P3ST
724BL37, 724PU9698, 724PU9699, 724TK1941
AMMONIA STORAGE
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: RMB	10/23/2003	NUMBER ASSIGNED	DATE
DESIGN BY:		AND TRACING	
CHECKED BY:	CHARGE:	RECORD ENTERED	
JOB NO.		YC- 176625	REVISION 5

- ## REFERENCE DRAWINGS

00401706.XLSX LINE LIST

3	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
2S	08/01/14	SMARTPLANT CONVERSION		
1	09/22/06	AS-BUILT - CONVERTED TO CAD	ARS	4372
				4164
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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MOTIVA
ENTERPRISES LLC

PORT ARTHUR
REFINERY

P3ST
724BL9703, 724BL9704, 724FIL05, 724FIL06, 724VES36114
AQUEOUS AMMONIA VAPORIZATION SKID
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: RMG	10/21/2003	NUMBER ASSIGNED	DATE
DESIGN BY:		AND TRACING	
CHECKED BY:	CHARGE: 4164	RECORD ENTERED	REVISION
JOB NO.		YC- 176626	3

CONFIDENTIAL

YC- 176626

REVISION



SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SPPID REV 0



CHKD BY:	CHARGE: 4164
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CHKED BY:	CHARGE: 4164
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YC- 176627

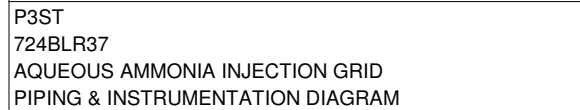
REVISION
0

- ## SAFETY--A LIFETIME COMMITMENT

00401706.XLSX LINE LIST

NOTE	
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YC- 176627

- ## SAFETY---A LIFETIME COMMITMENT



SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

SAFETY---A LIFETIME COMMITMENT

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1. SEE DWG. YC-143764, 5, 6 & 7 FOR GENERAL NOTES AND SYMBOLS.
2. DRAWING ONLY COVERS INSTRUMENT WINTERIZATION STEAM TRACING FOR THE MOTIVA PORT ARTHUR P53 HRSG NO. 37 PROJECT.
3. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
4. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
5. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

00401706.XLSX LINE LIST

2	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
1S	08/05/14	SMARTPLANT CONVERSION		
				4164
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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P3ST
724BLR37
INSTRUMENT WINTERIZATION STEAM TRACING
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: RMGB	10/6/2004	NUMBER ASSIGNED AND TRACING RECORD ENTERED	DATE
DESIGN BY:			
CHKD BY:	CHARGE: 4164		REVISION
JOB NO.		YC- 176630	2

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SPPID REV 0

00401706.XLSX LINE LIST

REV	DATE	DESCRIPTION	BY	CHARGE
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CHKD BY: LDM	CHARGE: 4372	YC- 180542	REVISION
JOB NO.			6

-

00401706.XLSX LINE LIST

00401706.XLSX LINE LIST

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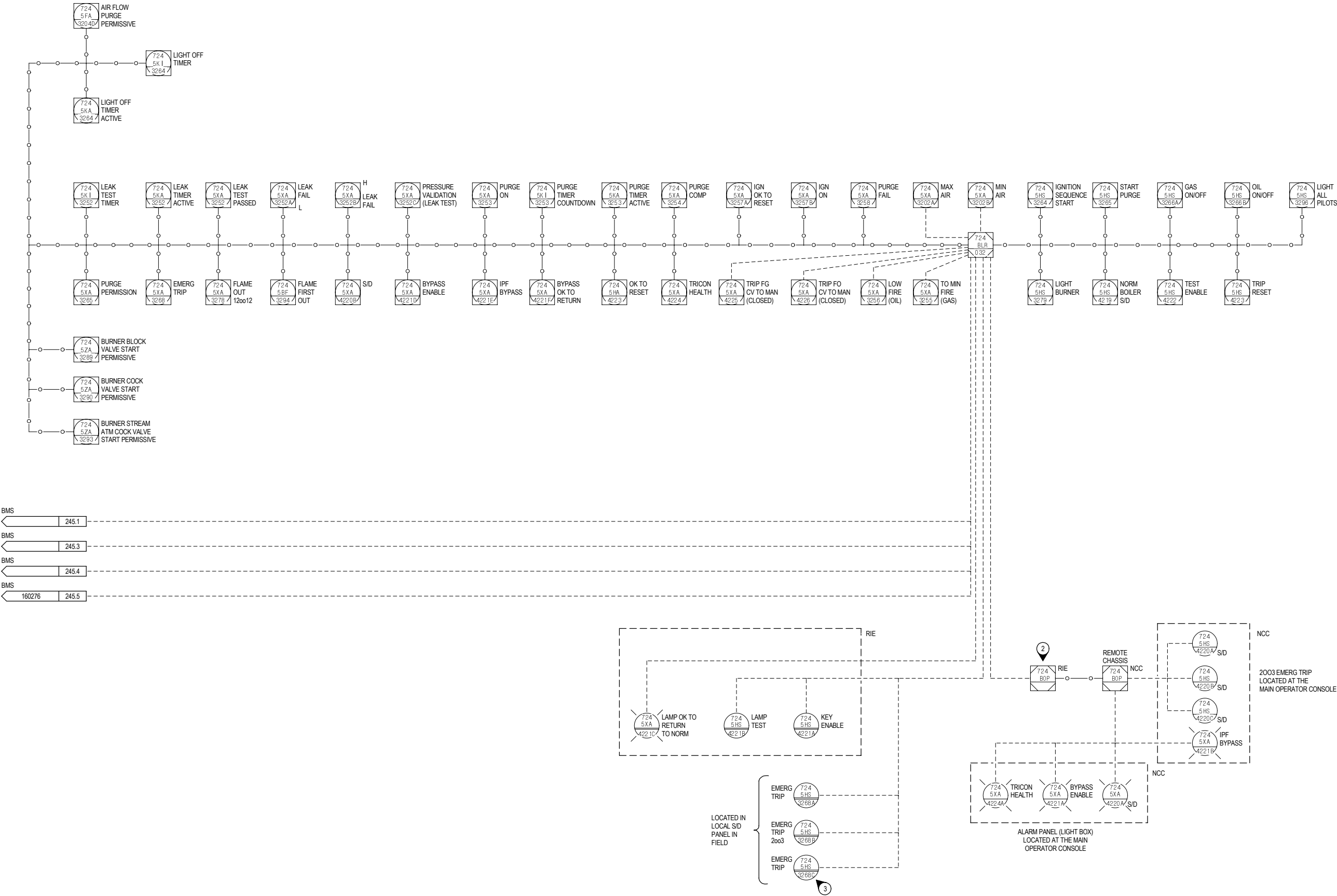


P3ST
723BLR32
BURNER CONTROLS - ALARMS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE	
DRAWN BY: LAP	12/20/2005	NUMBER ASSIGNED	DATE
DESIGN BY: ARS	12/20/2005	AND TRACING	
CHECKED BY: JW	CHARGE: 4372	RECORD ENTERED	
JOB NO.		YC- 182885	

YC- 182885 3

SPPID REV 0



SAFETY---A LIFETIME COMMITMENT

SPPID REV 0



1. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
2. REPRESENTS BOTH DIAGNOSTIC AND TRIP SIGNALS.
3. TRIPLICATED CONTACTS.
4. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199989 TO YC-199999.
5. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

00401706.XLSX LINE LIST

NOTE

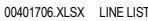
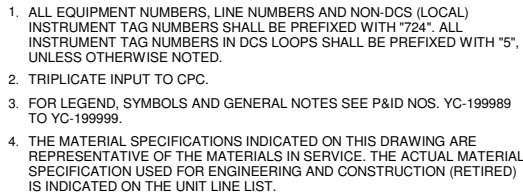
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MOTIVA
ENTERPRISES LLC

PORT ARTHUR
REFINERY

P3ST
723BLR33
BURNER CONTROLS - ALARMS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE		
DRAWN BY: LAP	12/20/2005	NUMBER ASSIGNED AND TRACING RECORD ENTERED	DATE	
DESIGN BY: ARS	12/20/2005			
CHECKED BY: JW	CHARGE: 4372	YC- 182886		REVISION
JOB NO.				3



3	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
2S	08/01/14	SMARTPLANT CONVERSION		
1	04/09/07	GENERAL REVISION	ARS	4372
0	10/06/06	ISSUE FOR BOILER 36 INSTR IFC	ARS	4372
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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ENTERPRISES LLC

PORT ARTHUR
REFINERY

P3ST
724BLR36
BURNER CONTROLS - ALARMS
PIPING & INSTRUMENTATION DIAGRAM

	DATES	SCALE: NONE
DRAWN BY: LAP	12/21/2005	NUMBER ASSIGNED AND TRACING RECORD ENTERED
DESIGN BY: LDM	12/21/2005	DATE 10/06/06

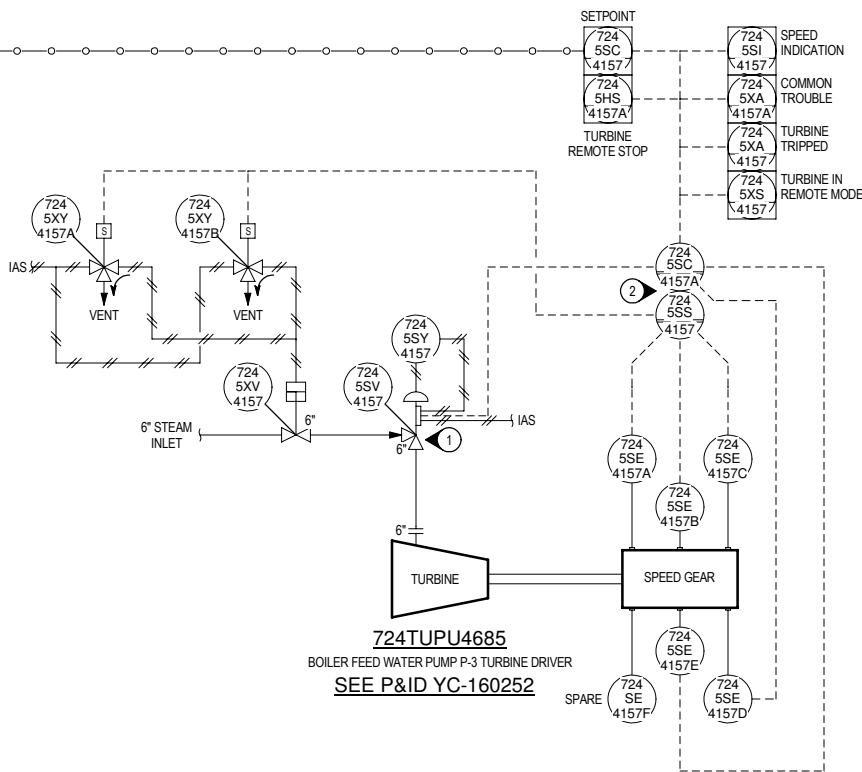
CONFIDENTIAL

YC- 182889 3

9335	160319
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SIGNAL FROM 5PC-4174

SIGNAL FROM 5PC-4174

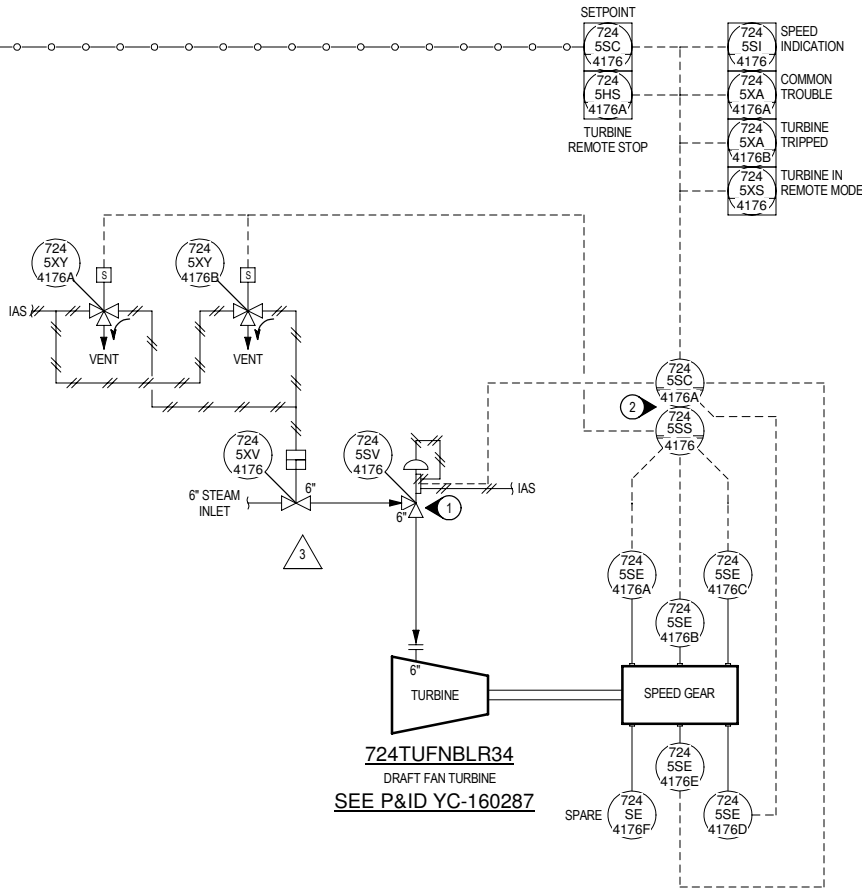


9333	160319
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SIGNAL FROM 5PC-4174

9334	160319
------	--------

SIGNAL FROM 5PC-4174



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1. TURBINE TRIP AND THROTTLE VALVES.
2. SPEED CONTROLLER (SC) IS TRISEN GOVERNOR. SPEED SWITCH (SS) IS TRISEN TURBOSAFE OVERSPEED DETECTION / TRIP DEVICE. BOTH ARE LOCATED IN FIELD PANEL NEAR TURBINE.
3. ALL EQUIPMENT NUMBERS, LINE NUMBERS AND NON-DCS (LOCAL) INSTRUMENT TAG NUMBERS SHALL BE PREFIXED WITH "724". ALL INSTRUMENT TAG NUMBERS IN DCS LOOPS SHALL BE PREFIXED WITH "5", UNLESS OTHERWISE NOTED.
4. FOR LEGEND, SYMBOLS AND GENERAL NOTES SEE P&ID NOS. YC-199999 TO YC-199999.
5. FUTURE MODIFICATIONS
6. THE MATERIAL SPECIFICATIONS INDICATED ON THIS DRAWING ARE REPRESENTATIVE OF THE MATERIALS IN SERVICE. THE ACTUAL MATERIAL SPECIFICATION USED FOR ENGINEERING AND CONSTRUCTION (RETIRED) IS INDICATED ON THE UNIT LINE LIST.

REFERENCE DRAWINGS

00401706.XLSX LINE LIST

4	06/06/25	MOC REVISION	ATE	MOC 25-329
3	12/02/24	MOC REVISION	JTF	MOC 24-1659
2	11/23/21	MOC REVISION (PER USMF/03155)	JTF	MOC 19-703
1	04/01/20	REVISED PER P3ST WALK DOWN	WCT	MOC 19-1361
0	10/18/18	MOC REVISION (PER USMF/3155)	JTF	MOC 19-703
REV	DATE	DESCRIPTION	BY	CHARGE

NOTE

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ENTERPRISES LLC

PORT ARTHUR
REFINERY

P3ST
724TUFNBLR34, TUFNBLR35, 724TUPU4685, 724TUPU4686
TURBINE SPEED AND TRIP CONTROLS
PIPING & INSTRUMENTATION DIAGRAM

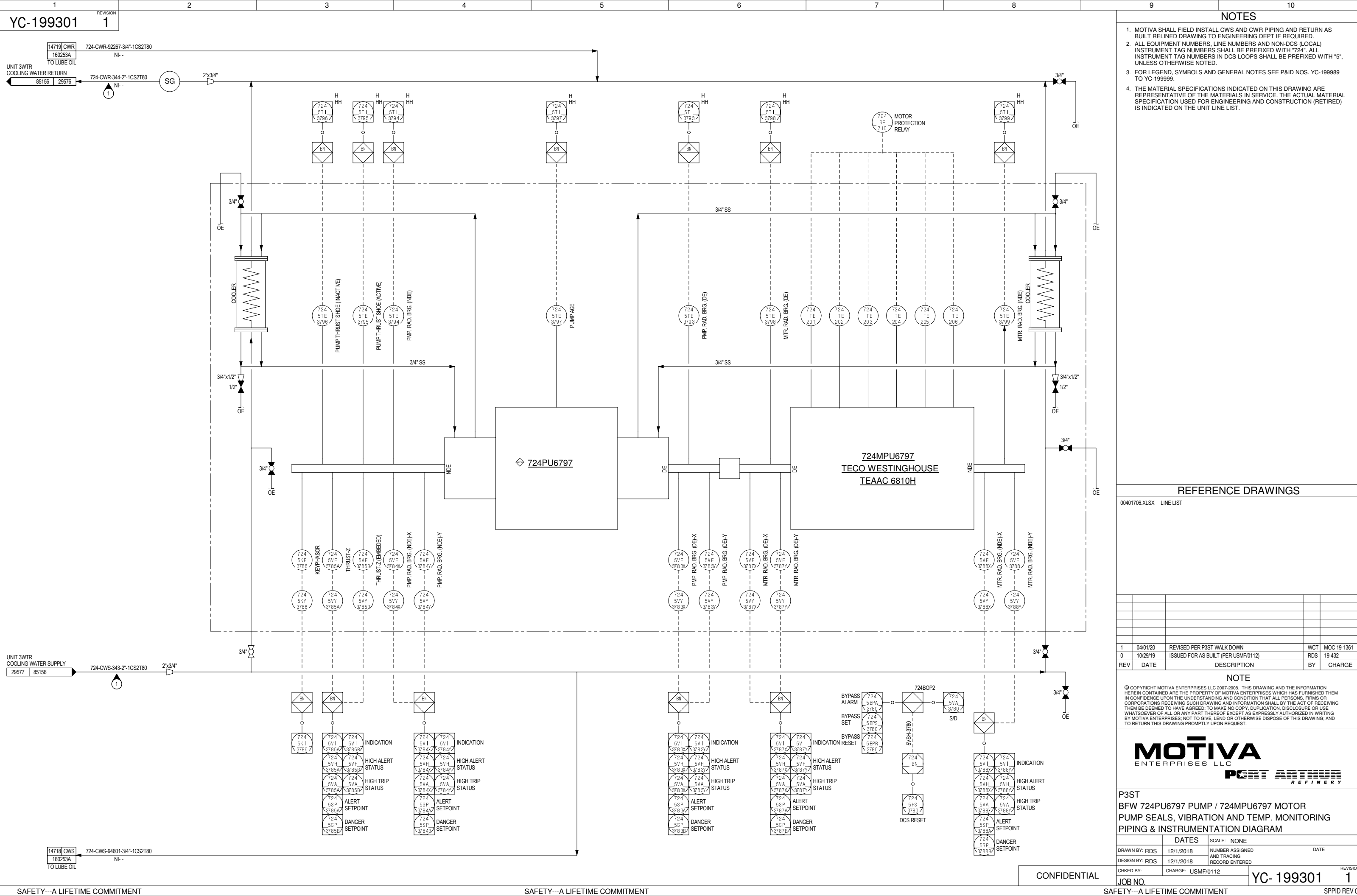
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DESIGN BY: JTF	10-18-2018	AND TRACING	
CHECKED BY:	CHARGE: 4500	RECORD ENTERED	
JOB NO.		YC- 197885	REVISION 4

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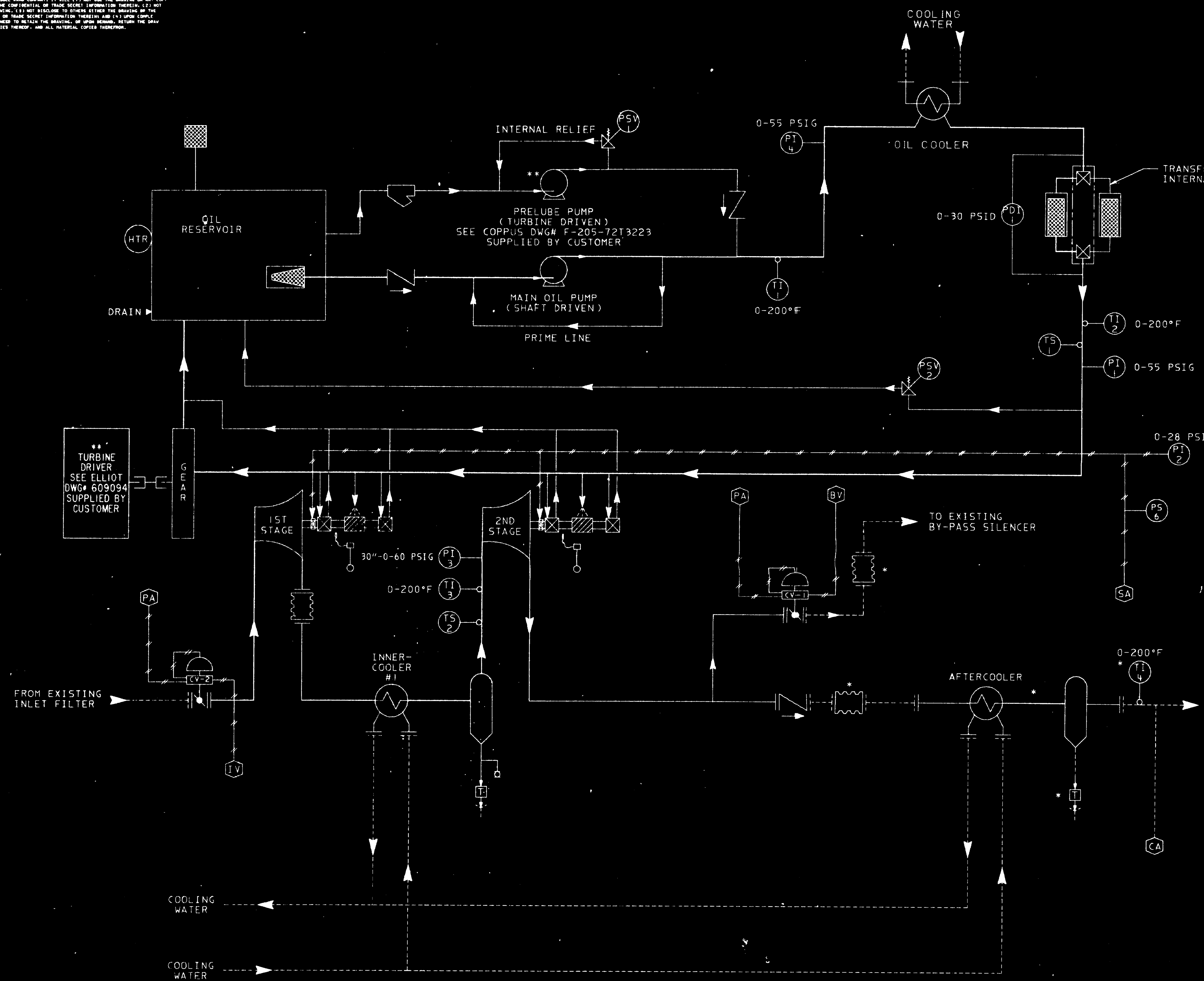
SPPID REV 0

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FIDENCE. THE RECIPIENT BY RECEIVING AND RETENTION OF THE DRAWING ACCEPTS
THE DRAWING IN CONFIDENCE AND AGREES THAT, EXCEPT AS AUTHORIZED IN WRIT-
ING BY INGERSOLL RAND COMPANY, IT WILL (1) NOT USE THE DRAWING OR ANY COPY
THEREOF OR THE CONFIDENTIAL OR TRADE SECRET INFORMATION THEREIN (2) NOT
COPY THE DRAWING, (3) NOT DISCLOSE TO OTHERS EITHER THE DRAWING OR THE
CONFIDENTIAL OR TRADE SECRET INFORMATION THEREIN AND (4) UPON COMPLE-
TION OF THE NEED TO RETAIN THE DRAWING, OR UPON DEMAND, RETURN THE DRAW-
ING, ALL COPIES THEREOF, AND ALL MATERIAL COPIES THEREFROM.

REVISIONS				
ZONE	SYN.	DESCRIPTION	DRAWN	CHECKED
A		REMOVED: L5-2, LQ, NOTE PS REVISED: NOTE #4, 15-1, 15-2 ADDED: PS-8, CORRECTED DWG. NO.	REX 3-15-85	JRB 3-15-85

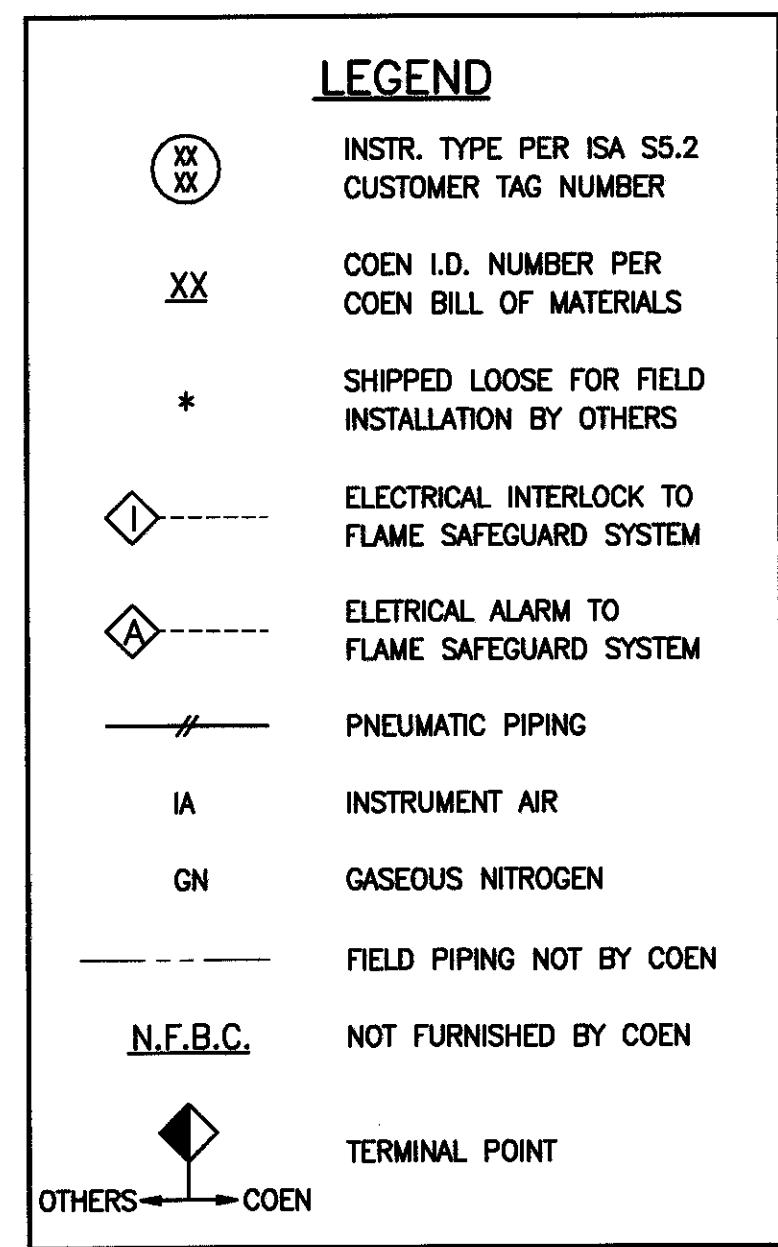


- NOTE:
1. SEE 8X3628 FOR PROCESS AND INSTRUMENTATION DIAGRAM SYMBOLS.
 2. ITEMS MARKED * ARE SUPPLIED BY IR AND MOUNTED BY CUSTOMER.
 3. ITEMS MARKED ** ARE SUPPLIED BY THE CUSTOMER AND MOUNTED BY IR.
 4. DENOTES BULKHEADS THAT ARE SHOWN ON PNEUMATIC SCHEMATIC DRAWING NUMBER, 8X6528

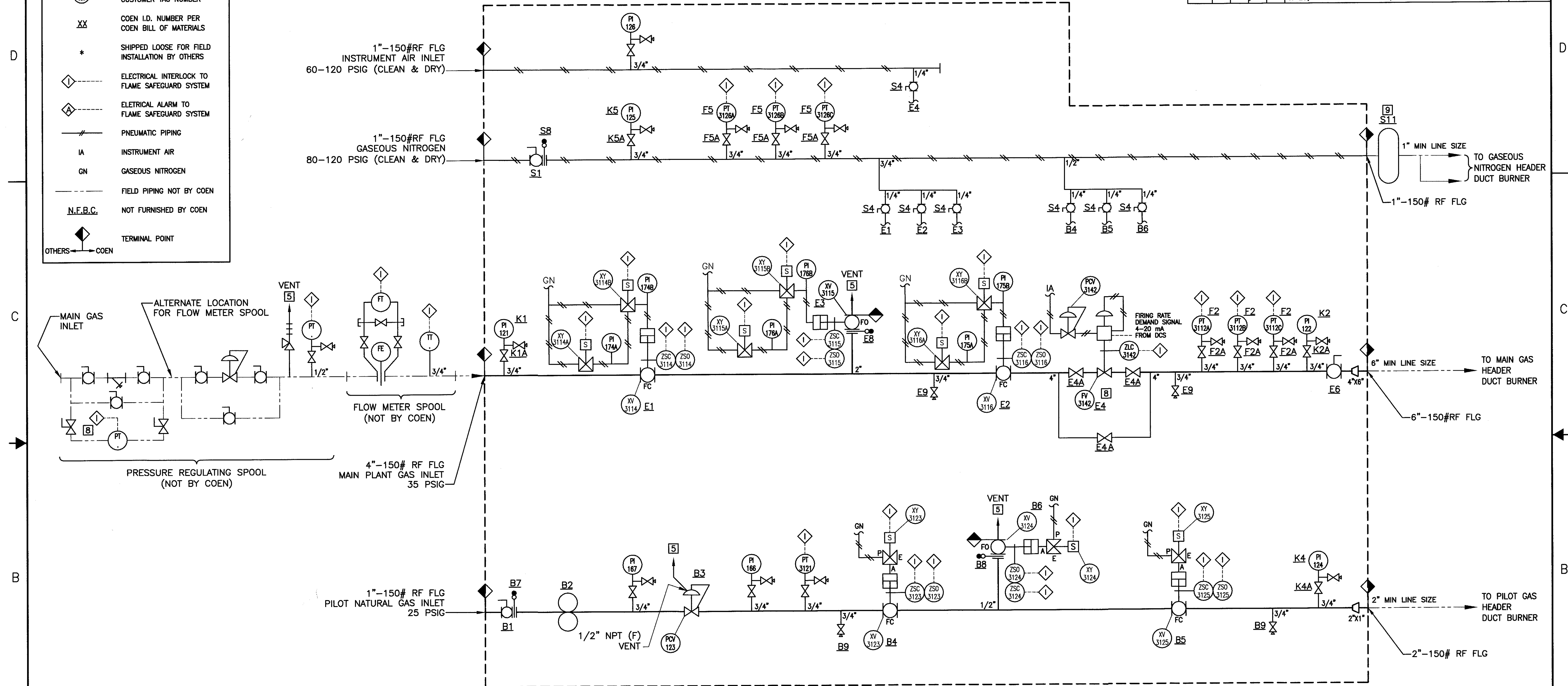
MAR 21 1985

R.PID.3493.8X6399
DRA DWG1

FIRST FOR: TEXACO MB4-3493 IAC11212		INGERSOLL-RAND MAYFIELD, KENTUCKY 42066	
DRAWN T.CIRIO		DATE 12-03-84	
CHECKED JRB		DATE 1-2-85	
APPROVED D.K.		DATE 1-2-85	
CODE IDENT. NO.		DWG. NO. 8X6399	
SCALE		SHEET OF	



		REVISIONS								
MASTER REV		ZONE	REV	BY	CKD	APP	DESCRIPTION		DATE	
2	1		A	LEB	DTM	JR	REVISED PER CUSTOMERS FLOW DIAGRAM		12/19/03	
B	B	Date:		B	LEB	<i>jr</i> <i>dt</i>	ADDED NOTE 8; GASEOUS NITROGEN WAS NITROGEN AIR; MOVED AIR TANK OFF SKID; ADDED SWITCHES TO E1, E2 & E3; ADDED CUST. BLOCK; MINOR REVISIONS		2/9/04	



SECTION II
PIPING MODULE ASSEMBLY
SEE COEN DWG D-0186-368

NOTES:

- SEE COEN BILL OF MATERIALS FOR ITEM DESCRIPTIONS, SYSTEM SPECIFICATIONS AND ASSOCIATED DRAWINGS.
- PIPING AND DUCTING BETWEEN ALL COEN EQUIPMENT IS NOT BY COEN. COEN RECOMMENDS LOCATING THE PIPING MODULE AND BLOWER SKID WITHIN 20 FEET OF BURNER PIPING INLETS. IN ADDITION, THE PRESSURE DROP IN THE MAIN GAS LINE BETWEEN THE PIPING MODULE AND THE DUCT BURNER INLET MAY NOT EXCEED 2.0 PSI.
- ADEQUATE PROVISIONS MUST BE MADE IN THE INTERCONNECTING PIPING FROM PIPING MODULE OUTLETS TO BURNER PIPING INLETS TO AVOID STRESSES DUE TO THERMAL EXPANSION OR VIBRATION. INTERCONNECTING PIPING IS NOT BY COEN.
- A DRIP LEG WILL BE INSTALLED AT THE LOWEST POINT IN ALL FIELD PIPING RUNS, NOT BY COEN.

- FUEL VENTS, INCLUDING PRV SPRING CHAMBER VENTS AND VENTS BETWEEN DOUBLE DIAPHRAGMS ON LIMIT SWITCHES MUST BE ROUTED TO AN APPROPRIATE LOCATION DEPENDING ON FLUID (GAS OR OIL), NOT BY COEN.
- PILOT / COOLING AIR BLOWER IS TO BE ON WHENEVER THE TURBINE IS RUNNING AND/OR THE DUCT TEMPERATURE IS GREATER THAN 150°F. CONTROLS AND MOTOR STARTERS ARE NOT BY COEN.
- ALL FIELD ERECTED PIPING BETWEEN AND UPSTREAM OF COEN'S PIPING MUST BE FLUSHED OUT AND CLEANED PRIOR TO START-UP, NOT BY COEN.
- FLOW CONTROL VALVE, E4 (FV3142), FAIL CLOSED WITH MECHANICAL DOWNSTOP FOR MINIMUM FLOW.
- NITROGEN SURGE TANK BY COEN. ALL ACCESSORIES (RELIEF VALVES, DRAIN VALVES, GAUGES, ETC) NOT BY COEN.

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ACCORD WITH THE CONTRACT OR OTHER WRITTEN PERMISSION OF
COEN COMPANY. ANY REPRODUCTIONS IN WHOLE OR IN PART SHALL
INCLUDE THIS NOTATION.

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES			DRAWN BY: RRL		DATE: 09/30/03	
TOLERANCES ARE:			CHECKED BY: LEM		DATE: 10/01/03	
DECIMAL XX = ±			APPROVED BY: JR		DATE: 10/3/03	
FRACTION XX = ±			COEN FILE NO. 40D-14341-1		RELATED DRAWINGS / NEXT ASSEMBLY:	
ANGLE XX = ±			FINISH		THIRD ANGLE PROJECTION	

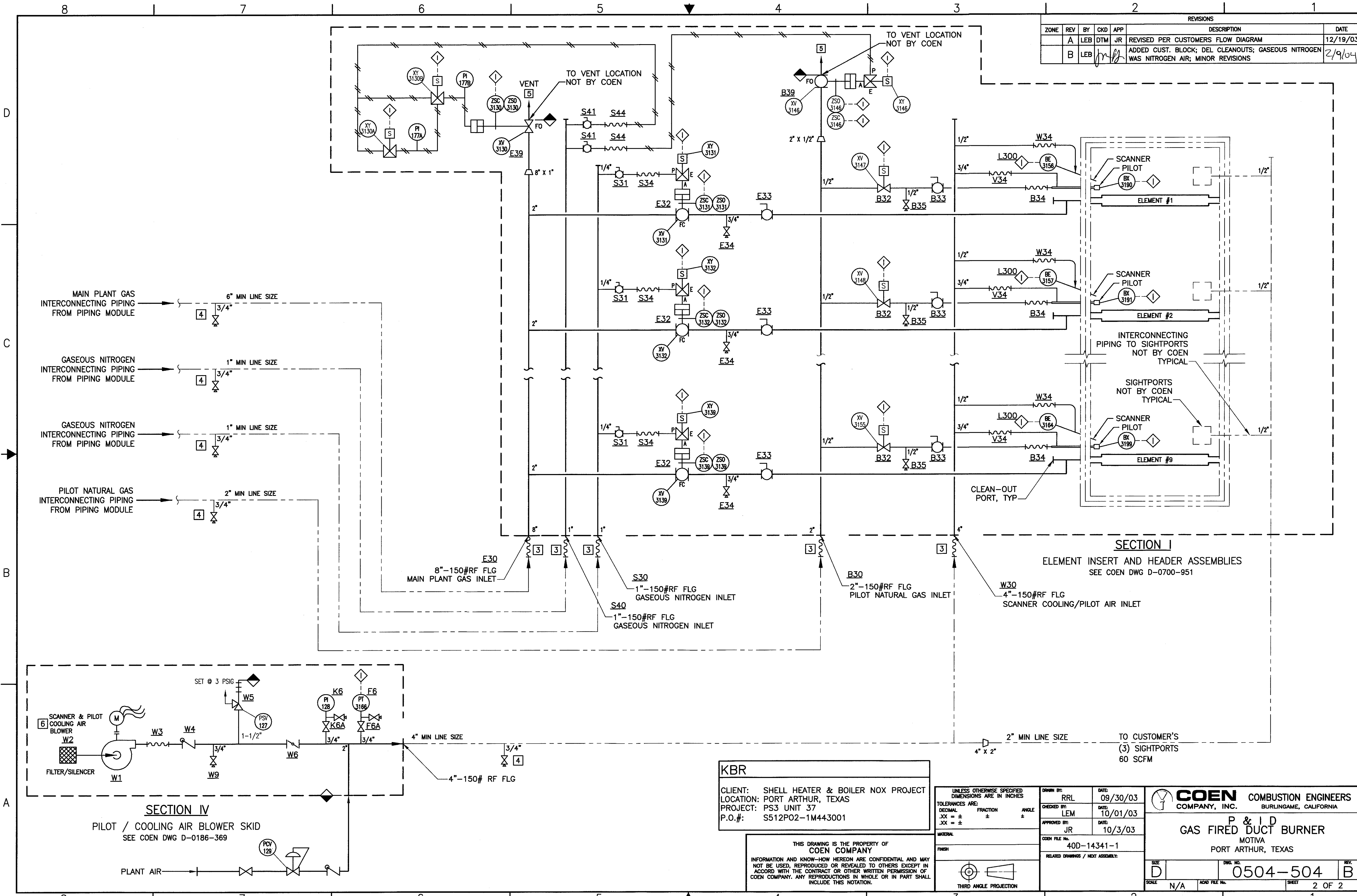
KBR

CLIENT: SHELL HEATER & BOILER NOX PROJECT
LOCATION: PORT ARTHUR, TEXAS
PROJECT: PS3 UNIT 37
P.O.#: S512P02-1M443001

COEN COMBUSTION ENGINEERS
COMPANY, INC. BURLINGAME, CALIFORNIA

P & I D
GAS FIRED DUCT BURNER
MOTIVA
PORT ARTHUR, TEXAS

SIZE: **D** DWG. NO.: **0504-504** REV.: **B**
SCALE: **N/A** ADD. FILE NO.: **1 OF 2**



REVISIONS						DATE
ZONE	REV	BY	CHKD	APP	DESCRIPTION	
	A	LEB	DTM	JR	REVISED PER CUSTOMERS FLOW DIAGRAM	12/19/03
	B	LEB			ADDED CUST. BLOCK; DEL CLEANOUTS; GASEOUS NITROGEN WAS NITROGEN AIR; MINOR REVISIONS	2/9/04

KBR
CLIENT: SHELL HEATER & BOILER NOX PROJECT
LOCATION: PORT ARTHUR, TEXAS
PROJECT: PS3 UNIT 37
P.O.#: S512P02-1M443001

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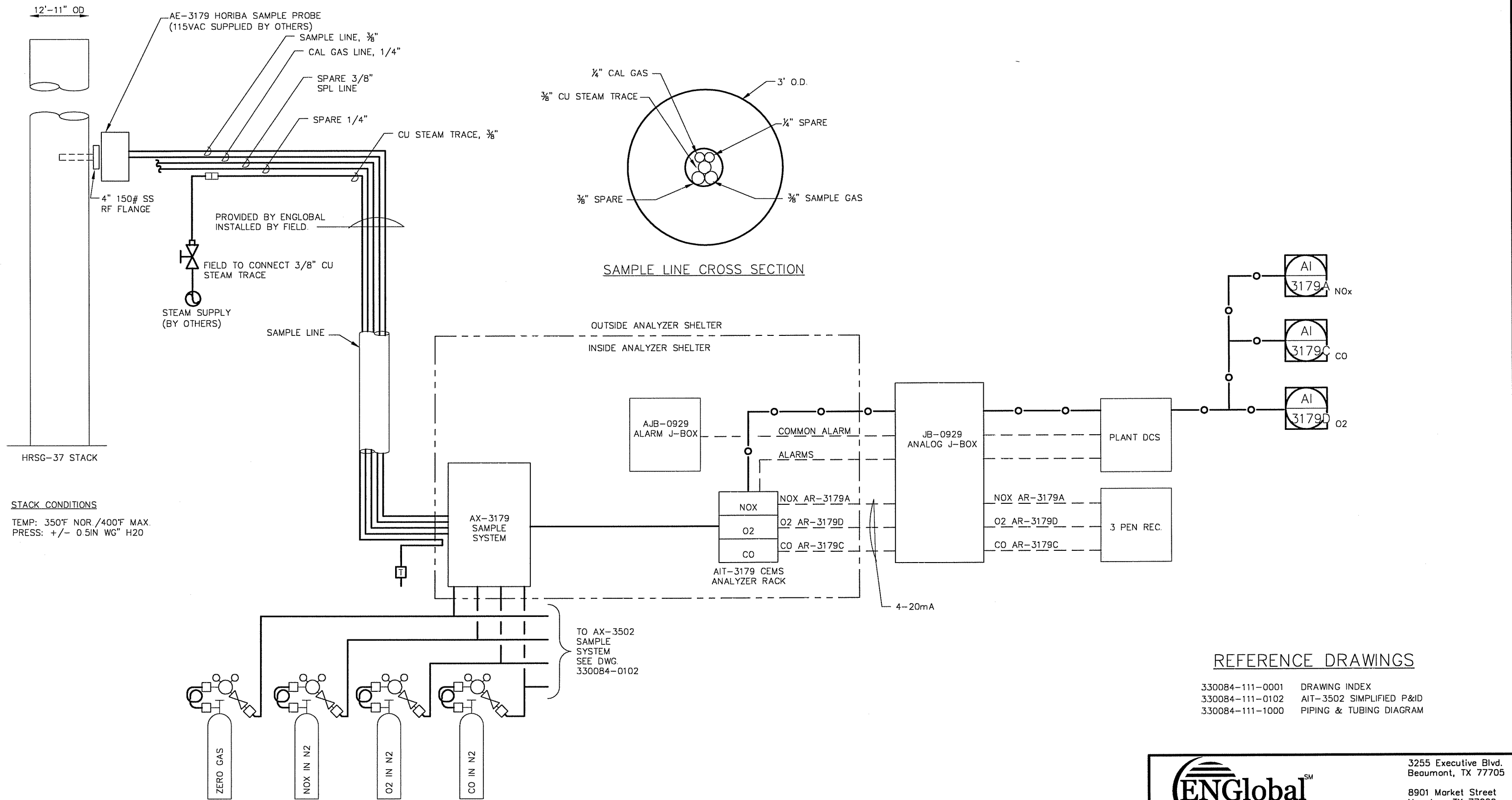
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES			DRAWN BY: RRL		DATE: 09/30/03
TOLERANCES ARE:			CHECKED BY: LEM		DATE: 10/01/03
DECIMAL	FRACTION	ANGLE	APPROVED BY: JR		DATE: 10/3/03
.XX = ±	±	±	COEN FILE NO. 40D-14341-1		
.XX = ±			RELATED DRAWINGS / NEXT ASSEMBLY:		
MATERIAL					
FINISH					

**COEN** COMBUSTION ENGINEERS
COMPANY, INC. BURLINGAME, CALIFORNIA

P & I D
GAS FIRED DUCT BURNER
MOTIVA
PORT ARTHUR, TEXAS

SIZE **D** DWG. NO. **0504-504** REV. **B**
SCALE **N/A** ACID FILE NO. SHEET **2 OF 2**

MFG.	MODEL	ANALYZER	RANGE	TAG NO.
HORIBA	ENDA-XE	NOx	0-25 PPM	AIT-3179
HORIBA	ENDA-XE	O2	0-25 %	AIT-3179
HORIBA	ENDA-XE	CO	0-200 PPM	AIT-3179



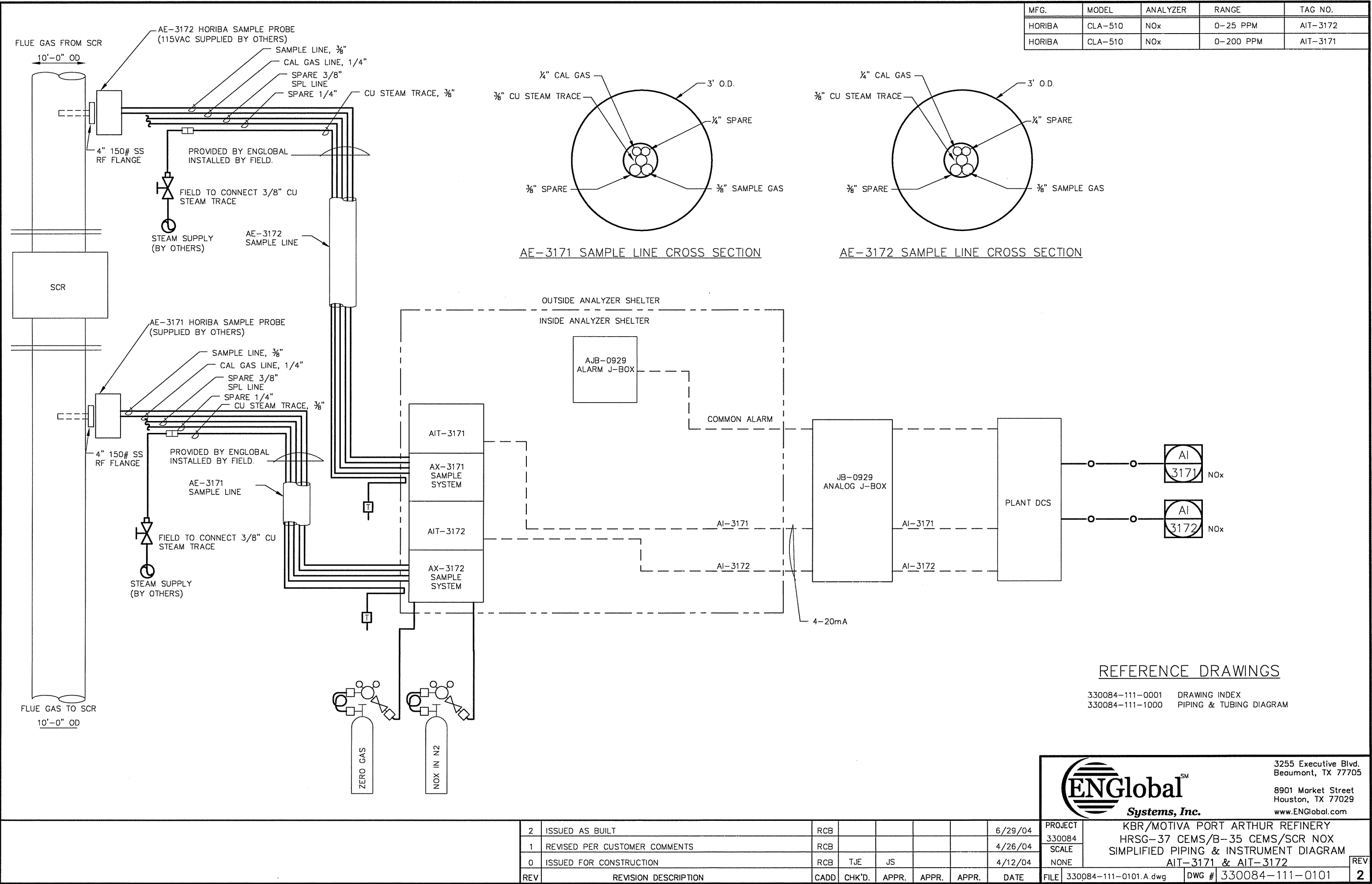
3255 Executive Blvd.
Beaumont, TX 77705

8901 Market Street
Houston, TX 77029

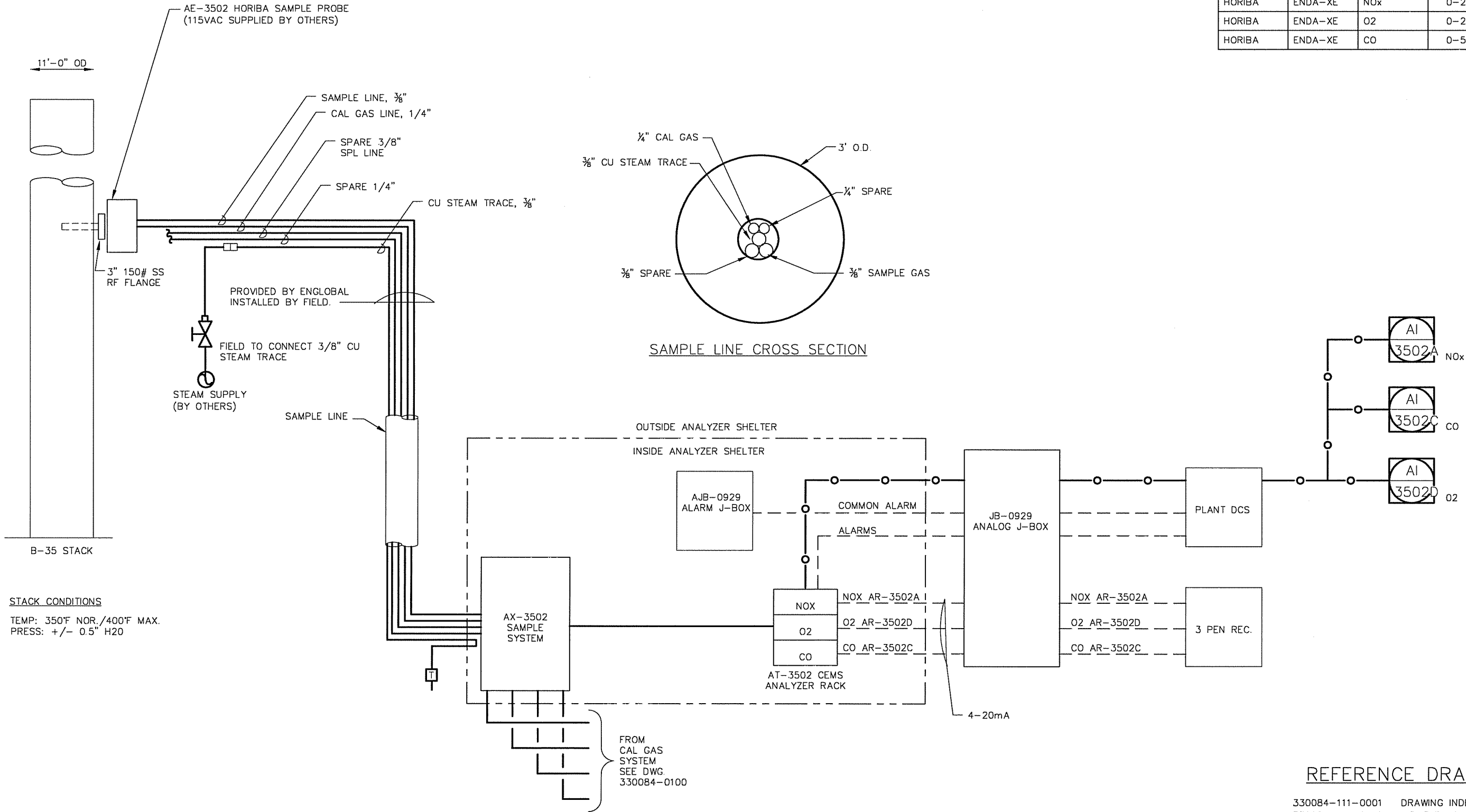
www.ENGlobal.com

2	ISSUED AS BUILT	RCB					6/29/04
1	REVISED PER CUSTOMER COMMENTS	RCB					4/26/04
0	ISSUED FOR CONSTRUCTION	RCB	TJE	JS			4/12/04
REV	REVISION DESCRIPTION	CADD	CHK'D.	APPR.	APPR.	APPR.	DATE

PROJECT	330084	KBR/MOTIVA PORT ARTHUR REFINERY HRSG-37 CEMS/B-35 CEMS/SCR NOX SIMPLIFIED PIPING & INSTRUMENT DIAGRAM AIT-3179	REV
SCALE	NONE		
FILE	330084-111-0100.A.dwg	DWG #	330084-111-0100
			2



MFG.	MODEL	ANALYZER	RANGE	TAG NO.
HORIBA	ENDA-XE	NOx	0-200 PPM	AIT-3502
HORIBA	ENDA-XE	O2	0-25 %	AIT-3502
HORIBA	ENDA-XE	CO	0-500 PPM	AIT-3502



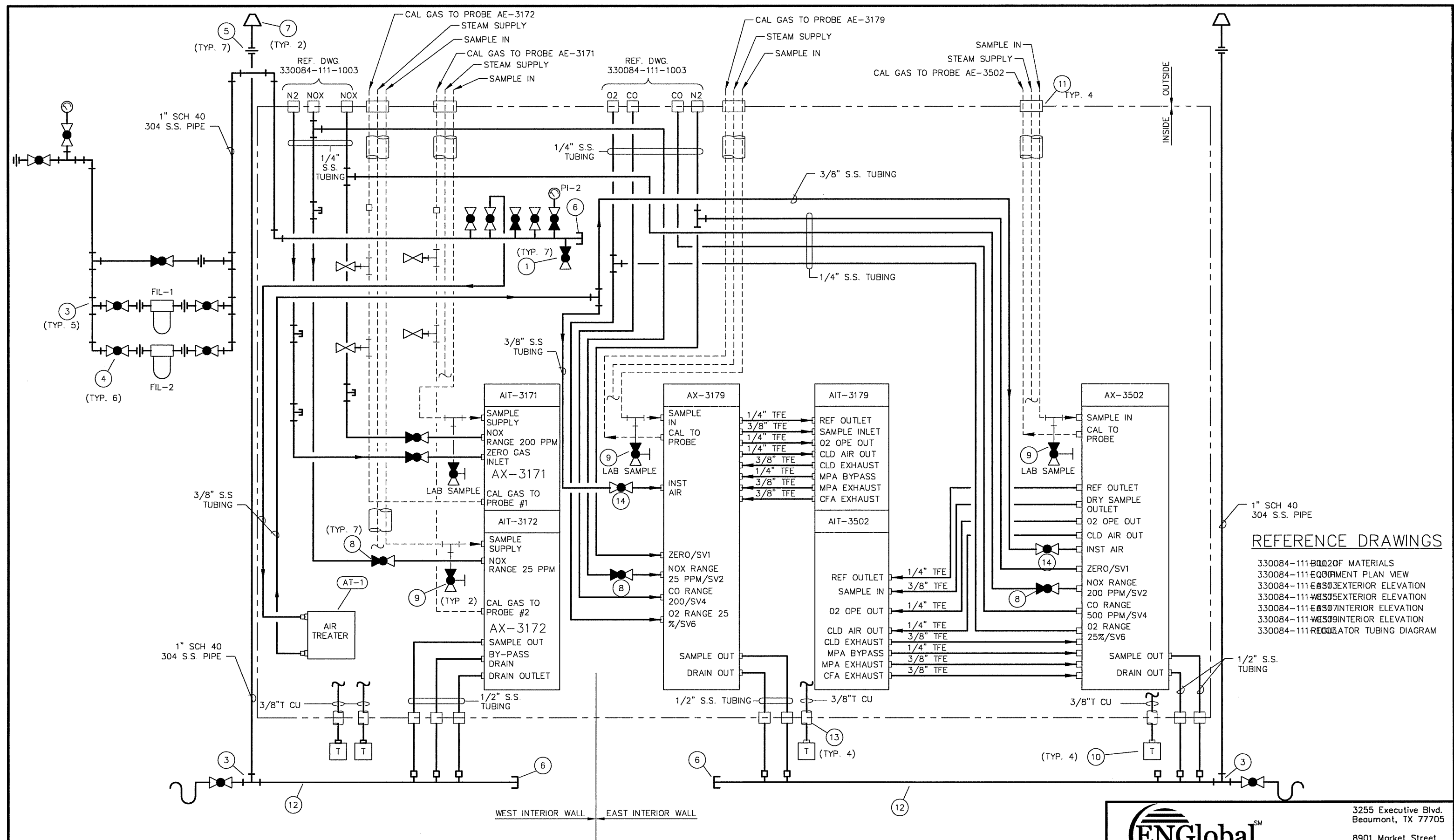
3255 Executive Blvd.
Beaumont, TX 77705

8901 Market Street
Houston, TX 77029

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2	ISSUED AS BUILT	RCB					6/29/04
1	REVISED PER CUSTOMER COMMENTS	RCB					4/26/04
0	ISSUED FOR CONSTRUCTION	RCB	TJE	JS			4/12/04
REV	REVISION DESCRIPTION	CADD	CHK'D.	APPR.	APPR.	APPR.	DATE

PROJECT	330084	KBR/MOTIVA PORT ARTHUR REFINERY HRSG-37 CEMS/B-35 CEMS/SCR NOx SIMPLIFIED PIPING & INSTRUMENT DIAGRAM AIT-3502	REV
SCALE	NONE		
FILE	330084-111-0102.A.dwg	DWG #	330084-111-0102
			2



REFERENCE DRAWINGS

330084-111EQ202 OF MATERIALS
330084-111EQ303 EXTERIOR ELEVATION
330084-111EQ305 EXTERIOR ELEVATION
330084-111EQ307 INTERIOR ELEVATION
330084-111EQ309 INTERIOR ELEVATION
330084-111EQ310 REGULATOR TUBING DIAGRAM



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Houston, TX 77029
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2	ISSUED AS BUILT	RCB				6/29/04	PROJECT	KBR/MOTIVA/PORT ARTHUR REFINERY			
1	REVISED PER CUSTOMER COMMENTS	RCB				4/26/04	330084	HRSG-37 CEMS/B-35 CEMS/SCR NOX			
0	ISSUED FOR CONSTRUCTION	RCB			TJE	4/21/04	SCALE	PIPING & TUBING DIAGRAM			
REV	REVISION DESCRIPTION	CADD	CHK'D.	APPR.	APPR.	DATE	NONE	AIT-3171, AIT-3172, AIT-3179, AIT-3502			REV
							FILE	330084-111-1001.0.dwg	DWG #	330084-111-1001	2

File: M:\PROJECTS\2003\PROJECTS\330084\100 Product\110 AS-0929\Working\111 Drawings\330084-111-1001.0.dwg

User: RHONDA.BOX

Date: Jul 02, 2004 - 10:51am

D

C

B

A

PROCESS PIPING SYMBOLS
VALVES

- GATE VALVE, NORMALLY OPEN
- GATE VALVE, NORMALLY CLOSED
- BUTTERFLY VALVE
- CHECK VALVE
- BALL VALVE, NORMALLY OPEN
- BALL VALVE, NORMALLY CLOSED
- GLOBE VALVE, NORMALLY CLOSED
- GLOBE VALVE, NORMALLY OPEN
- GLOBE VALVE, LOCKED THREAD
- DIAPHRAM VALVE, NORMALLY OPEN
- DIAPHRAM VALVE, NORMALLY CLOSED
- PLUG VALVE, NORMALLY OPEN
- PLUG VALVE, NORMALLY CLOSED
- NEEDLE VALVE, NORMALLY OPEN
- NEEDLE VALVE, NORMALLY CLOSED
- PINCH VALVE
- STOPCHECK VALVE, NORMALLY OPEN
- STOPCHECK VALVE, NORMALLY CLOSED
- 3-WAY VALVE NORMALLY OPEN
- 3-WAY VALVE MID CLOSED
- 3-WAY VALVE SIDE CLOSED
- 4-WAY VALVE, NORMALLY OPEN
- 3-WAY PLUG VALVE
- 4-WAY PLUG VALVE
- ANGLE VALVE, NORMALLY OPEN
- ANGLE VALVE / SAFETY VALVE, NORMALLY CLOSED
- ANGLE CHECK VALVE, NORMALLY OPEN
- ANGLE CHECK VALVE, NORMALLY CLOSED

VALVE OPERATOR SYMBOLS

- DIAPHRAM OPERATOR
- HAND OPERATOR
- PRESSURE RELIEF
- MOTOR OPERATOR
- ELECTRONIC TRANSDUCER
- SELF REGULATING OPERATOR
- DIGITAL OPERATOR
- SOLENOID OPERATOR
- PRESSURE BALANCE OPERATOR
- CYLINDER OPERATOR
- EXTERNAL REGULATING OPERATOR
- DIAPHRAGM W/ HAND OPERATOR

PROCESS PIPING SYMBOLS
FITTINGS

- CONCENTRIC REDUCER
- ECCENTRIC REDUCER
- WELDED CAP
- SCREWED CAP
- HOSE CONNECTOR
- QUICK DISCONNECT
- BREATHAR CAP
- FLANGE
- UNION
- LOOP SEAL
- VENT
- REMOVABLE SPOOL
- FLEXIBLE CONNECTOR
- SPECTACLE BLIND
- BASKET STRAINER
- TWIN STRAINER
- Y STRAINER
- STARTUP STRAINER
- SCREEN STRAINER
- TRAP
- FILTER
- EXPANSION JOINT
- SAMPLE COOLER
- EJECTOR
- EXHAUST
- MIXING TEE

ELECTRICAL LINES

- ELECTRICAL SIGNAL
- ELECTRICAL BINARY SIGNAL
- DIGITAL OR SOFTWARE LINK IN DCS (BY OTHERS)

INSTRUMENT LINES

- PROCESS
- PNEUMATIC SIGNAL

LINE LEGEND

- PIPING DESIGNED BY VPI, NOT IN VPI SCOPE OF SUPPLY
- PRIMARY - STEAM
- SECONDARY - WATER
- DRAIN
- PIPING AND COMPONENTS NOT IN VPI SCOPE OF SUPPLY
- FLOW ARROW
- LINE NUMBER BREAK
- TERMINAL POINT (INTERFACE POINT)
- SHEET CONNECTOR ONE DIRECTION
- SHEET CONNECTOR BOTH DIRECTIONS

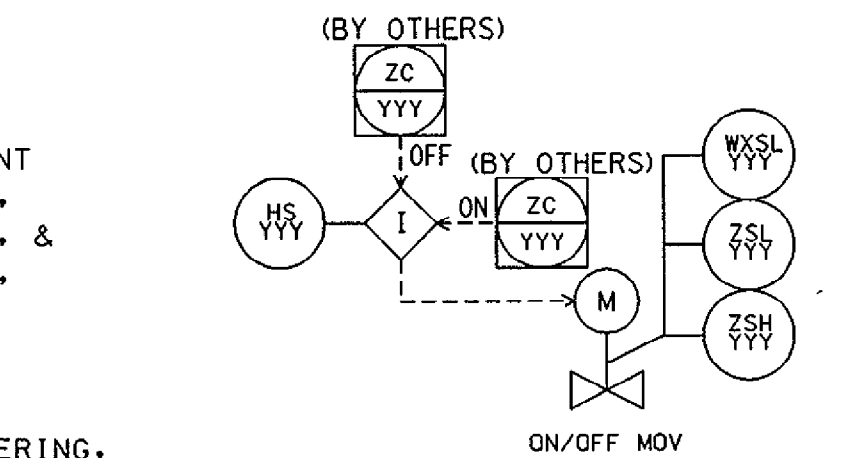
INSTRUMENT SYMBOLS

- MOUNTED ON MAIN PANEL
- LOCALLY MOUNTED IN FIELD
- TEST INSTRUMENT BY CUSTOMER
- MOUNTED ON MAIN PANEL W/SHARED DISPLAY/CONTROL, I.e. DCS
- ORIFICE PLATE
- ORIFICE PLATE IN QUICK CHANGE FITTING
- PITOT TUBE
- DOUBLE PITOT TUBE
- VENTURI FLOW ELEMENT
- CONDENSATE POT
- FLUME
- SIPHON
- WEIR
- FLOW TURBINE PRIMARY ELEMENT
- POSITIVE DISPLACEMENT
- FLOW STRAIGHTENING VANES
- ROTAMETER TYPE FLOW INDICATOR

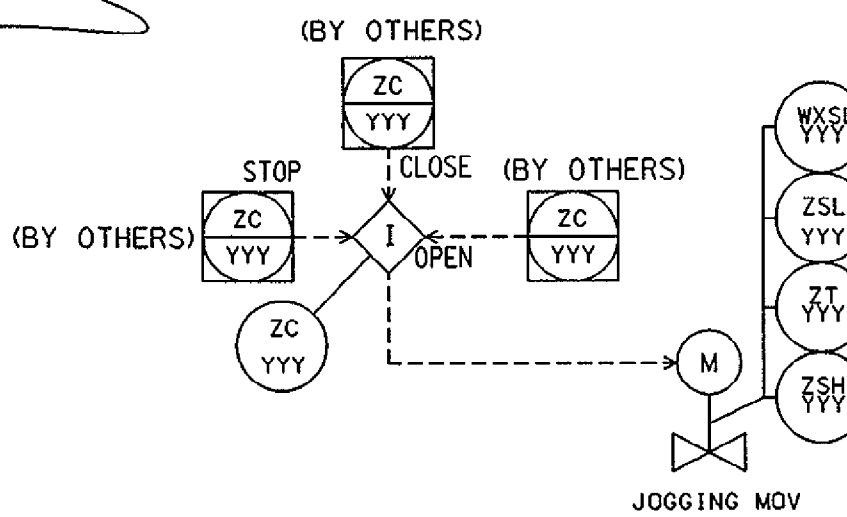
EQUIPMENT SYMBOLS

- MODULE SINGLE ROW
- MODULE 2 ROW
- MODULE 3 ROW
- DISTRIBUTION GRID
- BURNER
- AMMONIA INJECTION GRID
- EXPANSION JOINT
- SILENCER
- VENT STACK
- CENTRIFUGAL PUMP
- PRESSURE VESSEL (STEAM DRUM)
- NOZZLE DOUBLE
- NOZZLE SINGLE
- AIR COMPRESSOR
- VACUUM PUMP
- HAND PUMP
- ROTARY PUMP
- SOOTBLOWER
- MIXER
- CANNED PUMP
- WET PIT PUMP
- SUMP PUMP
- POSITIVE DISP PUMP
- PLATE HEAT EXCHANGER
- STRAIGHT TUBE HEAT EXCHANGER
- U-TUBE HEAT EXCHANGER
- BOILER FEED PUMP
- CONE ROOF TANK
- DOUBLE TURBINE
- SINGLE TURBINE

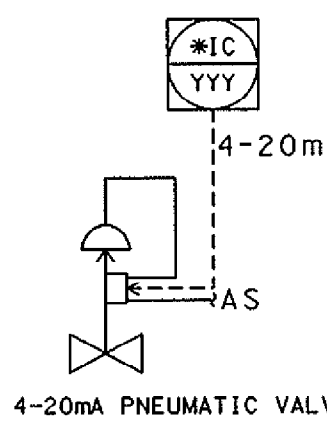
- ON MOTOR OPERATED VALVES INSTRUMENT TAGGING IDENTIFICATION LETTERING, "W" IS DEFINED AS "FORCE" PER ISA, & "X" IS DEFINED "ANGULAR DISTANCE". "WX" IS DEFINED AS A NEW MEASURED VARIABLE "TORQUE"
- ON PNEUMATIC VALVES INSTRUMENT "IC" TAGGING IDENTIFICATION LETTERING, "IC" IS DEFINED AS "PRESS (P)", "LEVEL (L)", OR "TEMP (T)" CORRESPONDING TO CONTROL FUNCTION PER CONTROL VALVE TAG NAME.
- ALL TRANSMITTERS TO BE MOUNTED ON MANIFOLDS WITH FACILITY TO VENT AND/OR BLEED.



DETAIL A



DETAIL B



DETAIL C

LINE NUMBER INFO:

- LXXX-BCD-Y
- LXXX=LINE NUMBER IDENTIFIER
- XXX- 001-099 HP SYSTEM
- 101-199 IP SYSTEM
- 201-299 LP SYSTEM
- 301-399 BLOWDOWN TANK SYSTEM
- 401-499 REHEATER SYSTEM
- 601-699 EXHAUST GAS SYSTEM
- B=PRESSURE CLASS IDENTIFIER (FLANGE RATING)
- A=4500#
- B=2500#
- C=1500#
- D=900#
- E=600#
- G=300#
- H=150#
- X=NO CLASS REQUIRED (NON PRESSURE PART)

C=MATERIAL TYPE IDENTIFIER

- A=ALLOY
- B=CARBON STEEL
- C=STAINLESS STEEL

D=CODE CLASS IDENTIFIER

- K=ASME SECTION I (SECTION I ADMINISTRATIVE & TECHNICAL JURISDICTION - BOILER PROPER)
- H=ASME SECTION I (SECTION I ADMINISTRATIVE JURISDICTION, ASME B31.1 TECHNICAL JURISDICTION - BOILER EXTERNAL)
- D=ASME B31.1
- X=NON-CODE

Y=NOMINAL PIPE SIZE

VALVE NUMBER INFO:

- V-YYYY
- V-YYYY=VALVE NUMBER IDENTIFIER
- YYYY = WILL CORRESPOND TO LINE NO. VALVE IS PART OF.

CODE BREAKS

- ASME SECT. I BOILER PROPER (BP)
- ASME SECT. I / B31.1 BOILER EXTERNAL PIPING (BEP)
- ASME B31.1 NON-BOILER EXTERNAL PIPING (NBEP)

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- ACCEPTED WITH COMMENTS (AC-C) ☐
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DIVISION Mach SIGNED JR
DATE 29Mar04

REFERENCE DRAWINGS:

- D-415-040 P&ID LEGEND
- D-415-041 P&ID HRSG EXHAUST GAS SYSTEM
- D-415-042 P&ID HRSG HP SYSTEM
- D-415-043 PROCESS FLOW DIAGRAM
- D-415-044 P&ID SCR SYSTEM
- 415-D001 LINE LIST
- 415-D002 TERMINAL POINT LIST
- 415-D003 VALVE LIST
- 415-D004 INSTRUMENT LIST

Rev.	Description	Date	Drawn	Chkd.	Appr.
5					
4	ADDED NOTE	2/24/03	ACU		
3	REVISED INSTRUMENT TAGS PER CUSTOMER COMMENTS (EOM15-015)	01/13/04	TLN	PH	KH
2	ADDED PNEUMATIC LINE & REV. ANGLE VALVE TAG (EOM15-008)	10/13/03	WSP	PH	KH
1	ADDED ELECTRICAL LINE LEGEND (EOM15-005)	8/01/03	WSP	PH	KH
0	FIRST ISSUE	7/28/03	WSP	B.W.	PH MF

Revisions

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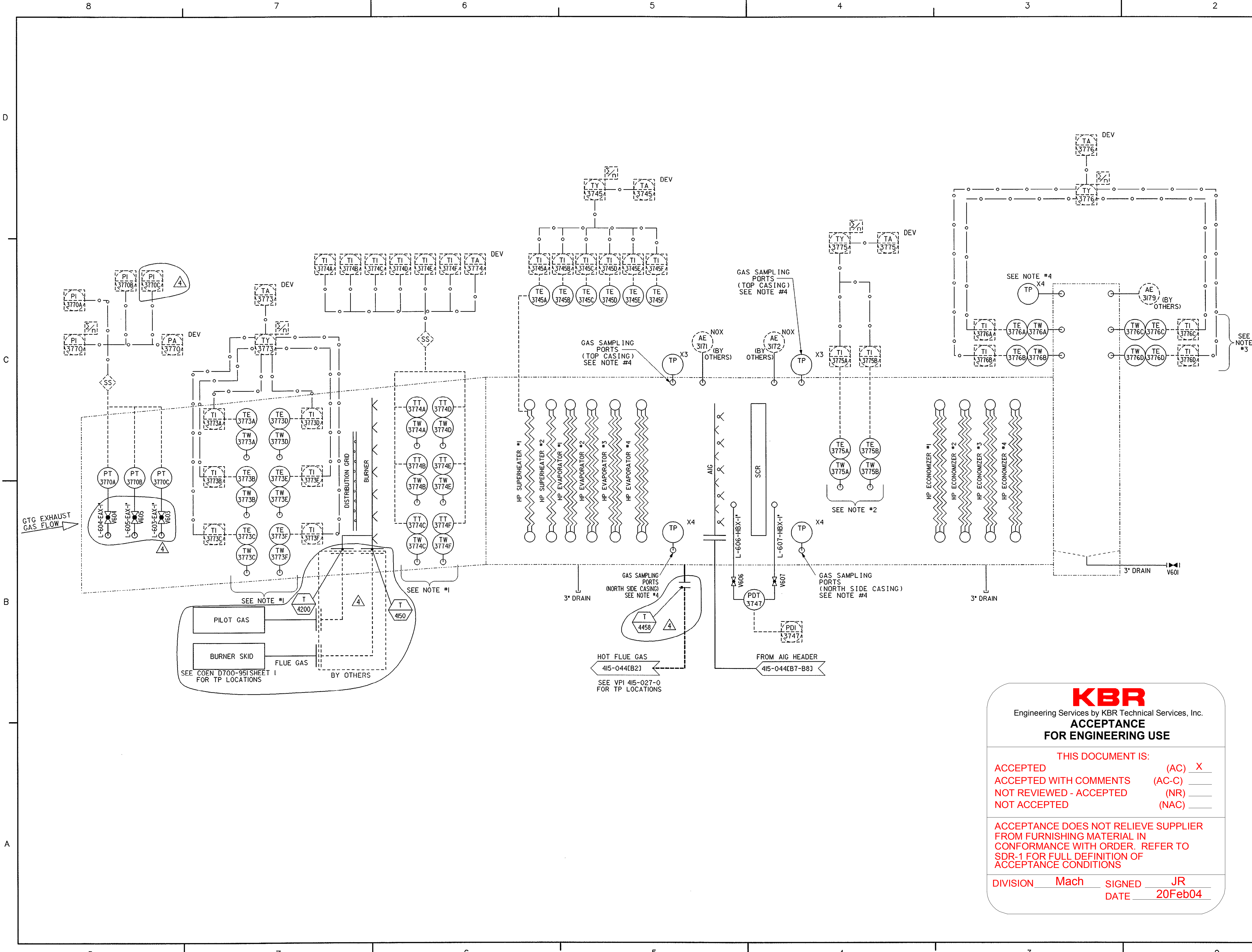
Vogt Power Job No. 17415
FOR
KBR Job No. S512
P.O. S512P02-1MU43001
PS 3 HRSG No. 31 Project
Motive Enterprises LLC
Port Arthur Refinery
Powerhouse No. 3
724BLR37 - HRSG Package

Scale: NONE

Title: P&ID LEGEND

Vogt Power International 4000 DuPont Circle
Louisville, Ky. 40207 USA
www.VogtPower.com

Internal Drawing Status	Size	Drawing No.	Rev.
FOR RECORD	D	415-040-0	04



- NOTES:
- TE/TW 3773 A-C & 3774 A-C LOCATED ON HRSG DUCT SOUTH SIDE.
TE/TW 3773 D-F & 3774 D-F LOCATED ON HRSG DUCT NORTH SIDE.
 - TE/TW 3775 A LOCATED ON HRSG DUCT SOUTH SIDE.
TE/TW 3775 B LOCATED ON HRSG DUCT NORTH SIDE.
 - TE/TW 3776 A-D LOCATED @ 90° INTERVALS ON HRSG OUTLET STACK.
 - PROVISIONS BY VOGT. INSTRUMENTS BY OTHERS

- REFERENCE DRAWINGS:
- | | |
|-----------|------------------------------|
| D-415-040 | P&ID LEGEND |
| D-415-041 | P&ID HRSG EXHAUST GAS SYSTEM |
| D-415-042 | P&ID HRSG HP SYSTEM |
| D-415-043 | PROCESS FLOW DIAGRAM |
| D-415-044 | P&ID SCR SYSTEM |
| 415-0001 | LINE LIST |
| 415-0002 | TERMINAL POINT LIST |
| 415-0003 | VALVE LIST |
| 415-0004 | INSTRUMENT LIST |

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DIVISION Mach

SIGNED JR

DATE 20Feb04

5						
4	ADDED INSTRUMENT & TERMINAL POINTS PER CUSTOMER COMMENT	02/14/04	FRY	TF		SG
3	ADDED INSTRUMENT PER CUSTOMER COMMENT (ECON415-015)	01/13/04	WSP	PH		KH
2	REV. INSTRUMENT TAGS PER CUSTOMER COMMENT (ECON415-008)	10/13/03	WSP	PH		KH
1	ADDED INSTRUMENT TAGS PER CUSTOMER COMMENT (ECON415-005)	8/01/03	WSP	PH		KH
0	FIRST ISSUE	7/28/03	WSP	B.W.	PH	MF
Rev.	Description	Date	Drawn	Chkd.	Chkd.	Appr.
Revisions						
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Voigt-Power Job No. 17415 for KBR Job No. S512 P.O. S512P02-1MU43001 PS 3 HRSG No. 37 Project Motiva Enterprises LLC Port Arthur Refinery Powerhouse No. 3 724BLR37 - HRSG Package						
Title P&ID HRSG EXHAUST GAS SYSTEM						
VoigtPower International 4000 DuPont Circle Louisville, Ky. 40207 USA www.VoigtPower.com						
Internal Drawing Status FOR RECORD	Size D	Drawing No. 415-041-0	Rev. 04			

KBR

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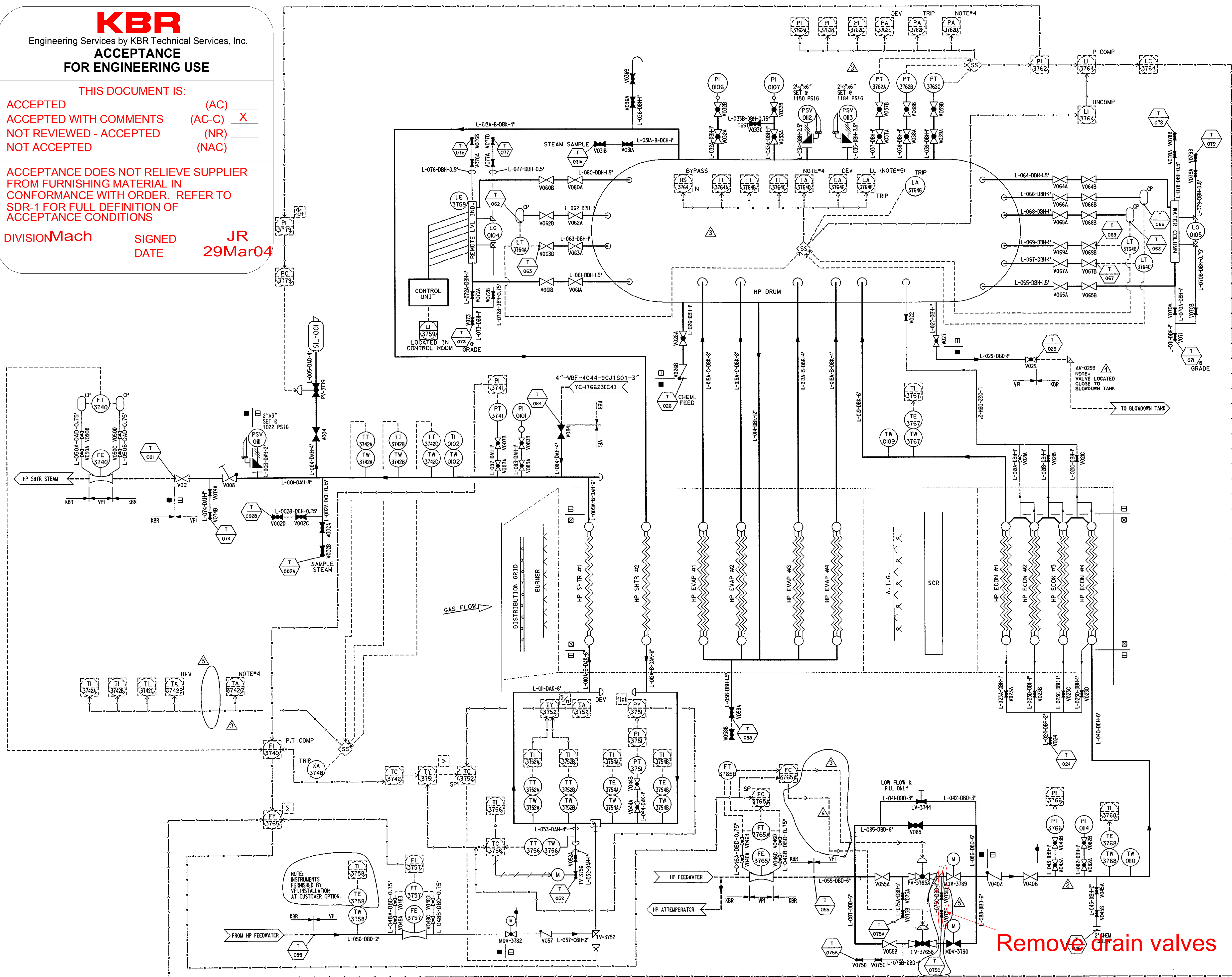
**ACCEPTANCE
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ACCEPTED (AC) _____
ACCEPTED WITH COMMENTS (AC-C) **X**
NOT REVIEWED - ACCEPTED (NR) _____
NOT ACCEPTED (NAC) _____

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CONFORMANCE WITH ORDER. REFER TO
SDR-1 FOR FULL DEFINITION OF
ACCEPTANCE CONDITIONS

DIVISION **Mach** SIGNED **JR**
DATE **29Mar04**



ON/OFF MOTOR OPERATED VALVE (DETAIL A)

VALVE NUMBER	INSTRUMENT NUMBER
MOV-3782	3782
MOV-3789	3789
MOV-3790	3790

JOGGING MOTOR OPERATED VALVE (DETAIL B)

VALVE NUMBER	INSTRUMENT NUMBER
TV-3756	3756

4-20mA PNEUMATIC VALVE (DETAIL C)

VALVE NUMBER	INSTRUMENT NUMBER
FV-3765A	3765A
FV-3765B	3765B
PV-3779	3779
TV-3752	3752

- NOTE:
- SEE DRAWING 415-040-0 FOR VALVE OPERATOR DETAILS.
 - DGS & SAFETY SYSTEM (SS) SUPPLIED BY OTHERS.
 - ALL PSV VALVES & VENT STACKS TO HAVE DRAINS.
 - ALARMS WHEN ANY ONE OF THREE TRANSMITTERS ARE IN SHUTDOWN STATE.
 - TWO OUT OF THREE VOTING FOR TRIP.

REFERENCE DRAWINGS:

D-415-040	P&ID LEGEND
D-415-041	P&ID HRSG EXHAUST GAS SYSTEM
D-415-042	P&ID HRSG HP SYSTEM
D-415-043	PROCESS FLOW DIAGRAM
D-415-044	P&ID SCR SYSTEM
415-0001	LINE LIST
415-0002	TERMINAL POINT LIST
415-0003	VALVE LIST
415-0004	INSTRUMENT LIST

REV.	INSTRUMENT TAGS & ADDED INSTRUMENTS PER CUST. COMMENTS (ECN#15-022)	2/24/04	ACU		
5	ADDED AV-029B IN CUSTOMERS LINE	01/26/04	FRY	PH	KH
4	REV. INSTRUMENT TAGS & ADDED INSTRUMENTS PER CUST. COMMENTS (ECN#15-016)	01/13/04	TLH	PH	KH
3	REV. INSTRUMENT TAGS & ADDED INSTRUMENTS PER CUST. COMMENTS (ECN#15-008)	10/13/03	WSP	PH	KH
2	ADDED INSTRUMENT TAGS & ADDED INSTRUMENTS PER CUST. COMMENTS (ECN#15-005)	8/01/03	WSP	PH	KH
1	FIRST ISSUE	7/28/03	WSP	B.W.	PH
0	Description	Date	Drawn	Chkd.	Appr.

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Vogt Power Job No. 17415
KBR Job No. 5512
P.O. 5512P02-1MUA3001
PS 3 HRSG No. 37 Project
Motiva Enterprises LLC
Port Arthur Refinery
Powerhouse No. 3
724BLR37 - HRSG Package

Title
P&ID HRSG
HP SYSTEM

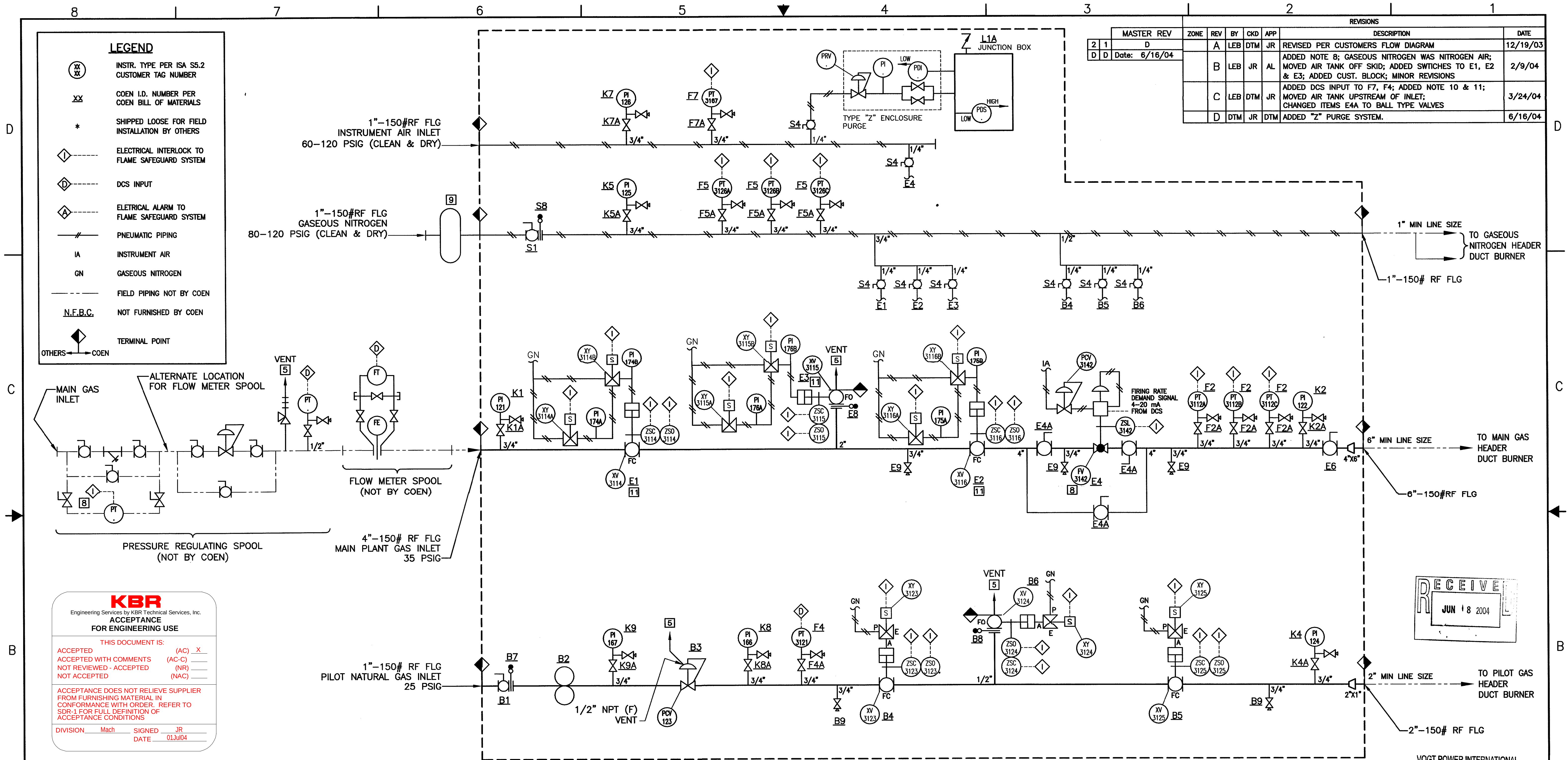
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4000 DuPont Circle
Louisville, Ky. 40207 USA
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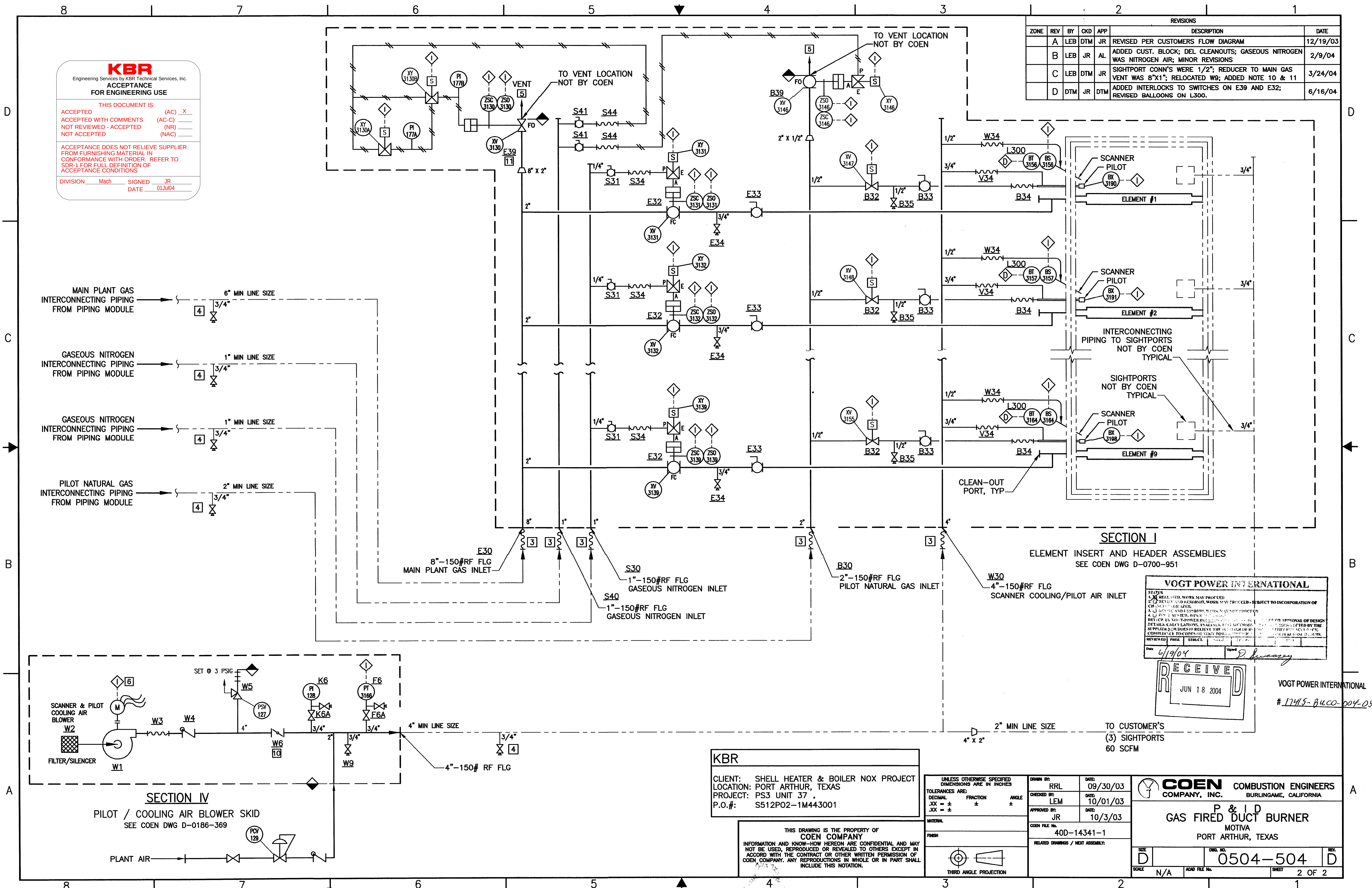
Internal Drawing Status
FOR RECORD

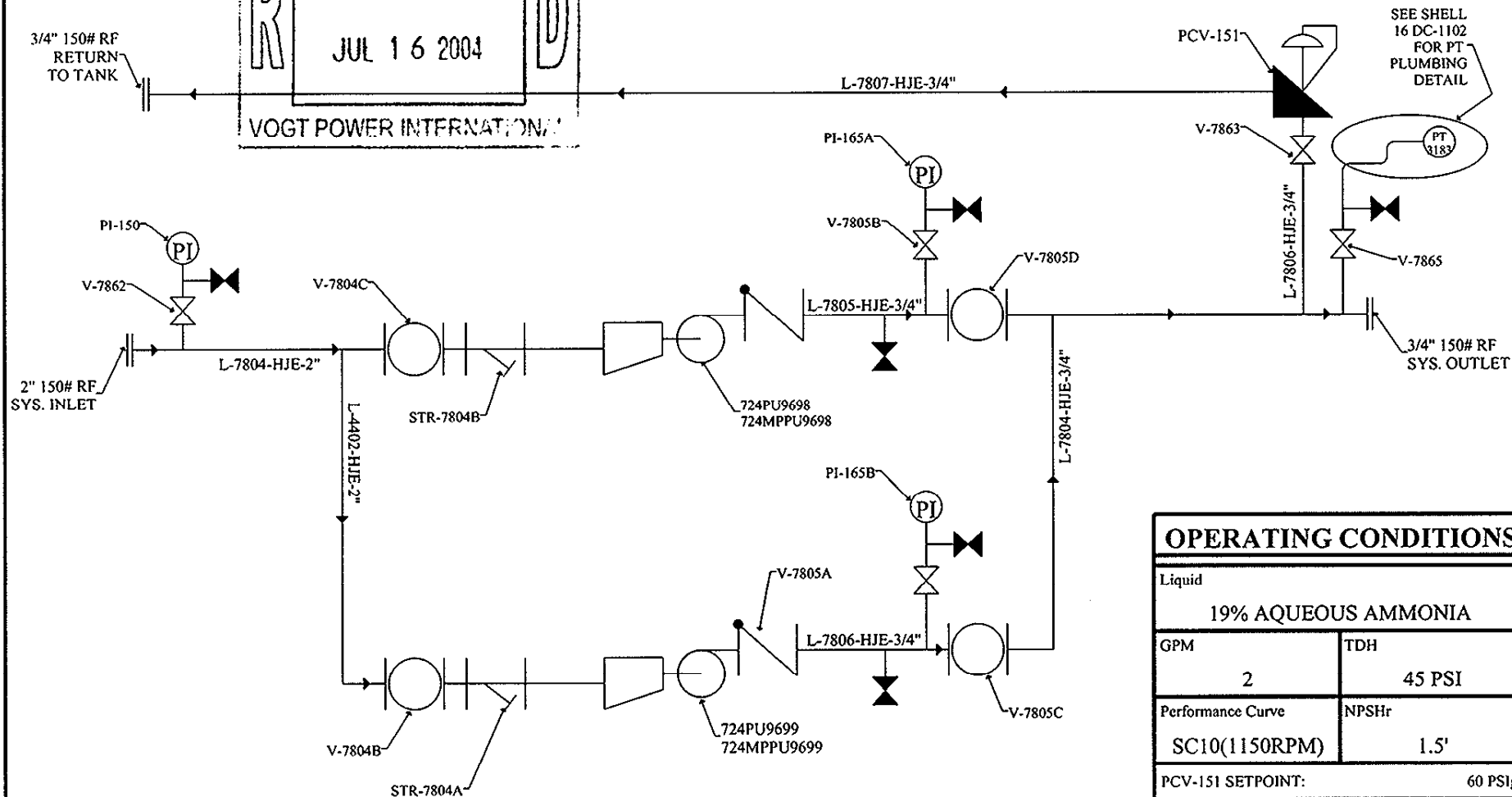
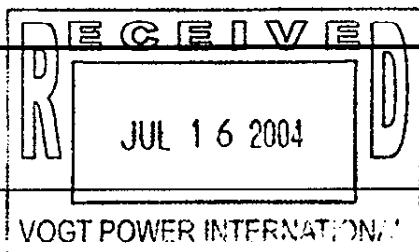
Size
D

Drawing No.
415-042-0

Rev.
05







OPERATING CONDITIONS			
Liquid			
19% AQUEOUS AMMONIA			
GPM	2	TDH	45 PSI
Performance Curve	SC10(1150RPM)	NPSHr	1.5'
PCV-151 SETPOINT:		60 PSig	

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ACCEPTANCE FOR ENGINEERING USE

THIS DOCUMENT IS:

ACCEPTED	(AC)	X
ACCEPTED WITH COMMENTS	(AC-C)	
NOT REVIEWED - ACCEPTED	(NR)	
NOT ACCEPTED	(NAC)	

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DIVISION Mach SIGNED JR DATE 27Jul04

VOGT POWER INTERNATIONAL

#17415-SRML-009-05

REV #	DESCRIPTION	DATE
Δ	CHANGED OR ADDED	2/9/04
Δ	CHANGED CONSISTENT W/CUST DOCS	2/11/04
Δ	CHANGED OR ADDED	3/5/04
Δ	CHG PER CUST COMMENTS 6/28/04	7/16/04

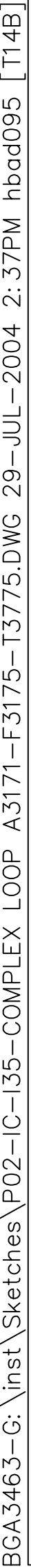
DRAWN BY:	RRB
CHECKED BY:	RRB
FOR:	VOGT-NEM
PO#:	8622881
VOGT JOB #:	17415
RAM JOB NUMBER	0326640
DRAWING NUMBER	0326640-PID



RA MUELLER INC. FLUID HANDLING EQUIPMENT

KBR Job No. S512
P.O. S512P02-1MJ43001
PS 3 HRSG No. 37
Project Motiva Enterprises LLC
Port Arthur Refinery
Powerhouse No. 3
724BLR37 - HSRG Package

SCALE:	NTS
DATE:	11/26/03



LEGEND

XX INSTR. TYPE PER ISA S5.2
CUSTOMER TAG NUMBER

XX COEN I.D. NUMBER PER
COEN BILL OF MATERIALS

* SHIPPED LOOSE FOR FIELD
INSTALLATION BY OTHERS

◇ ELECTRICAL INTERLOCK TO
FLAME SAFEGUARD SYSTEM

◇ ELECTRICAL ALARM TO
FLAME SAFEGUARD SYSTEM

— PNEUMATIC PIPING

IA INSTRUMENT AIR

GN GASEOUS NITROGEN

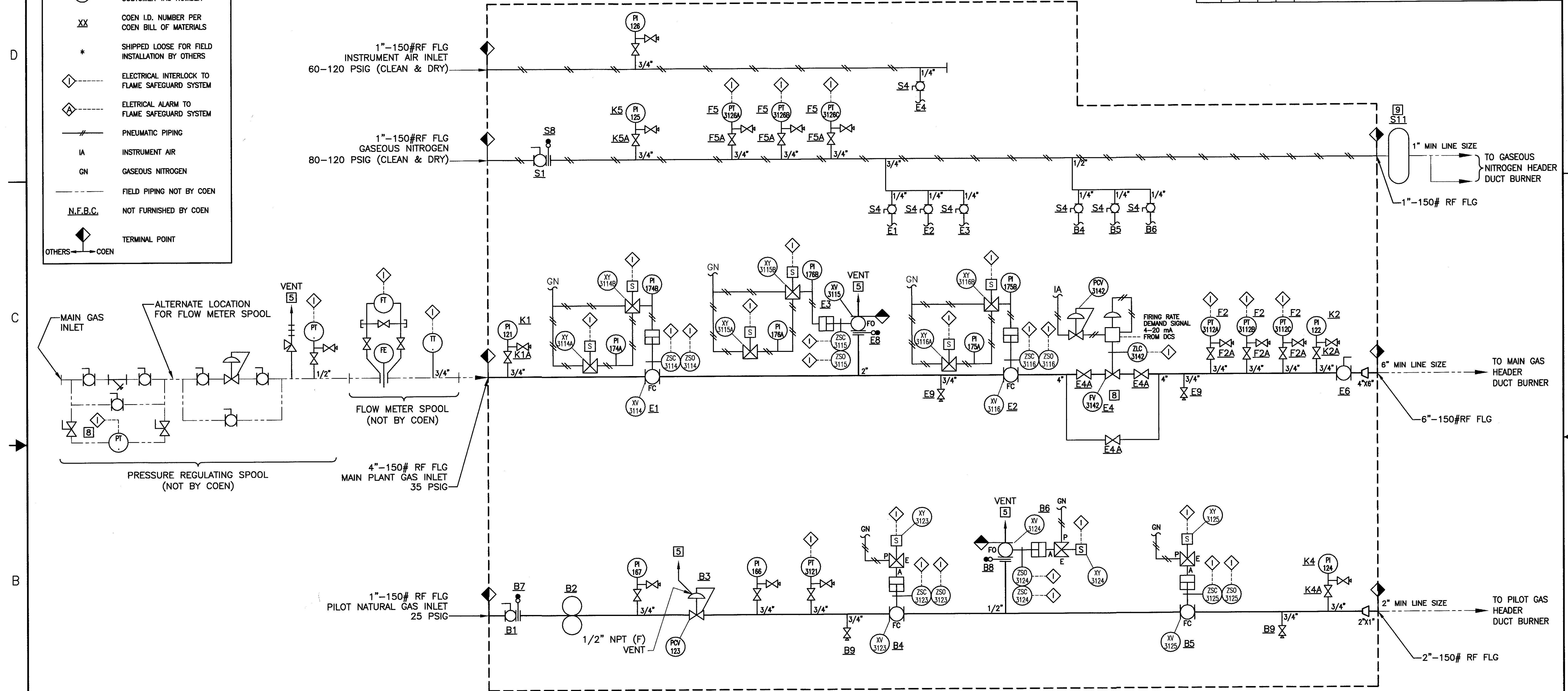
--- FIELD PIPING NOT BY COEN

N.F.B.C. NOT FURNISHED BY COEN

◇ TERMINAL POINT

OTHERS → COEN

		REVISIONS								
MASTER REV		ZONE	REV	BY	CKD	APP	DESCRIPTION		DATE	
2	1		A	LEB	DTM	JR	REVISED PER CUSTOMERS FLOW DIAGRAM		12/19/03	
B	B	Date:		B	LEB	<i>jr</i>	ADDED NOTE 8; GASEOUS NITROGEN WAS NITROGEN AIR; MOVED AIR TANK OFF SKID; ADDED SWITCHES TO E1, E2 & E3; ADDED CUST. BLOCK; MINOR REVISIONS		2/9/04	



- NOTES:**
- SEE COEN BILL OF MATERIALS FOR ITEM DESCRIPTIONS, SYSTEM SPECIFICATIONS AND ASSOCIATED DRAWINGS.
 - PIPING AND DUCTING BETWEEN ALL COEN EQUIPMENT IS NOT BY COEN. COEN RECOMMENDS LOCATING THE PIPING MODULE AND BLOWER SKID WITHIN 20 FEET OF BURNER PIPING INLETS. IN ADDITION, THE PRESSURE DROP IN THE MAIN GAS LINE BETWEEN THE PIPING MODULE AND THE DUCT BURNER INLET MAY NOT EXCEED 2.0 PSI.
 - ADEQUATE PROVISIONS MUST BE MADE IN THE INTERCONNECTING PIPING FROM PIPING MODULE OUTLETS TO BURNER PIPING INLETS TO AVOID STRESSES DUE TO THERMAL EXPANSION OR VIBRATION. INTERCONNECTING PIPING IS NOT BY COEN.
 - A DRIP LEG WILL BE INSTALLED AT THE LOWEST POINT IN ALL FIELD PIPING RUNS, NOT BY COEN.

- FUEL VENTS, INCLUDING PRV SPRING CHAMBER VENTS AND VENTS BETWEEN DOUBLE DIAPHRAGMS ON LIMIT SWITCHES MUST BE ROUTED TO AN APPROPRIATE LOCATION DEPENDING ON FLUID (GAS OR OIL), NOT BY COEN.
- PILOT / COOLING AIR BLOWER IS TO BE ON WHENEVER THE TURBINE IS RUNNING AND/OR THE DUCT TEMPERATURE IS GREATER THAN 150°F. CONTROLS AND MOTOR STARTERS ARE NOT BY COEN.
- ALL FIELD ERECTED PIPING BETWEEN AND UPSTREAM OF COEN'S PIPING MUST BE FLUSHED OUT AND CLEANED PRIOR TO START-UP, NOT BY COEN.
- FLOW CONTROL VALVE, E4 (FV3142), FAIL CLOSED WITH MECHANICAL DOWNSTOP FOR MINIMUM FLOW.
- NITROGEN SURGE TANK BY COEN. ALL ACCESSORIES (RELIEF VALVES, DRAIN VALVES, GAUGES, ETC) NOT BY COEN.

SECTION II
PIPING MODULE ASSEMBLY
SEE COEN DWG D-0186-368

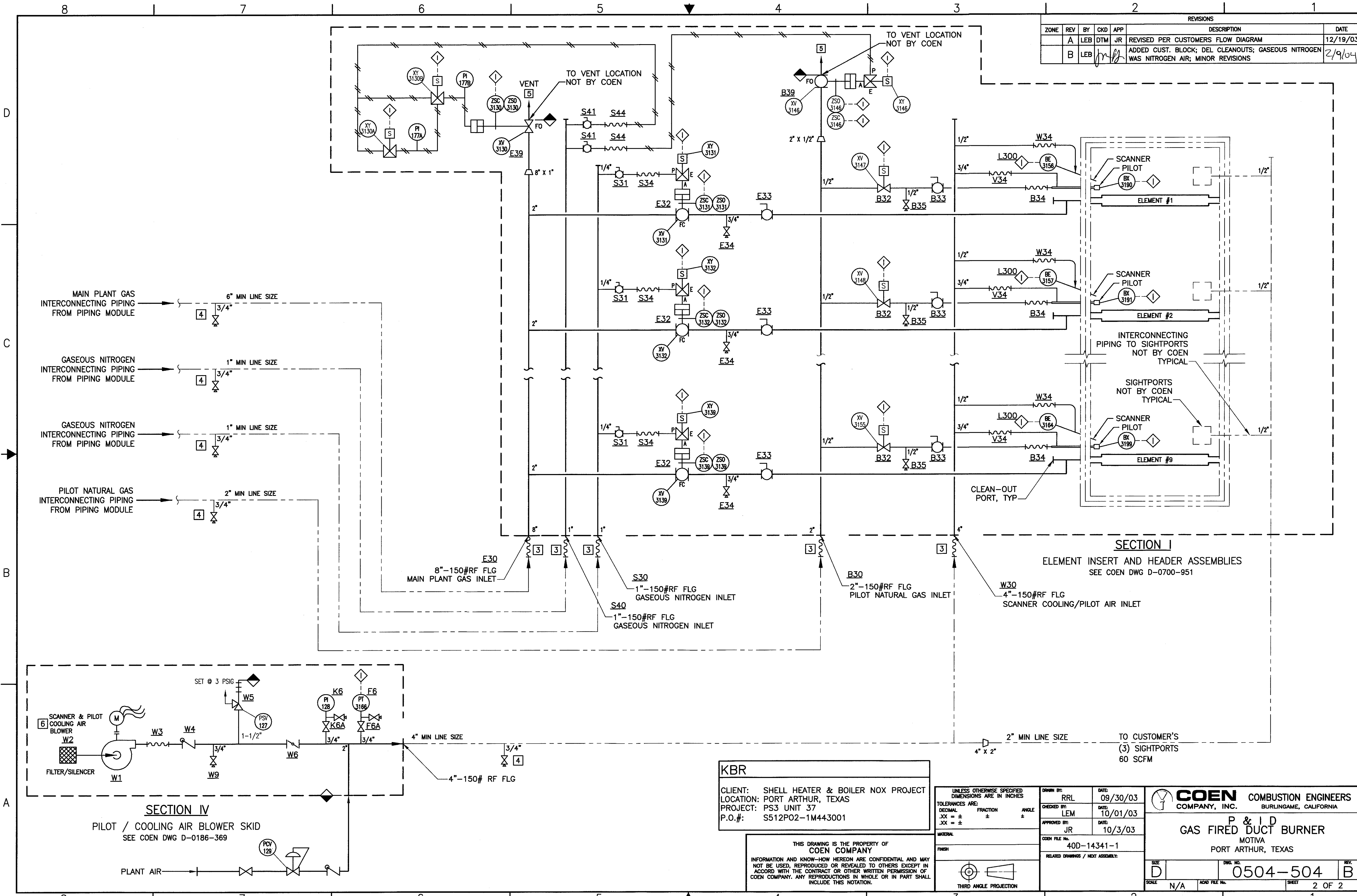
THIS DRAWING IS THE PROPERTY OF
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UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES			DRAWN BY: RRL	DATE: 09/30/03
TOLERANCES ARE:			CHECKED BY: LEM	DATE: 10/01/03
DECIMAL XX = ±	FRACTION ±	ANGLE ±	APPROVED BY: JR	DATE: 10/3/03
MATERIAL			COEN FILE NO. 40D-14341-1	
FINISH			RELATED DRAWINGS / NEXT ASSEMBLY:	
THIRD ANGLE PROJECTION				

KBR

CLIENT: SHELL HEATER & BOILER NOX PROJECT
LOCATION: PORT ARTHUR, TEXAS
PROJECT: PS3 UNIT 37
P.O.#: S512P02-1M443001

COEN COMBUSTION ENGINEERS COMPANY, INC. BURLINGAME, CALIFORNIA	
P & I D GAS FIRED DUCT BURNER MOTIVA PORT ARTHUR, TEXAS	
SIZE D	DWG. NO. 0504-504
SCALE N/A	SHEET 1 OF 2



REVISIONS						DATE
ZONE	REV	BY	CKD	APP	DESCRIPTION	
	A	LEB	DTM	JR	REVISED PER CUSTOMERS FLOW DIAGRAM	12/19/03
	B	LEB			ADDED CUST. BLOCK; DEL CLEANOUTS; GASEOUS NITROGEN WAS NITROGEN AIR; MINOR REVISIONS	2/9/04

KBR
CLIENT: SHELL HEATER & BOILER NOX PROJECT
LOCATION: PORT ARTHUR, TEXAS
PROJECT: PS3 UNIT 37
P.O.#: S512P02-1M443001

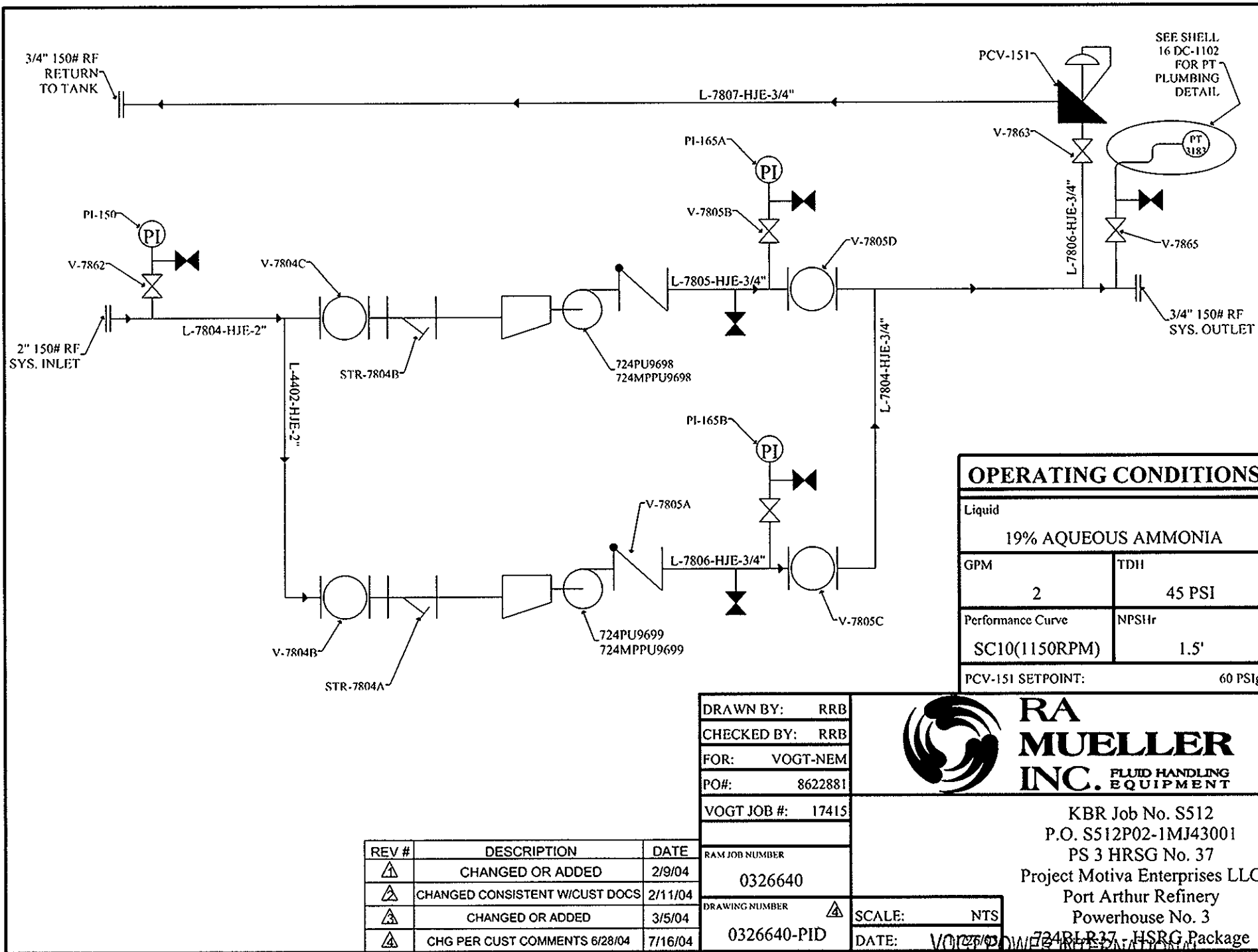
THIS DRAWING IS THE PROPERTY OF
COEN COMPANY
INFORMATION AND KNOW-HOW HEREON ARE CONFIDENTIAL AND MAY NOT BE USED, REPRODUCED OR REVEALED TO OTHERS EXCEPT IN ACCORD WITH THE CONTRACT OR OTHER WRITTEN PERMISSION OF COEN COMPANY. ANY REPRODUCTIONS IN WHOLE OR IN PART SHALL INCLUDE THIS NOTATION.

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES			DRAWN BY: RRL	DATE: 09/30/03
TOLERANCES ARE:			CHECKED BY: LEM	DATE: 10/01/03
DECIMAL	FRACTION	ANGLE	APPROVED BY: JR	DATE: 10/3/03
.XX = ±	±	±	COEN FILE NO. 40D-14341-1	
.XX = ±			RELATED DRAWINGS / NEXT ASSEMBLY:	
MATERIAL			FINISH	
			THIRD ANGLE PROJECTION	

**COEN** COMBUSTION ENGINEERS
COMPANY, INC. BURLINGAME, CALIFORNIA

P & I D
GAS FIRED DUCT BURNER
MOTIVA
PORT ARTHUR, TEXAS

SIZE	DWG. NO.	REV.
D	0504-504	B
SCALE	ACID FILE NO.	SHEET
N/A		2 OF 2



OPERATING CONDITIONS	
Liquid	
19% AQUEOUS AMMONIA	
GPM	TDH
2	45 PSI
Performance Curve	NPSHr
SC10(1150RPM)	1.5'
PCV-151 SETPOINT: 60 PSIG	

DRAWN BY: RRB
 CHECKED BY: RRB
 FOR: VOGT-NEM
 PO#: 8622881
 VOGT JOB #: 17415



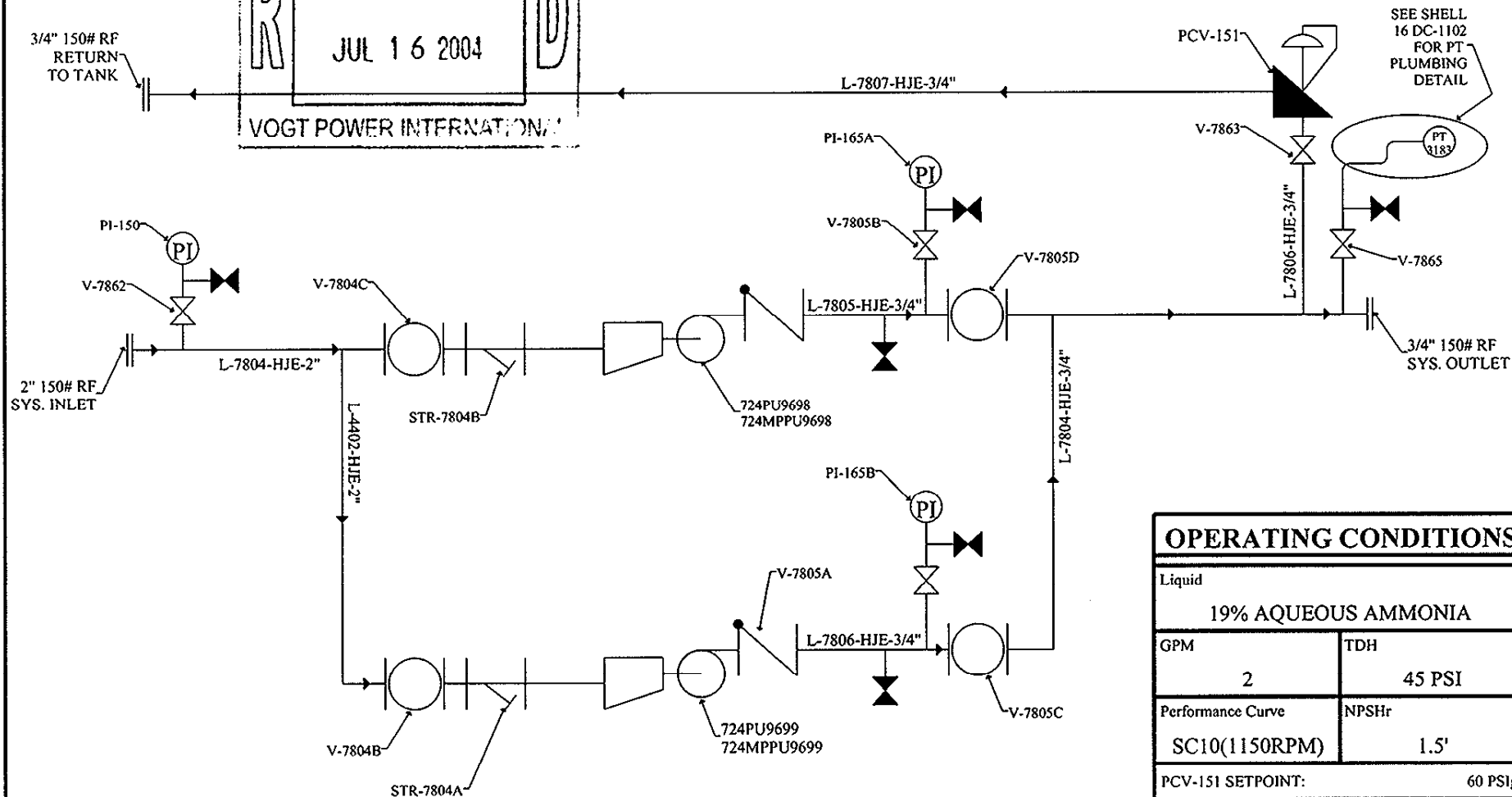
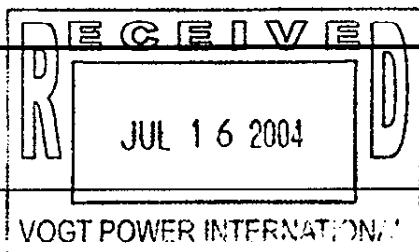
KBR Job No. S512
 P.O. S512P02-1MJ43001
 PS 3 HRSG No. 37
 Project Motiva Enterprises LLC
 Port Arthur Refinery
 Powerhouse No. 3
 724BLR37 - HRSG Package

REV #	DESCRIPTION	DATE
△	CHANGED OR ADDED	2/9/04
△	CHANGED CONSISTENT W/CUST DOCS	2/11/04
△	CHANGED OR ADDED	3/5/04
△	CHG PER CUST COMMENTS 6/28/04	7/16/04

RAM JOB NUMBER	0326640
DRAWING NUMBER	0326640-PID
SCALE:	NTS
DATE:	VOGT POWER INTERNATIONAL

17415-SRML-009-06

JUL 28 2004



OPERATING CONDITIONS			
Liquid			
19% AQUEOUS AMMONIA			
GPM	2	TDH	45 PSI
Performance Curve	SC10(1150RPM)	NPSHr	1.5'
PCV-151 SETPOINT:		60 PSig	

KBR
Engineering Services by KBR Technical Services, Inc.

ACCEPTANCE FOR ENGINEERING USE

THIS DOCUMENT IS:

ACCEPTED	(AC)	X
ACCEPTED WITH COMMENTS	(AC-C)	
NOT REVIEWED - ACCEPTED	(NR)	
NOT ACCEPTED	(NAC)	

ACCEPTANCE DOES NOT RELIEVE SUPPLIER FROM FURNISHING MATERIAL IN CONFORMANCE WITH ORDER. REFER TO SDR-1 FOR FULL DEFINITION OF ACCEPTANCE CONDITIONS

DIVISION Mach SIGNED JR DATE 27Jul04

VOGT POWER INTERNATIONAL

#17415-SRML-009-05

REV #	DESCRIPTION	DATE
Δ	CHANGED OR ADDED	2/9/04
Δ	CHANGED CONSISTENT W/CUST DOCS	2/11/04
Δ	CHANGED OR ADDED	3/5/04
Δ	CHG PER CUST COMMENTS 6/28/04	7/16/04

DRAWN BY:	RRB
CHECKED BY:	RRB
FOR:	VOGT-NEM
PO#:	8622881
VOGT JOB #:	17415
RAM JOB NUMBER	0326640
DRAWING NUMBER	0326640-PID

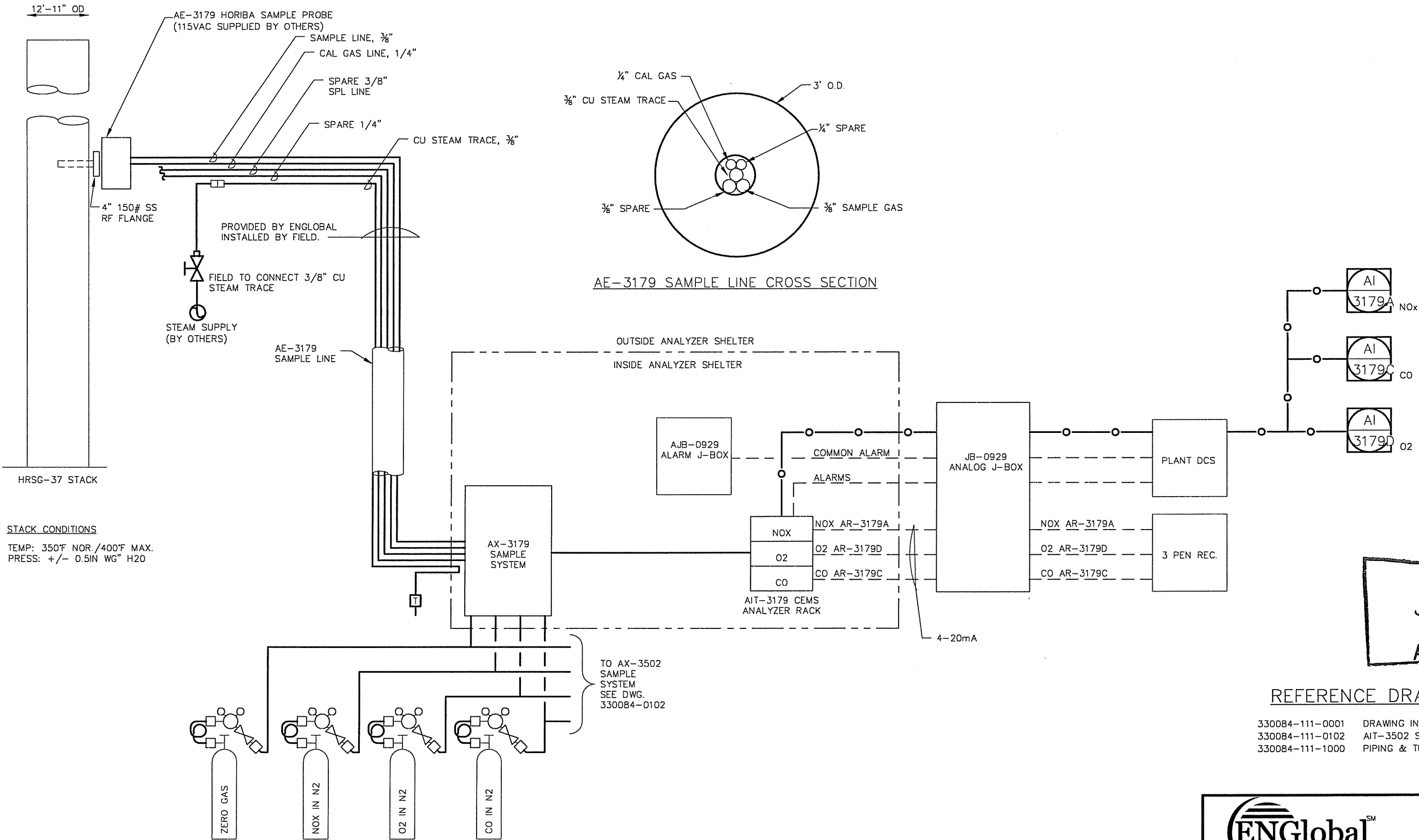


RA MUELLER INC. FLUID HANDLING EQUIPMENT

KBR Job No. S512
P.O. S512P02-1MJ43001
PS 3 HRSG No. 37
Project Motiva Enterprises LLC
Port Arthur Refinery
Powerhouse No. 3
724BLR37 - HSRG Package

SCALE:	NTS
DATE:	11/26/03

MFG.	MODEL	ANALYZER	RANGE	TAG NO.
HORIBA	ENDA-XE	NOx	0-25 PPM	AIT-3179
HORIBA	ENDA-XE	O2	0-25 %	AIT-3179
HORIBA	ENDA-XE	CO	0-200 PPM	AIT-3179



ISSUED
JUL 28 2004
AS BUILT

REFERENCE DRAWINGS

330084-111-0001	DRAWING INDEX
330084-111-0102	AIT-3502 SIMPLIFIED P&ID
330084-111-1000	PIPING & TUBING DIAGRAM



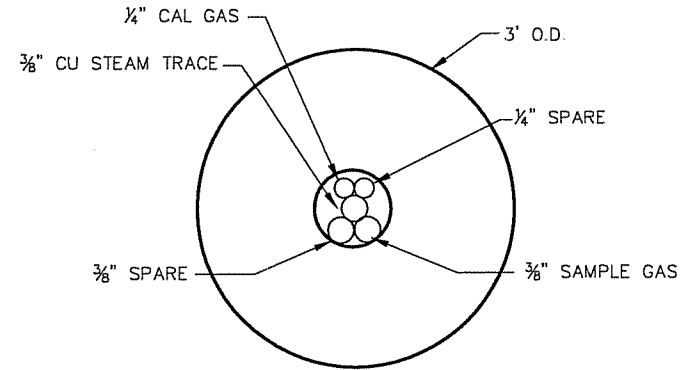
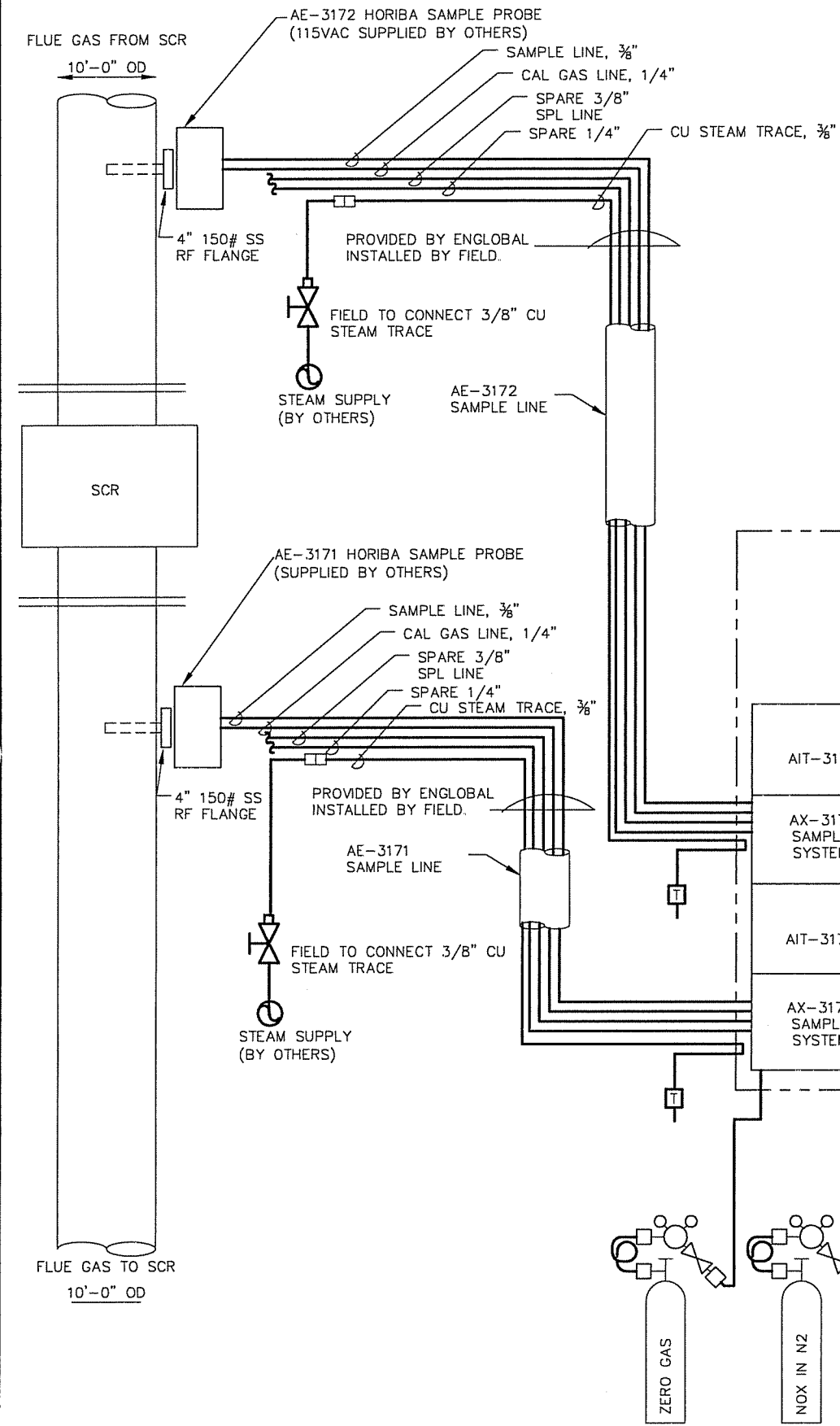
3255 Executive Blvd.
Beaumont, TX 77705

8901 Market Street
Houston, TX 77029
www.ENGGlobal.com

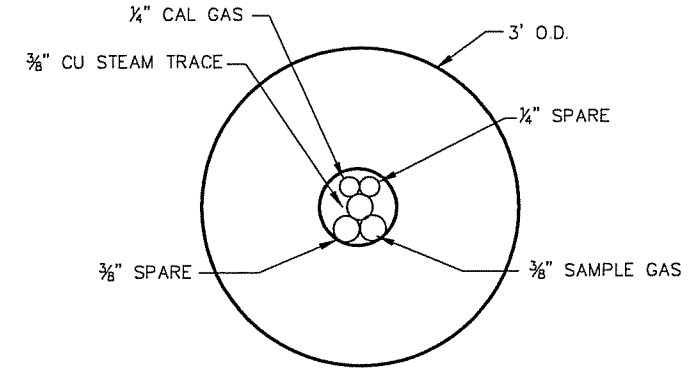
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1	REVISED PER CUSTOMER COMMENTS	RCB					4/26/04
0	ISSUED FOR CONSTRUCTION	RCB	TJE	JS			4/12/04
REV	REVISION DESCRIPTION	CADD	CHK'D.	APPR.	APPR.	APPR.	DATE

PROJECT	330084	SCALE	NONE	FILE	330084-111-0100.A.dwg	DWG #	330084-111-0100	REV	2
KBR/MOTIVA PORT ARTHUR REFINERY HRSG-37 CEMS/B-35 CEMS/SCR NOX SIMPLIFIED PIPING & INSTRUMENT DIAGRAM AIT-3179									

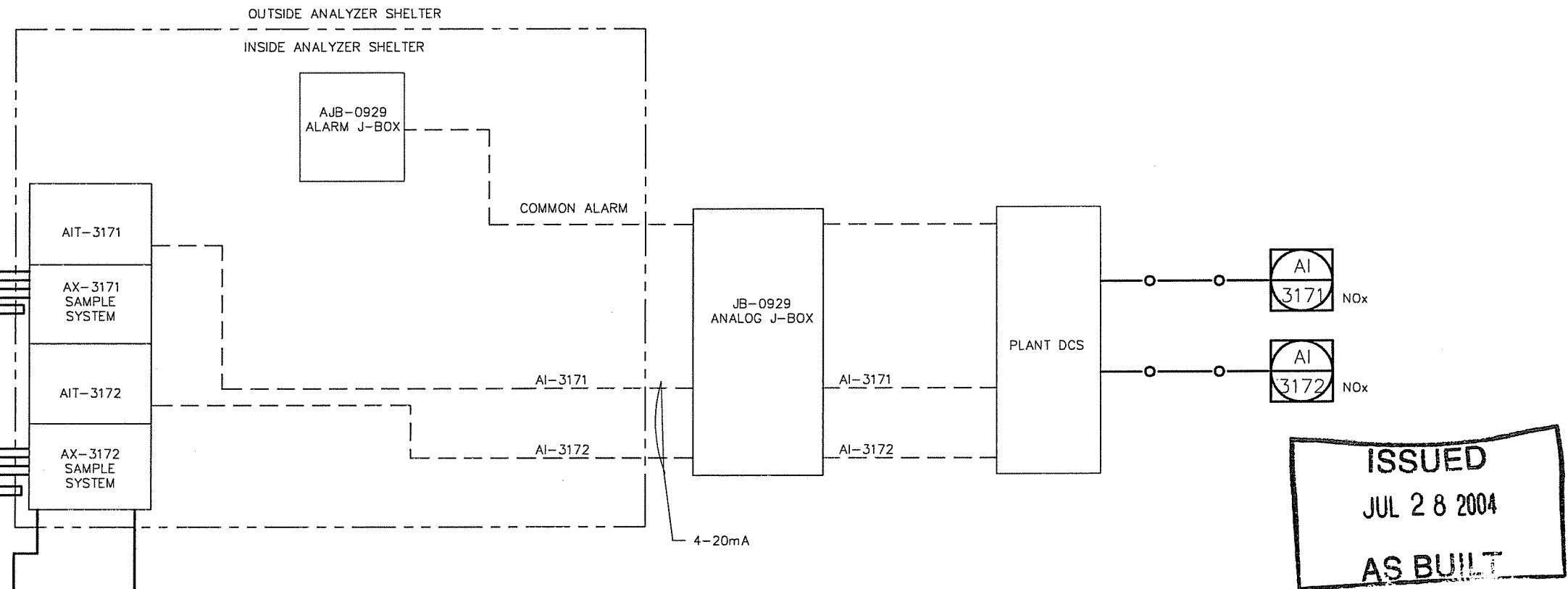
MFG.	MODEL	ANALYZER	RANGE	TAG NO.
HORIBA	CLA-510	NOx	0-25 PPM	AIT-3172
HORIBA	CLA-510	NOx	0-200 PPM	AIT-3171



AE-3171 SAMPLE LINE CROSS SECTION



AE-3172 SAMPLE LINE CROSS SECTION



ISSUED
JUL 28 2004
AS BUILT

REFERENCE DRAWINGS

330084-111-0001 DRAWING INDEX
330084-111-1000 PIPING & TUBING DIAGRAM

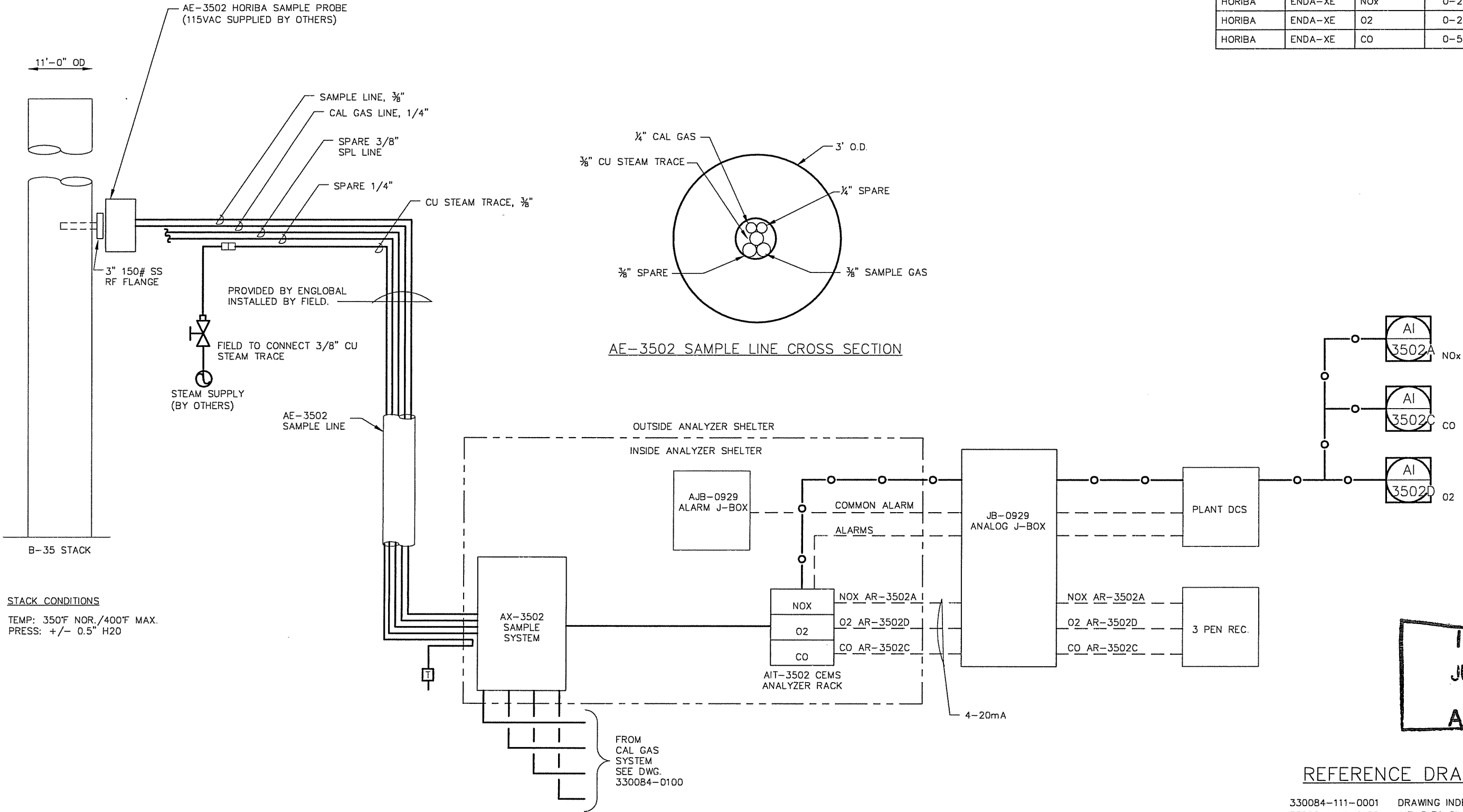
3255 Executive Blvd.
Beaumont, TX 77705

8901 Market Street
Houston, TX 77029

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2	ISSUED AS BUILT	RCB					6/29/04	PROJECT	KBR/MOTIVA PORT ARTHUR REFINERY HRSG-37 CEMS/B-35 CEMS/SCR NOX SIMPLIFIED PIPING & INSTRUMENT DIAGRAM AIT-3171 & AIT-3172			REV
1	REVISED PER CUSTOMER COMMENTS	RCB					4/26/04	SCALE				
0	ISSUED FOR CONSTRUCTION	RCB	TJE	JS			4/12/04	NONE				
REV	REVISION DESCRIPTION	CADD	CHK'D.	APPR.	APPR.	APPR.	DATE	FILE	330084-111-0101.A.dwg	DWG #	330084-111-0101	2

MFG.	MODEL	ANALYZER	RANGE	TAG NO.
HORIBA	ENDA-XE	NOx	0-200 PPM	AIT-3502
HORIBA	ENDA-XE	O2	0-25 %	AIT-3502
HORIBA	ENDA-XE	CO	0-500 PPM	AIT-3502



ISSUED
JUL 28 2004
AS BUILT

REFERENCE DRAWINGS

330084-111-0001	DRAWING INDEX
330084-111-0100	AIT-3179 SIMPLIFIED P&ID
330084-111-1000	PIPING & TUBING DIAGRAM

ENGlobal
Systems, Inc.

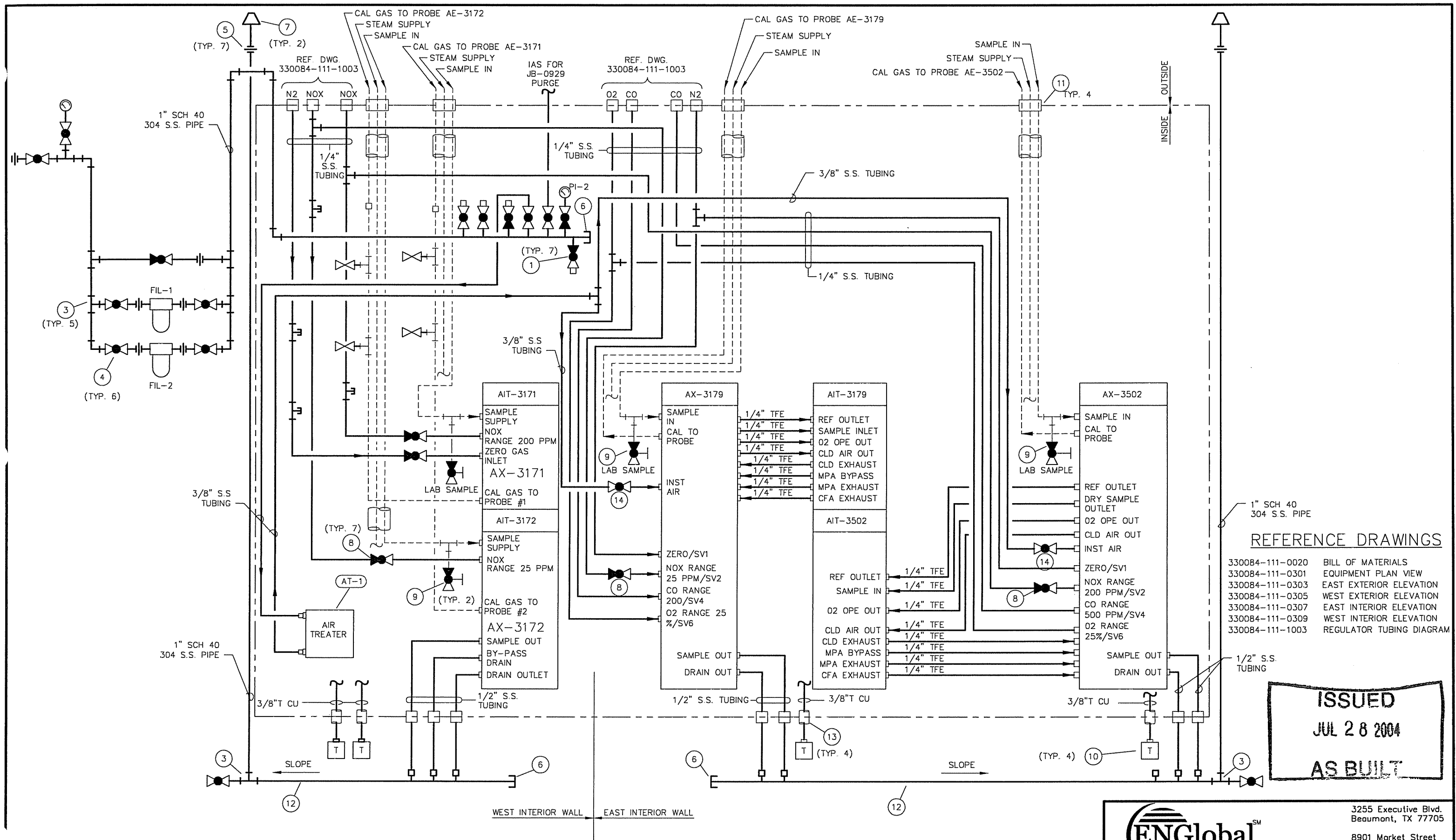
3255 Executive Blvd.
Beaumont, TX 77705

8901 Market Street
Houston, TX 77029

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2	ISSUED AS BUILT	RCB				6/29/04	PROJECT	KBR/MOTIVA PORT ARTHUR REFINERY	REV
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REV	REVISION DESCRIPTION	CADD	CHK'D.	APPR.	APPR.	DATE	NONE	AIT-3502	2

FILE	330084-111-0102.A.dwg	DWG #	330084-111-0102
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REFERENCE DRAWINGS

330084-111-0020	BILL OF MATERIALS
330084-111-0301	EQUIPMENT PLAN VIEW
330084-111-0303	EAST EXTERIOR ELEVATION
330084-111-0305	WEST EXTERIOR ELEVATION
330084-111-0307	EAST INTERIOR ELEVATION
330084-111-0309	WEST INTERIOR ELEVATION
330084-111-1003	REGULATOR TUBING DIAGRAM

ISSUED
JUL 28 2004
AS BUILT



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2	ISSUED AS BUILT	RCB				6/29/04	PROJECT	KBR/MOTIVA/PORT ARTHUR REFINERY
1	REVISED PER CUSTOMER COMMENTS	RCB				4/26/04	330084	HRSG-37 CEMS/B-35 CEMS/SCR NOX
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REV	REVISION DESCRIPTION	CADD	CHK'D.	APPR.	APPR.	DATE	NONE	AIT-3171, AIT-3172, AIT-3179, AT3502
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							REV	2

